

Inherited Disorders of the Renal Tubule

Alain Bonnardeaux | Daniel G. Bichet

CHAPTER OUTLINE

INHERITED DISORDERS ASSOCIATED WITH GENERALIZED DYSFUNCTION OF THE PROXIMAL TUBULE (RENAL FANCONI SYNDROME), 1450

INHERITED DISORDERS OF RENAL AMINO ACID TRANSPORT, 1461

INHERITED DISORDERS OF RENAL PHOSPHATE TRANSPORT, 1465

INHERITED DISORDER OF URATE TRANSPORT, 1469

INHERITED DISORDERS OF RENAL GLUCOSE TRANSPORT, 1469

INHERITED DISORDERS OF ACID-BASE TRANSPORTERS, 1470

BARTTER AND GITELMAN SYNDROMES, 1473

INHERITED DISORDERS WITH HYPERTENSION AND HYPOKALEMIA, 1475

PSEUDOHYPOALDOSTERONISM, 1479

INHERITED DISORDERS OF RENAL MAGNESIUM PROCESSING, 1481

DIABETES INSIPIDUS, 1483

Considerable progress has been made in understanding the molecular basis of inherited renal tubule disorders. These advances have allowed the identification of genes expressed in the renal tubule (Table 44.1), increasing our knowledge of basic renal physiology and pathobiology. Other benefits include potential prenatal and postnatal screening and better phenotype characterization and knowledge. Diseases described in this section are relatively rare (1:2000 or less; affecting fewer than 200,000 persons in the United States) and some were previously restricted to pediatric nephrology; however, advances in therapy have increased longevity for many patients, thus confronting the adult nephrologist with new challenges.

INHERITED DISORDERS ASSOCIATED WITH GENERALIZED DYSFUNCTION OF THE PROXIMAL TUBULE (RENAL FANCONI SYNDROME)

Renal Fanconi syndrome is a generalized dysfunction of the proximal tubule with no primary glomerular involvement. It is characterized by variable degrees of phosphate, glucose, amino acid, and bicarbonate wasting. Isolated or partial defects are described in other sections of this chapter. The clinical presentation in children consists of rickets and impaired growth. In adults, bone disease manifests as osteomalacia and osteoporosis. Polyuria, renal sodium and potassium wasting,

metabolic acidosis, hypercalciuria, and low-molecular-weight proteinuria may be part of the clinical spectrum.

There are hereditary and acquired variants of Fanconi syndrome. Acquired forms in adults are often associated with urinary excretion of protein corresponding to a paraproteinemic disorder or the nephrotic syndrome, with residual cases being secondary to tubular damage caused by toxic or immunologic factors.¹ Hereditary Fanconi syndrome occurs principally by one of two mechanisms: primary proximal tubule transport defects or accumulation of toxic metabolic products in the kidneys (Table 44.2).

PATHOGENESIS

The proximal tubule is responsible for reclaiming most of the filtered load of bicarbonate, glucose, urate, amino acids and low-molecular-weight proteins as well as an important fraction of the filtered load of sodium, chloride, phosphate, and water. It exhibits a very extensive apical endocytic apparatus consisting of an elaborate network of coated pits and small, coated and noncoated endosomes. In addition, the cells contain a large number of late endosomes, prelysosomes, lysosomes, and so-called dense apical tubules involved in receptor recycling from the endosomes to the apical plasma membrane. This endocytic apparatus is involved in the reabsorption of molecules filtered by the glomeruli (Fig. 44.1). The process is very effective, as demonstrated by the fact that although several grams of proteins are filtered daily,

the urine is virtually devoid of protein under physiologic conditions. Reabsorption of solutes by proximal tubule cells is achieved by transport systems at the brush border membrane that are directly or indirectly coupled to sodium movement, by energy production and transport from the mitochondria, and by the $\text{Na}^+\text{--K}^+\text{--adenosine triphosphatase}$ (ATPase) at the basolateral membrane. The $\text{Na}^+\text{--K}^+\text{--ATPase}$ lowers intracellular Na^+ concentration and provides the electrochemical gradient that allows Na^+ -coupled solute to enter the cell. A second route, the paracellular pathway, is responsible for reclaiming up to half of the sodium and most of the water through tight junctions.

Table 44.1 Impact of DNA Variation on Protein Function

Loss of function	A mutation that reduces or abolishes a normal physiologic function (likely to be recessive)
Gain of function	A mutation that increases the function of a protein (likely to be dominant)
Dominant negative	A mutation that dominantly affects the phenotype by means of a defective protein or RNA molecule that interferes with the function of the normal gene product in the same cell (likely to be dominant)

There are several genetic forms of Fanconi syndrome, the majority of which are associated with multisystem disorders (Table 44.2). In general, the disease mechanism for these disorders can be categorized as either (i) accumulation of a toxic metabolite (e.g., cystinosis, tyrosinemia, galactosemia, Fanconi–Bickel, congenital fructose intolerance, and Wilson disease), (ii) disruption of energy provision (e.g., mitochondrial cytopathies), or (iii) disruption of endocytosis and intracellular transport (e.g., Lowe syndrome, Dent disease, and arthrogryposis-renal dysfunction-cholestasis [ARC] syndrome).² The field continues to evolve and a novel mechanism secondary to monoallelic mutations in the gene encoding glycine amidinotransferase (GATM), a proximal tubular enzyme in the creatine biosynthetic pathway, characterized by renal Fanconi syndrome and kidney failure has just been published.³ All disease-related GATM mutations create an additional interaction interface within the GATM protein, promoting its linear aggregation. Aggregate-containing mitochondria in proximal tubular cells are associated with elevated production of reactive oxygen species, initiation of an inflammatory response, and increased cell death. These data establish a link between intramitochondrial GATM aggregates, renal Fanconi syndrome, and chronic kidney disease.⁴

Diseases associated with the accumulation of toxic metabolites are potentially reversible. For example, the defect is reversed after dietary restriction of tyrosine and phenylalanine in tyrosinemia,⁵ fructose in hereditary fructose intolerance,⁶ and galactose in galactosemia.⁷ The duration of the exposure

Table 44.2 Inherited Causes of Fanconi Syndrome

Gene	OMIM	Disorders/Systemic Disorders	Associated Features
Systemic Disorders			
<i>GALT</i>	230400	Galactosemia	Liver dysfunction, jaundice, encephalopathy, sepsis
Multiple nuclear DNA variants	Multiple	Mitochondrial cytopathies	Usually multisystem dysfunction (brain, muscle, liver, heart)
<i>FAH</i>	276700	Tyrosinemia	Poor growth, hepatic enlargement and dysfunction, liver cancer
<i>ALDOB</i>	229600	Congenital fructose intolerance	Rapid onset after fructose ingestion, vomiting, hypoglycemia, hepatomegaly
<i>CTNS</i>	219800	Cystinosis	Poor growth, vomiting, rickets ± corneal cystine crystals, kidney failure
<i>GLUT2</i>	227810	Fanconi–Bickel syndrome	Failure to thrive, hepatomegaly, hypoglycemia, rickets
<i>OCRL</i>	309000	Lowe syndrome	Males (X-linked), cataracts, hypotonia, developmental delay
<i>CLCN5, OCRL</i>	300009, 300555	Dent disease I, II	Males (X-linked), hypercalciuria, nephrocalcinosis
<i>ATP7B</i>	277900	Wilson disease	Hepatic and neurological disease, Kayser–Fleischer rings
<i>VPS33B, VIPAR</i>	208085, 613404	ARC syndrome	Arthrogryposis, platelet abnormalities, cholestasis
<i>HNF4A</i>	125850	MODY1	Neonatal hyperinsulinism, maturity-onset of diabetes in the young, mutation R76W shows RFS
Isolated Renal Fanconi Syndrome			
<i>GATM</i>	602360	FRTS1	Kidney failure
<i>SLC34A1</i>	613388	FRTS2	Phosphaturia dominating
<i>EHHADH</i>	615605	FRTS3	No kidney failure

ARC, Arthrogryposis-renal dysfunction-cholestasis; OMIM, Online Mendelian Inheritance in Man; RFS, renal Fanconi syndrome.

Modified from Klootwijk ED, Reichold M, Unwin RJ, et al. Renal Fanconi syndrome: taking a proximal look at the nephron. *Nephrol Dial Transplant*. 2015;30:1456–1460.

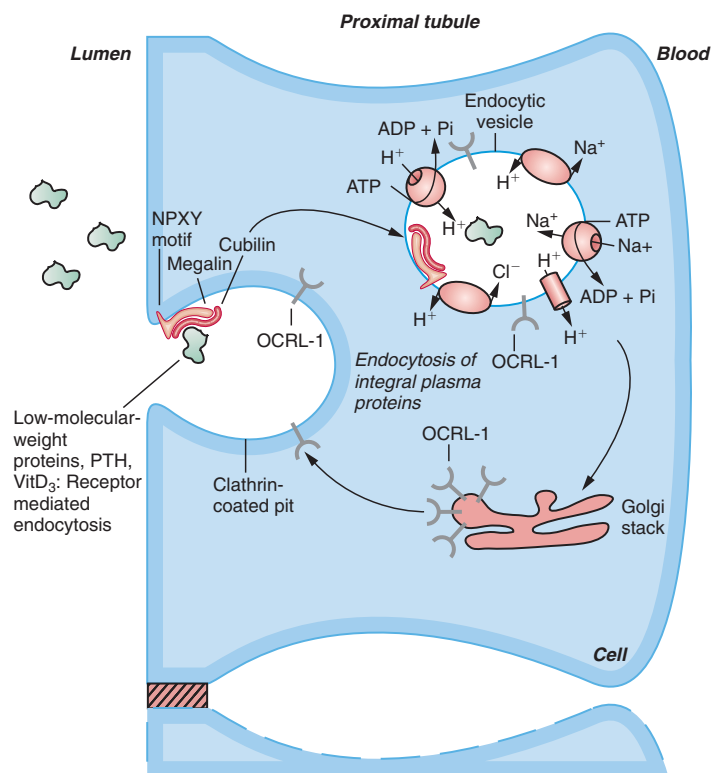


Fig. 44.1 The glomerular filtration barrier restricts the passage of large molecules, particularly those that are negatively charged molecules. The proteins appearing in the proximal tubule are reabsorbed by endocytosis (see the luminal part of the schematic representation). Vitamins and iron that are complexed to carrier proteins bind to megalin and/or cubilin followed by endocytosis. The ligands are released from the receptors by the low pH in the endosomes, and receptors recycle through the membrane recycling compartment. The protein component is degraded, whereas the vitamin, as well as iron, is transported across the epithelial cell (not represented). 1,25-Dihydroxyvitamin D₃ is activated by mitochondrial 1 α -hydroxylase, implying that the vitamin is transported to the cytoplasm, possibly by diffusion, before its release to the basolateral membrane. The regulation and maintenance of an endocytic vesicle pH are represented here according to Weisz,³⁵² whereby protons are pumped into the organelle by vacuolar H⁺-ATPase and can leave by the chloride proton exchanger ClC-5. The *ClC5* gene is mutated in patients with type 1 Dent disease, and the *OCRL1* gene is mutated in patients with type 2 Dent disease. OCRL1 encodes a phosphatase that is present on the trans-Golgi network and is important for regulating the traffic between the network, early endosomes, and clathrin-coated intermediate particles. ADP, Adenosine diphosphate; ATP, adenosine triphosphate; ATPase, adenosine triphosphatase; NPXY, asn-pro-any amino acid-tyrosine motif; OCRL-1, phosphatidylinositol 4,5-bisphosphate 5-phosphatase encoded by OCRL1 gene; P_i, inorganic phosphate; PTH, parathyroid hormone; Vit, vitamin.

is also important for the disorder to be expressed and is protracted in cadmium intoxication⁸ or brief in fructose intolerance following a fructose load.⁹

Considerable progress has been made in understanding the molecular basis of inherited renal tubule disorders. These advances have allowed the identification of genes expressed in the renal tubule (Table 44.1), increasing our knowledge of basic renal physiology and pathobiology. Other benefits include potential prenatal and postnatal screening and better phenotype characterization and knowledge. Diseases described in this section are relatively rare ($\leq 1:2000$; affecting $<200,000$ persons in the United States) and some were previously restricted to pediatric nephrology; however, advances in therapy have increased longevity for many patients, thus confronting the adult nephrologist with new challenges.

CLINICAL PRESENTATION OF RENAL FANCONI SYNDROME

AMINOACIDURIA

Amino acids are filtered by the glomerulus with more than 98% subsequently reabsorbed by multiple proximal tubule

transporters. In Fanconi syndrome, all amino acids are excreted in excess. The pattern of excretion of amino acids parallels that in physiologic conditions, so those excreted at the highest levels are histidine, serine, cystine, lysine, and glycine. Aminoaciduria is usually quantified by one of several chromatographic methods in specialized centers. Clinically, losses are relatively modest and do not lead to specific deficiencies. There is no need to supplement amino acids to affected subjects.

PHOSPHATURIA AND BONE DISEASE

Phosphate wasting is a cardinal manifestation but bone features are variable. Serum phosphate levels are usually decreased, and tubular reabsorption of phosphate (TRP) and maximal capacity—calculated by dividing maximum tubular reabsorption of phosphate (Tm_P) by glomerular filtration rate (GFR) or Tm_P/GFR—are systematically reduced. Rickets and osteomalacia, which are caused by increased urinary losses of phosphate as well as by impaired 1 α -hydroxylation of 1,25-dihydroxyvitamin D₃, can compose the dominant clinical picture. Rickets manifests as the bowing deformity of the lower limbs with metaphyseal widening of

the proximal and distal tibia, distal femur, ulna, and radius. Bone manifestations in subjects with adult-onset renal Fanconi syndrome are severe bone pain and spontaneous fractures.

RENAL TUBULAR ACIDOSIS

Hyperchloremic metabolic acidosis is a frequent finding and is caused by defective bicarbonate reabsorption. Hence, renal acidification by the distal tubule is normal, as demonstrated by the ability to acidify urine at a pH below 5.5 when plasma bicarbonate is below the threshold. Because the more distal segments have substantial bicarbonate reabsorptive capacity, the plasma bicarbonate concentration is usually maintained between 12 and 20 mmol/L. The diagnosis can be established by raising the plasma bicarbonate concentration with an intravenous sodium bicarbonate infusion (0.5–1 mmol/kg per hour) to 18 to 20 mmol/L. The fractional excretion of bicarbonate will usually rise to 15% to 20% in proximal renal tubular acidosis (pRTA) but will remain lower (3%) in distal RTA (dRTA). Treatment with large doses of alkali may be necessary to correct the acidosis.

GLUCOSURIA

Glucosuria is a common manifestation although the serum glucose level is normal, and the amount of glucose lost in the urine varies from 0.5 to 10 g/day. Glucosuria (and hypoglycemia) may be massive in glycogenosis type I.¹⁰

POLYURIA, SODIUM, AND POTASSIUM WASTING

Polyuria, polydipsia, and dehydration may be prominent features. Hypokalemia can contribute to decreased concentrating ability of the kidney from abnormal tubule function of the distal tubule and collecting duct. Recent studies suggest that autophagic loss of aquaporin-2 (AQP2) may contribute to the decreased urinary concentrating ability complicating hypokalemia.¹¹ A similar mechanism has also been proposed for the urinary concentrating defect accompanying hypercalcemia.¹² Renal sodium losses may be significant, leading to hypotension, hyponatremia, and metabolic alkalosis. Supplementation with sodium chloride is indicated and achieves clinical improvement. Potassium losses are secondary to increased delivery of sodium to the distal tubule and activation of the renin–angiotensin–aldosterone system from hypovolemia.

PROTEINURIA

Low-molecular-weight proteinuria is almost always present in low to moderate amounts. Endosomal machinery proteins, including the vacuolar type H⁺-ATPase (V-ATPase), ClC-5 chloride/proton exchanger channels, and the endocytotic receptors megalin and cubilin,¹³ are important in protein reabsorption. The urine dipstick test result is frequently positive because of the presence of albuminuria, typically at levels of 1 g/day. Rates of β_2 -microglobulin excretion are also elevated.

HYPERCALCIURIA

Hypercalciuria is a frequent manifestation in patients with renal Fanconi syndrome. The pathogenesis is not known but could be related to abnormal recycling of proteins involved in calcium reabsorption by the proximal tubule, natriuresis, and increased vitamin D synthesis from hypophosphatemia.

Hypercalciuria is only occasionally associated with nephrolithiasis, possibly because of the presence of polyuria.

DENT DISEASE

PATHOGENESIS

Dent disease, X-linked recessive hypophosphatemic rickets, and X-linked recessive nephrolithiasis (Online Mendelian Inheritance in Man [OMIM] entry #300009)¹⁴ are clinical manifestations of the same disease, reviewed elsewhere.¹⁵ Clinical features include primary Fanconi syndrome, low-molecular-weight proteinuria, hypercalciuria with calcium nephrolithiasis, nephrocalcinosis, rickets, and progressive renal failure. Dent disease 1 is caused by mutations in the *CLCN5* gene located on chromosome Xp11.22 encoding a lysosomal transport protein, ClC-5, a 2 chloride/1 proton antiporter.¹⁶ Defects in the *OCRL1* gene (oculocerebrorenal syndrome of Lowe), encoding a phosphatidylinositol 4,5-bisphosphate (PIP₂) 5-phosphatase (Ocr1) and usually found mutated in patients with Lowe syndrome, can also induce a Dent-like phenotype (type 2 Dent disease [OMIM #300555]; see later)^{17,18} (Fig. 44.1). Of the 32 families with the clinical diagnosis of Dent disease reported by Hoopes and colleagues,¹⁹ 19 (60%) had mutations in *CLCN5* and 5 families (16%) had mutations in *OCRL1* (for whom the diagnosis of Lowe syndrome had been excluded because of absence of cataracts). It is possible that additional genes are involved in families not linked to *CLCN5* and *OCRL1*.¹⁹

CLCN5 mutations expressed in vitro lead to a disorder manifesting as defects in receptor-mediated endocytosis and/or endosomal acidification.²⁰ ClC-5 colocalizes with the proton pump and internalized proteins early after uptake and is also expressed in type A intercalated cells. ClC-5 forms a dimer of two identical subunits, each of which contains a complete ion conduction pathway and is composed of 18 α -helices. It is thought to provide an electrical shunt for the acidification of vesicles of the endocytotic pathway required for receptor–ligand interactions and cell sorting events.²¹ Inhibition of the acidification interferes with cell-surface receptor recycling, reducing endocytosis of albumin and resulting in mistargeting of megalin, cubilin, Na⁺–H⁺-exchanger isoform 3 (NHE3), and the sodium/phosphate transporter NPT2a.

CLINICAL PRESENTATION

The clinical presentation is explained by the predominant expression of ClC-5 in the proximal tubule. Patients have varying degrees of low-molecular-weight proteinuria, hypercalciuria with calcium nephrolithiasis, rickets, nephrocalcinosis, and renal failure.²² There is considerable intrafamilial variability,²³ probably because of genetic and/or environmental modifiers. The disease affects male patients predominantly, and female patients have an attenuated phenotype. However, renal failure develops only in male patients.

The excretion of low-molecular-weight proteins in the urine such as albumin, β_2 -microglobulin, and α_1 -microglobulin is thought to be the most reliable marker for the disease. Affected male patients usually excrete β_2 -microglobulin in amounts that are more than 100-fold the upper limit of normal. Female carriers can also have low-molecular-weight proteinuria, but it is usually less pronounced than in male

patients and sometimes absent. Low-molecular-weight proteinuria is not a specific finding because it can be seen in tubulointerstitial diseases as well. The degree of proteinuria is relatively constant and amounts to 0.5 to 2 g/day in adults and up to 1 g/day in children.^{24–26} The nephrotic syndrome does not occur, and albumin excretion represents less than half of the proteins excreted. An attenuated form of the disease with low-molecular-weight proteinuria as the only or predominant feature appears to be prevalent in Japan.²⁷

Hypercalciuria is also a common finding in this disorder and is present in most cases, beginning in childhood. It is usually overt and predominant in male patients (>7.5 mmol/day). Female patients are also frequently hypercalciuric, but their values are usually closer to the upper limit of the normal range. Nephrolithiasis is frequent, with 50% of male patients affected. Stones are composed of calcium phosphate or a mixture of calcium phosphate and oxalate.²⁶ Multiple episodes starting during the teenage years are common. Radiologic nephrocalcinosis of the medullary type is seen in most affected male patients and occasionally in female patients. Serum phosphate levels are usually below normal values or at the lower limit of the normal range. Tm_P/GFR is decreased, indicating defective reabsorption by the proximal tubule. Rickets may be present in children and is cured by the administration of pharmacologic doses of vitamin D. Osteomalacia occurs in adults and is also corrected after administration of vitamin D. Serum levels of 1,25-dihydroxyvitamin D₃ are normal or slightly raised, whereas 25-hydroxyvitamin D levels are normal. The cause of hypercalciuria and renal stones is not known, but one possible explanation is that renal phosphate leak and reduced degradation of luminal parathyroid hormone (PTH) results in increased 1 α -hydroxylase activity and 1,25-dihydroxycholecalciferol production. Other explanations are abnormal trafficking (recycling) of transporters or channels necessary for calcium transport to the apical membrane, and decreased reabsorption of a regulatory protein by the proximal tubule.²⁸

Systemic acidosis is typically not seen before renal function deteriorates significantly. Male patients usually have urinary acidification defects detectable by an ammonium chloride load, but these are not consistent features of the phenotype. Spontaneous hypokalemia is common in male patients, and there is inability to concentrate urine maximally. Aminoaciduria and glucosuria are also frequent. Half the male patients have raised serum creatinine with progressive renal failure. End-stage kidney disease occurs at age 47 ± 13 years. Renal biopsy specimens show a pattern of a chronic interstitial nephritis with scattered calcium deposits.^{24–26} The glomeruli are normal or hyalinized. There is prominent tubular atrophy with diffuse inflammatory infiltrate composed of lymphocytes, and foci of calcification around and within epithelial cells. Molecular genetic testing is available and confirms the diagnosis.

TREATMENT

Renal stones and hypercalciuria are treated with supportive measures (and in particular, increasing fluid intake). Dietary restriction of calcium reduces calcium excretion but is not recommended because it might contribute to the bone disease.²⁹ Thiazide diuretics can be given in small doses and may decrease calciuria,³⁰ but in patients with Dent disease, who have a tendency for salt wasting, response to these agents

seems to consist of excessive diuresis, kaliuresis, and blood pressure reduction.²⁶ Rickets are treated with small doses of vitamin D, but this treatment should be given with caution as it might increase urine calcium excretion and the risk of nephrolithiasis. Verifying urine calcium excretion before and after vitamin D therapy might be appropriate.²⁹ There is no specific treatment for preventing progression of renal failure, although citrate has been shown to delay progression in a *Clec-5* knockout mouse model.³¹ This goal has to be balanced against the possibility of increasing calcium phosphate supersaturation and stone formation by urine alkalization.

OCULOCEREBRORENAL DYSTROPHY (LOWE SYNDROME)

The oculocerebrorenal syndrome (OCRL) of Lowe (OMIM #309000) is an X-linked recessive multisystem disorder characterized by congenital cataracts, mental retardation, and incomplete renal Fanconi syndrome with renal failure occurring later in life (reviewed elsewhere^{32,33}).

PATHOGENESIS

Mutations in the *OCRL1* gene are responsible for OCRL.^{34,35} *OCRL1* encodes a 105-kDa Golgi protein with PIP₂ 5-phosphatase activity. Phosphorylation of phosphatidylinositol at the 3, 4, or 5 position of the inositol ring generates seven phosphoinositides (PIs) that play a central role in the regulation of diverse cellular processes, including gene expression, cytokinesis, cell motility, actin cytoskeleton remodeling, membrane trafficking, and cell signaling.³⁶ Therefore OCRL1 is mainly a lipid phosphatase that may control cellular levels of a critical metabolite, PIP₂,³⁷ and is involved in the inositol phosphate signaling pathway. OCRL1 is likely to regulate cellular trafficking. It is present in the trans-Golgi network, in early endosomes, and in clathrin-coated intermediates that move between these compartments³⁷ that play key regulatory functions in cell physiology reviewed by Pirruccello and colleagues.³⁸ The related soluble inositol polyphosphates and pyrophosphates, generated from inositol 1,4,5-trisphosphate (IP₃)—a product of phosphatidylinositol 4,5-bisphosphate [PI(4,5)P₂] cleavage by phospholipases—are also important signaling molecules. Phosphorylated headgroups of PIs, which are localized on the cytosolic leaflets of membranes and help define membrane identity, interact with a variety of amino acid motifs or protein domains, and regulate protein–bilayer interactions. Thus OCRL1 deficiency may result in reductions in recycling of receptors and delivery of cargo to the plasma membrane. This possibility is consistent with the apparent loss of megalin from the apical membrane and protein absorption defects seen in patients (Fig. 44.1). It is not clear why the loss of OCRL1, a ubiquitously expressed protein, should result in defects to only the eyes, brain, and kidney proximal tubule. Compensation by another enzyme with overlapping specificity in the unaffected tissues is a possible explanation.

CLINICAL PRESENTATION

OCRL is a multisystem disorder characterized by ocular, neurologic, and renal defects. Renal dysfunction (Fanconi syndrome) is a feature that may occur in the first year of life, but with great variability in severity and age of onset. It

is characterized by proteinuria (0.5–2 g of daily urinary protein per square meter of body surface area), generalized aminoaciduria (100–1000 mmol of daily urinary amino acid per kg of body weight), carnitine wasting (mean fractional excretion, 0.05–0.15), phosphaturia, and bicarbonaturia.³³ Glucosuria is variably present as well. Linear growth decreases after 1 year of age. Glomerular function decreases with age, with end-stage kidney disease predicted between the second and fourth decade of life.

Neurologic findings include infantile hypotonia, mental retardation, and areflexia. Prenatal development of cataracts is universal, and other ocular anomalies are glaucoma, microphthalmos, and corneal keloid formation. Visual acuity is frequently decreased. Mental retardation is very common. Cranial magnetic resonance imaging (MRI) shows mild ventriculomegaly and cysts in the periventricular regions. Status epilepticus is also frequent. Life span rarely exceeds 40 years.

Some patients with a phenotype indistinguishable from Dent disease (see previous discussion in the “[Dent Disease](#)” section) carry mutations in the *OCRL1* gene and have an attenuated form of OCRL.¹⁷

The diagnosis is established in affected individuals by the demonstration of reduced (<10% of normal) activity of inositol polyphosphate 5-phosphatase OCRL-1 in cultured skin fibroblasts. Molecular genetic testing of *OCRL* detects mutations in approximately 95% of affected males and a similar proportion of carrier females.³⁹ Carrier detection by slit-lamp examination has high but not absolute sensitivity. Concentrations of the muscle enzymes creatine kinase, aspartate aminotransferase, and lactate dehydrogenase, as well as of total serum protein, serum α_2 -globulin, and high-density lipoprotein cholesterol, are elevated.

TREATMENT

Treatment of OCRL is supportive and includes taking care of ocular (cataract extraction, treatment of glaucoma), neurologic (anticonvulsants, speech therapy), and renal complications. Nasogastric tube feedings or feeding gastrostomy and antipsychotic therapy may be required. Bicarbonate therapy is usually given at a dose of 2 to 3 mmol/kg/day every 6 to 8 hours, but may vary from 1 to 8 mmol/kg/day to maintain a serum bicarbonate concentration of 20 mmol/L. Sodium or potassium phosphate can be given in amounts of 1 to 4 g/day for phosphate depletion, and vitamin D may be added if that approach is unsuccessful.

MISTARGETING TO MITOCHONDRIA OF PEROXISOMAL EHHADH, AN ENZYME INVOLVED IN PEROXISOMAL OXIDATION OF FATTY ACIDS

Peroxisomes were first identified by Christian de Duve, and it was originally thought that the primary function of these organelles was the metabolism of hydrogen peroxide. One of their main metabolic functions is β -oxidation of very-long-chain fatty acids (for review see Lodhi and Semenkovich⁴⁰). An EHHADH (enoyl-CoA hydratase/3-hydroxyacyl-CoA dehydrogenase) dominant mutant has been shown to mistarget to mitochondria and not to peroxisomes, impair mitochondrial oxidative phosphorylation, and cause renal Fanconi syndrome ([Fig. 44.2](#)).⁴¹

IDIOPATHIC CAUSES OF RENAL FANCONI SYNDROME

Renal Fanconi syndrome occurs in the absence of both known inborn errors of metabolism and acquired causes. Sporadic and familial cases unlinked to *CLCN5* (Dent disease), *OCRL1* (Lowe syndrome), or EHHADH⁴¹ have been described,¹⁹ with variable progressive renal failure.^{33,42,43}

CYSTINOSIS

PATHOGENESIS

Cystinosis is a rare autosomal recessive disease of lysosomal transport of the disulfide amino acid cystine.^{44–46} Lysosomes are intracellular organelles containing enzymes responsible for the digestion of macromolecules. Lysosomal hydrolases are optimally active at low pH. The by-products of the hydrolytic digestion exit the lysosome through specific transporters. A defect in one of the lysosomal hydrolases results in the accumulation of macromolecules or by-products that lead to lysosomal, cellular, and eventually organ dysfunction. Inactivating mutations in the *CTNS* (cystinosis) gene on chromosome 17p13, encoding an integral lysosomal membrane protein termed *cystinosin*,⁴⁷ cause cystinosis. This lysosomal cystine transporter has seven transmembrane domains and at least two lysosomal targeting signals (GYDQL in the C terminus and YFPQA in the fifth intertransmembrane loop). Cystine is poorly soluble and forms crystals in the lysosomes, but the mechanisms leading to cellular dysfunction are not clear.⁴⁸

CLINICAL PRESENTATION

Cystinosis, which affects 1 in 100,000 to 1 in 200,000 newborns, is the most frequent cause of the inherited forms of renal Fanconi syndrome. The clinical presentation is variable^{49,50} and encompasses classic nephropathic cystinosis (OMIM #219800), a rare “adolescent” form (OMIM #219900), and also a mild adult-onset variant (OMIM #219750; [Table 44.3](#)). The most severe form manifests in the first year of life as failure to thrive, increased thirst, polyuria, and poor feeding. Affected children of Caucasian parents are frequently blond-haired and have white skin, which may be explained by the role of cystinosin in melanogenesis.⁵¹ Additional clinical findings include phosphaturia, rickets, aminoaciduria, glucosuria, hypouricemia, and bicarbonaturia. Renal wasting of sodium, calcium (leading to nephrocalcinosis), and magnesium and tubular proteinuria are usually present. If the cystinosis is not treated, progressive renal damage culminates in end-stage kidney disease by the end of the first decade of life.

Damage in multiple organ systems, including ocular, endocrine, hepatic, muscular, and central nervous, has been reported. Cystinosis can affect most of the structures of the eye, with variable rates of progression. In the cornea, crystal deposits are absent at birth and appear by the end of the first year of life. These can be seen by slit-lamp examination as fusiform crystals involving the anterior third of the central cornea and the full thickness of the peripheral cornea. Eventually, these deposits progress to develop a characteristic haziness. They can also be found in the iris and conjunctiva as well as in the retina, with consequent development of a characteristic peripheral retinopathy.

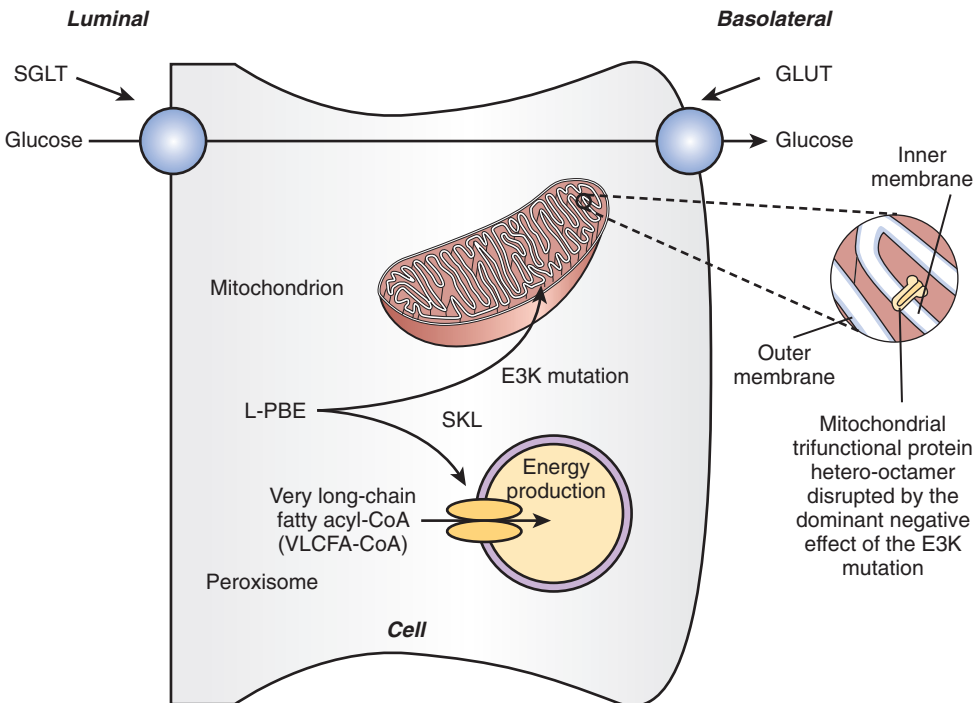


Fig. 44.2 Proposed model of a dominant negative effect of mutant peroxisomal L-bifunctional enzyme disrupting the mitochondrial trifunctional protein heterooctamer in the renal proximal tubule. Glucose is reabsorbed from the luminal (urine-side) membranes to the basolateral (blood-side) membranes. Most of the basolateral fatty acid uptake is used for mitochondrial energy production by means of trifunctional protein. A small fraction of the fatty acid uptake is used for peroxisomal energy production by means of bifunctional protein (D-bifunctional enzyme [D-PBE]). The physiologic targeting motif, SKL (serine–lysine–leucine), normally directs L-bifunctional enzyme (L-PBE) to the peroxisome. The E3K mutation instead creates a targeting motif that directs L-PBE to the mitochondrion, where its homology causes it to interfere with the assembly and function of trifunctional protein. *GLUT*, Glucose transporter; *SGLT*, sodium-glucose linked transporter. (Modified from Klotzwijk ED, Reichold M, Helio-Wooley A, et al. Mistargeting of peroxisomal EHHADH and inherited renal Fanconi's syndrome. *N Engl J Med*. 2014;370:129–138.)

Table 44.3 Age-Related Clinical Characteristics of Untreated Nephropathic Cystinosis

Age (years)	Symptom or Sign	Prevalence of Affected Patients (%)
6–12 months	Renal Fanconi syndrome (polyuria, polydipsia, electrolyte imbalance, dehydration, rickets, growth failure)	95
5–10	Hypothyroidism	50
8–12	Photophobia	50
8–12	Chronic renal failure	95
12–40	Myopathy, difficulty swallowing	20
13–40	Retinal blindness	10–15
18–40	Diabetes mellitus	5
18–40	Male hypogonadism	70
21–40	Pulmonary dysfunction	100
21–40	Central nervous system calcifications	15
21–40	Central nervous system symptomatic deterioration	2

From Gahl WA, Thoene JG, Schneider JA. Cystinosis. *N Engl J Med*. 2002;347:111-121.

Other features include hypothyroidism from cystine crystallization in the follicular cells of the thyroid gland. It is present in more than 70% of patients older than 10 years. Insulin-dependent diabetes mellitus develops from long-standing cystine crystal accumulation in the pancreas, particularly after renal transplantation.⁵² Hepatomegaly and splenomegaly with little clinical impact also occur in more than 40% of subjects older than 10 years.⁴⁹ A distal vacuolar myopathy is a late finding in 25% of cystinotic patients, with wasting in the small muscles of the hand. Facial weakness and dysphagia are often seen. In a previous study, muscle biopsies revealed marked fiber size variability, prominent acid phosphatase-positive vacuoles, and an absence of fiber-type grouping or inflammatory cells. Crystals of cystine were detected in cells of the perimysium but not within the muscle cell vacuoles. The muscle cystine content of clinically affected muscles is markedly elevated.⁵³ Central nervous system involvement has been described in the late stages of the disease,⁴⁵ and cystine crystal accumulation has been reported. The accumulation of intracellular cystine itself may also be a risk factor for vascular calcifications.⁵⁴

The diagnosis is usually established by measuring the cystine content in peripheral leukocytes. Patients usually have values higher than 2 nmol of half-cystine per milligram of protein (normal <0.2 nmol). Alternatively, the diagnosis can be made through recognition of the characteristic corneal crystals on slit-lamp examination.

Approximately half of patients who have cystinosis and are of northern European descent carry at least one allele that bears a specific 57-kb deletion that encompasses the *CTNS* gene. Molecular analysis of the cystinosis gene allows early diagnosis and can be used for prenatal diagnosis as well. Prenatal diagnosis of cystinosis can also be made by measuring S-labeled cystine accumulation in cultured amniocytes or chorionic villi samples,⁴³ and by a direct measurement of cystine in uncultured chorionic villi samples. Over 90 mutations have been reported, with a detection ratio close to 100%.⁵⁵

TREATMENT

Patients with cystinosis who are treated at an early age are now expected to live a nearly normal life, but a limited access to cysteamine still exists in developing nations.⁵⁶ Before development of the cystine-depleting drug cysteamine, dialysis had to be initiated on average in patients reaching age 10. Cysteamine enters the lysosome by a specific transporter for aminothiols or aminosulfides and cleaves cystine into cysteine and a cysteine–cysteamine mixed disulfide (Fig. 44.3). Oral cysteamine is given at doses of 60 to 90 mg/kg per day every 6 hours and generally achieves approximately 90% depletion of cellular cystine, as measured in circulating leukocytes.⁵⁷ A long-acting form appears to have similar efficacy.⁵⁸ It slows the rate of progression of renal failure and increases growth in affected subjects.^{59–61} Kidney function stabilizes on initiation of therapy, and even some recovery can be seen if therapy is begun in the first year or two of life.⁶⁰ The growth rate becomes normal, but there is no “catching up.” Topical cysteamine eye drops⁶² and a new gel formulation⁶³ are used to treat ocular complications of cystinosis; they cause dissolution of corneal crystals.

Symptomatic treatment also involves rehydration, particularly during episodes of gastroenteritis. Replacement of bicarbonate losses with citrate- or bicarbonate-containing salts is frequently necessary. Phosphate losses are replaced with phosphate salts and oral vitamin D therapy. Indomethacin has been used to decrease renal salt and water wasting. Recombinant human growth hormone increases growth but does not hasten the rate of progression of renal failure.⁶¹ Kidney transplantation is routinely performed, and most recipients do well, although extrarenal late-onset complications will occur.⁶⁴ Kidneys from heterozygous family donors are widely accepted because there is no evidence of cystine accumulation in kidney transplants from such donors.

GLYCOGENOSIS (VON GIERKE DISEASE, OMIM #232200)

Glycogen storage diseases are inherited disorders that affect glycogen metabolism (Fig. 44.4).⁶⁵ Both the liver and muscle store physiologically important quantities of glycogen. These organs are primarily affected, with the symptoms usually including hepatomegaly, hypoglycemia, muscle cramps and weakness, exercise intolerance, and fatigue. This section discusses type I glycogen storage disease because it is the only form associated with primary renal involvement. Type V glycogen storage disease (McArdle disease) as well as other rare glycogenoses associated with rhabdomyolysis, myoglobinuria, and acute tubular necrosis are not discussed further.

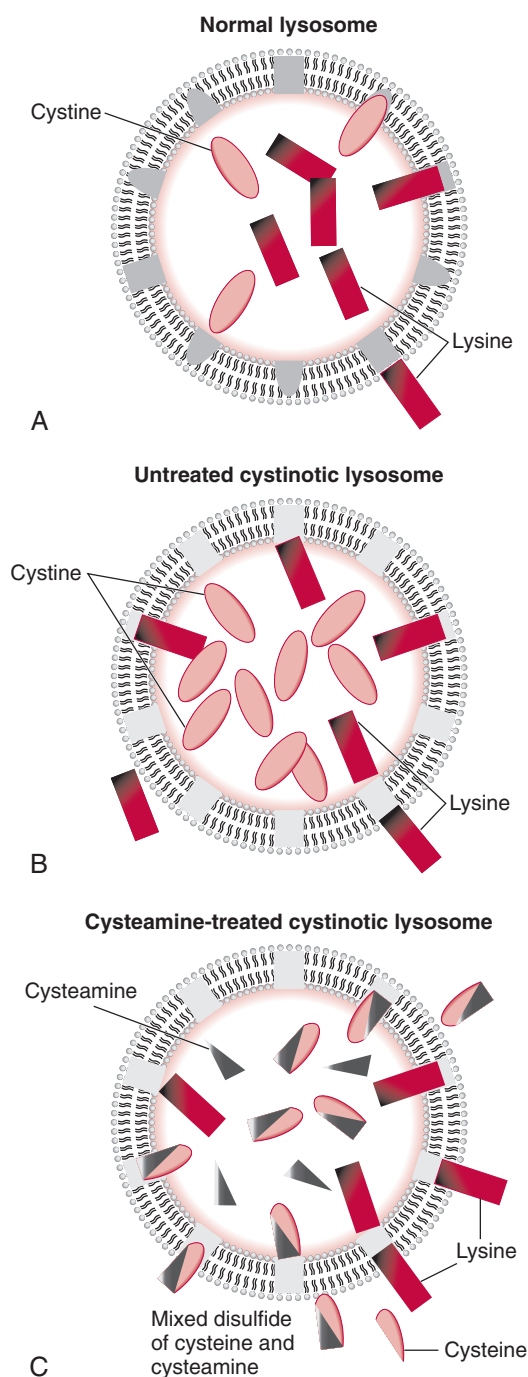


Fig. 44.3 Mechanism of cystine depletion by cysteamine. (A) In normal lysosomes, cystine and lysine freely traverse the lysosomal membrane through specific transporters (rectangles for the lysine transporter; ovals for the cystine transporter). (B) In cystinotic lysosomes (note the absence of specific cystine transporters), lysine can freely traverse through specific transporters in the lysosomal membrane, but cystine cannot and therefore accumulates inside the lysosome. (C) In cysteamine-treated lysosomes, cysteamine combines with half-cystine (i.e., cysteine) to form the mixed disulfide cysteine–cysteamine, which uses the lysine transporter to exit the lysosome. (Modified from Gahl WA, Thoene JG, Schneider JA. Cystinosis. *N Engl J Med*. 2002;347:111–121. Copyright © 2002 Massachusetts Medical Society. All rights reserved. With permission.)

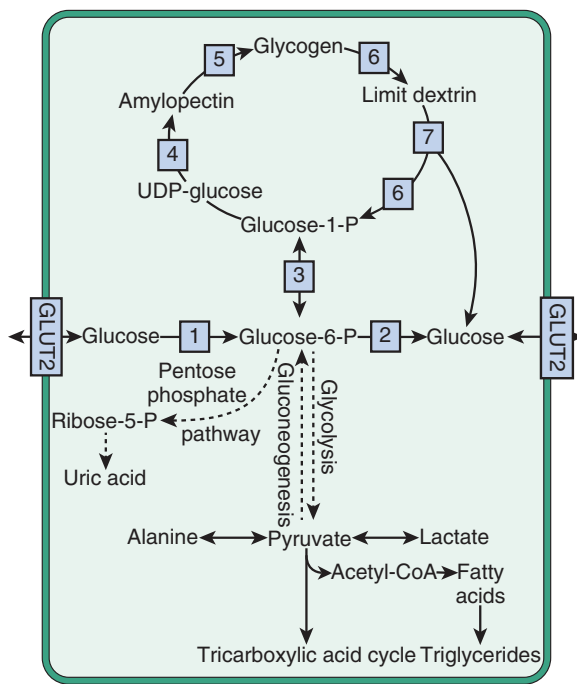


Fig. 44.4 Simplified scheme of glycogen synthesis and breakdown. 1, Hexokinase/glucokinase; 2, glucose-6-phosphatase (G6Pase); 3, phosphoglucomutase; 4, glycogen synthase; 5, branching enzyme; 6, glycogen phosphorylase; 7, debranching enzyme; CoA, coenzyme A; GLUT2, glucose transporter 2; P, phosphate; UDP-glucose, uridine diphosphoglucose. (From Wolfsdorf JI, Weinstein DA. Glycogen storage diseases. *Rev Endocr Metab Disord*. 2003;4:95–102.)

PATHOGENESIS

Glycogen storage disease type I (GSD-I), also known as von Gierke disease, is a group of autosomal recessive metabolic disorders caused by deficiencies in the activity of the glucose-6-phosphatase (G6Pase) system that consists of at least two membrane proteins, the glucose-6-phosphate transporter (G6PT) and G6Pase.⁶⁵ G6PT and G6Pase work in concert to maintain glucose homeostasis. G6Pase catalyzes the hydrolysis of glucose-6-phosphate (G6P) to produce glucose and phosphate. GSD-Ia (G6Pase deficiency, 80% of cases)^{66,67} is caused by deficiency in enzymatic activity. GSD-Ib is caused by mutations in G6PT, which translocates G6P from the cytoplasm to the lumen of the endoplasmic reticulum (ER). Other variants include GSD-Ic (defect in microsomal phosphate or pyrophosphate transport) and GSD-Id (defect in microsomal glucose transport). The molecular basis of these variants remains to be determined.

CLINICAL PRESENTATION

GSD-I manifests as functional G6Pase deficiency, characterized by growth retardation, hypoglycemia, hepatomegaly, kidney enlargement, hyperlipidemia, hyperuricemia, and lactic acidemia. Patients with type Ib disease also suffer from chronic neutropenia and functional deficiencies of neutrophils and monocytes, resulting in recurrent bacterial infections as well as ulceration of the oral and intestinal mucosae. GSD-Ia manifests in the first year of life with hepatomegaly and/or hypoglycemic seizures with lactic acidosis. Hypoglycemia occurs because of impaired gluconeogenesis, glycogenolysis, and recycling of

glucose through G6P to the glucose system. Adults usually present with hypoglycemic symptoms that are exacerbated by exercise and relieved by food. However, 48-hour fasting blood glucose levels are frequently normal.⁶⁸ The accumulation of G6P leads to an increase in glycolysis and lactic acidosis. Hyperuricemia and gout are caused by increased activity of hepatic adenosine monophosphate (AMP) deaminase and adenine nucleotide production, thus increasing uric acid production. Hyperuricemia also results from decreased renal excretion because urate competes with lactate for secretion in the proximal tubule. Gout occurs in adults but is rarely seen in children. An enlarged abdomen from hepatomegaly, short stature from impaired growth, and skin manifestations from dyslipidemia (xanthomas) are frequent findings. Dyslipidemia is a result of increased synthesis of very-low-density lipoprotein and low-density lipoprotein and decreased lipolysis. Impairment of platelet adhesion and aggregation with prolonged bleeding time lead to easy bruising and epistaxis.

A renal Fanconi syndrome occurs in GSD-I, with aminoaciduria, low-molecular-weight proteinuria, phosphaturia, and bicarbonaturia.^{69,70} Renal disease is common in adult patients⁷¹ with untreated GSD-I. It evolves slowly and is a late finding. In children, increased kidney size, hyperfiltration, and moderate proteinuria are common.⁷² dRTA, hypocitraturia, hypercalciuria, and nephrocalcinosis with calcium nephrolithiasis can be variably associated,⁷³ but virtually all patients have impaired distal tubular acidification. The most common finding is focal and segmental glomerulosclerosis with tubulointerstitial atrophy. Glomerular changes include thickening, lamellation, and glycogen deposition in the glomerular basement membrane.⁷⁴

The diagnosis of GSD-I is established by DNA testing of *G6PC* and *SLC37A4*, or rarely from liver biopsy.⁷⁵ A 1-mg intramuscular glucagon test can be used and the result is frequently abnormal (an increase in blood glucose level <4 mmol/L, usually at 30 minutes). Liver histology reveals prominent storage of glycogen and fat, large lipid vacuoles with hepatocyte distension, and steatosis with little fibrosis. Abnormally high glycogen levels are noted in liver biopsy samples. Electron microscopy shows moderate to large excesses of glycogen in the cytoplasm, often displacing the organelles in hepatocytes. Enzyme analysis of fresh and frozen samples can distinguish type Ia from type Ib GSD.

TREATMENT

Life expectancy for patients with GSD-I has improved considerably. The treatment goal is to maintain normoglycemia to avoid the metabolic complications that are secondary to hypoglycemia and lactic acidosis. This can be accomplished at night with nasogastric feeding of glucose⁷⁶ or with orally administered uncooked cornstarch. A single dose of cornstarch (1.75–2.5 g/kg) at bedtime will maintain serum glucose concentrations above 3.9 mmol/L for 7 hours or longer in most young adults.⁷⁷ Because hypoglycemia and lactic acidosis occur in adults as well, treatment might also be indicated after childhood.⁷⁸ Guidelines for the management of GSD-I published by the European Study on Glycogen Storage Disease Type I⁷⁹ include a preprandial blood glucose level higher than 3.5 to 4.0 mmol/L (60–70 mg/dL), a urine lactate-to-creatinine ratio lower than 0.06 mmol/mmol, a serum uric acid concentration in the high normal range for age, a venous blood bicarbonate level above 20 mmol/L (20 mEq/L), a

serum triglyceride concentration below 6.0 mmol/L (531 mg/dL), normal fecal alpha₁-antitrypsin concentration for GSD-Ib, and a body mass index within two standard deviations of normal. Enzyme replacement therapy is not currently available for GSD-I. Kidney transplantation has been successfully performed but does not correct the hypoglycemia.

TYROSINEMIA

Hepatorenal tyrosinemia (tyrosinemia type I, OMIM #276700) is a rare autosomal recessive disorder caused by deficiency of fumarylacetoacetate hydrolase (FAH),⁸⁰ leading to the accumulation of fumarylacetoacetate (FAA) in body fluids and tissues, affecting principally the liver, kidneys, and peripheral nerves. The disorder is characterized by severe liver disease, which either causes liver failure in infancy or may take a more protracted course, with death often occurring during childhood or adolescence because of hepatocarcinoma.⁸¹ Worldwide, the incidence is 1 in 100,000. The disorder is particularly prevalent in a genetic isolate in Canada (Saguenay-Lac-Saint-Jean), in which the carrier rate is 1 in 20 and the incidence is 1 in 2000.

PATHOGENESIS

Mutations in the gene encoding FAH on chromosome 15q23-q25 are responsible for tyrosinemia type I.⁸²⁻⁸⁶ The cellular toxicity is caused by accumulation of FAA, which apparently induces oxidative stress⁸⁷ and the release of cytochrome *c*, thus triggering the activation of the caspase cascade in hepatocytes as seen in affected animal models.⁸⁸ It is unlikely that tyrosine accumulation itself leads to the hepatic and renal manifestations of the disease, because hypertyrosinemia has been described in other settings without renal and liver involvement.

CLINICAL PRESENTATION

Most patients present clinically in regions without a specific newborn screening program for tyrosinemia type I, although a few may be identified because of affected siblings or abnormalities of plasma or urine amino acids. Renal involvement is almost always present in tyrosinemic subjects and is probably caused by succinylacetone toxicity.^{89,90} It ranges from mild tubular dysfunction to renal failure. Hypophosphatemic rickets is the principal sign of tubular dysfunction, and acute decompensation can exacerbate the dysfunction. Generalized aminoaciduria is frequent. Nephrocalcinosis and nephromegaly can often be seen on renal ultrasonography.⁹¹ Glucosuria and proteinuria are usually mild. GFR is frequently decreased. Tubular defects respond to diet but may be irreversible in chronic cases.

The liver is the main organ affected in tyrosinemia type I. Initially, liver dysfunction often affects coagulation factors, even before other signs of liver failure appear. In fact, jaundice and liver enzyme elevations are rare in the early stages of tyrosinemia. A common presentation mode is the “acute hepatic crisis,” in which ascites, jaundice, and gastrointestinal bleeding are precipitated by an acute event such as an infection. Acute hepatic crises usually resolve spontaneously but on occasion progress to complete liver failure and encephalopathy. Cirrhosis eventually develops in most patients with the disease, and hepatocellular carcinoma is frequent in tyrosinemic subjects with chronic liver disease.⁹² It is believed

that toxic metabolites that accumulate in tyrosinemia, such as FAA, are mutagenic and contribute to the elevated rate of liver carcinoma.⁹³ Serial liver imaging (ultrasonography and/or MRI) is indicated. Neurologic crises are acute episodes of peripheral neuropathy with painful paresthesias and, eventually, autonomic dysfunction.⁹⁴

The diagnosis of tyrosinemia type I is made with the detection of excess succinylacetone in urine or plasma. Succinylacetone can be detected in the biologic fluid of all untreated patients with the disorder. The increase in succinylacetone is typically detected by gas chromatography/mass spectroscopy of extracted organic acids. Confirmatory testing can be performed with the measurement of FAH in cultured skin fibroblasts or the documentation of pathogenic mutations from DNA. Enzymatic analysis of FAH is not readily available as a common clinical assay. Several laboratories perform DNA mutation analysis of the *FAH* gene. Molecular genetic studies may be needed for counseling, prenatal diagnosis, and family screening. One splice mutation is prevalent in French-Canadians, and more than 40 mutations are now known, reviewed by de Laet et al.⁸¹

TREATMENT

Since 1992, nitisinone has been used in tyrosinemia.⁹⁵ It is an effective inhibitor of *p*-hydroxyphenylpyruvic acid oxidase that blocks an enzyme proximal in the catabolic pathway of tyrosine and prevents substrate from reaching the FAH enzyme. Thus the intracellular accumulation of toxic FAA and its eventual extracellular conversion to succinylacetone are prevented. A nutritionist skilled in metabolic disorders is essential to assist with the use of special formulas for infants and low-protein diet for older patients. At least 90% of patients with the acute form of tyrosinemia type I show response to nitisinone therapy. Unresponsive children require orthotopic liver transplantation, but carries a significant mortality rate (10% to 20%). The decision to perform liver transplantation depends on the patient's liver status and neurologic symptoms.⁸¹ The renal dysfunction may persist following transplantation because the renal enzyme is still defective.⁹⁶

GALACTOSEMIA

Galactosemia is an autosomal recessive metabolic disorder in which the individual is unable to metabolize lactose to glucose.^{97,98} Three disorders of galactose metabolism resulting in galactosemia have been described. Clinical manifestations appear after exposure to galactose and can produce failure to thrive, vomiting, inanition, liver disease, cataracts, and developmental delays.

PATHOGENESIS

The genetic defect responsible for galactosemia may be a defect in galactose-1-phosphate uridylyltransferase, galactokinase, or uridine diphosphate galactose-4-epimerase. All of these enzymes catalyze the reactions in the unique pathway converting galactose to glucose. *Classic galactosemia* (OMIM #230400), which affects approximately 1 in 30,000 to 60,000 live births, is the most frequent form. The underlying pathogenic basis of galactosemia remains poorly understood. Untreated and treated patients with galactosemia experience abnormal accumulation and/or depletion of certain metabolites. Specific abnormalities of glycosylation can also

be demonstrated, suggesting that aberrant biosynthesis of glycoproteins and/or glycolipids may contribute not only to the acute but also to some of the long-term complications experienced by galactosemic patients.^{98,99}

CLINICAL PRESENTATION

The clinical manifestations of galactosemia range from cataracts caused by galactokinase deficiency to important toxicity syndromes inducing vomiting, diarrhea, jaundice, hepatomegaly, ascites, and sepsis. Tubular proteinuria, generalized aminoaciduria, and bicarbonaturia occur but may quickly disappear following withdrawal of galactose.

The diagnosis is suggested by elevation of galactose or galactose-1-phosphate in serum or of galactose in the urine. The definitive diagnosis is made by the demonstration of the enzyme deficiency in erythrocytes.^{100,101} Prenatal diagnosis can be offered to parents with a family history of the disease.¹⁰²

In the United States and many other industrialized nations, inclusion of galactosemia in the mandated panel of conditions tested during newborn screening has all but eliminated the acute presentation.

TREATMENT

With early diagnosis and simple dietary intervention, galactosemia is no longer a lethal condition. The only therapy is a galactose-restricted diet, and initially all galactose must be removed from the diet as soon as the diagnosis is suspected.^{101,103} A newborn with a positive screening result should immediately be started on a soy-based infant formula. In spite of the strict diet, long-term complications such as retarded mental development, verbal dyspraxia, motor abnormalities, and hypergonadotropic hypogonadism are frequently seen in patients with classic galactosemia.

WILSON DISEASE

Wilson disease (OMIM #277900) is an autosomal recessive disorder in which biliary excretion of copper and its incorporation into ceruloplasmin are impaired, leading to liver, kidney, and corneal damage from accumulation of copper.¹⁰⁴ The frequency of the disease is approximately 1 to 3 per 100,000 live births.

PATHOGENESIS

ATP7B, the gene responsible for Wilson disease, encodes a P-type ATPase, and is located on chromosome 13q14.3. More than 380 mutations in this gene have been identified in patients with Wilson disease worldwide. *ATP7B* protein regulates systemic copper levels by excreting excess copper into the bile^{105–107} and is targeted to the ER and exocytic vesicles.¹⁰⁸

CLINICAL PRESENTATION

The primary consequence is liver disease. Most patients present with liver dysfunction, neurologic symptoms, or a combination of the two. Liver symptoms can take multiple forms, that is, chronic and acute liver failure. Copper accumulates in the liver, with progressive damage, and overflow to the brain. This causes central nervous system anomalies such as dysarthria and coordination defects of voluntary movements. Pseudobulbar palsy is frequent and is a common cause of death in undiagnosed cases. Psychiatric symptoms

can encompass a spectrum of personality changes, depression, bipolar disorder, schizophrenia, and dementia.¹⁰⁹ Wilson disease should be suspected in all subjects with acute or chronic liver dysfunction.

Affected adults often show features of Fanconi syndrome, with aminoaciduria, bicarbonaturia, phosphaturia, glucosuria, and low-molecular-weight proteinuria,¹¹⁰ probably from copper accumulation. Children do not frequently have renal manifestations. Hypercalciuria is common, and kidney stones and nephrocalcinosis have been described in several cases.^{111–113} Ultrastructural findings on renal biopsies include electron-dense deposits in the tubular cytoplasm.¹¹⁴

The most useful laboratory tests for diagnostic purposes are those measuring 24-hour urinary copper excretion, and concentrations of hepatic copper, serum-free copper, and ceruloplasmin. Increased liver copper level (>300 mg/g dry weight) is a reliable finding. The Kayser–Fleischer ring is an important clinical sign for making the diagnosis; it is seen in 50% of patients with liver dysfunction and in 90% of those with neurologic disease.¹¹⁵ This ring is a yellow-brown (dull copper colored) granular deposit on Descemet membrane at the limbus of the cornea, usually seen earliest at the upper and lower poles. Given the variability of the biochemical and clinical features of Wilson disease, mutation analysis is becoming more and more essential to confirm a suspicion of the disorder.¹⁰⁴ Molecular genetic diagnosis is available.¹¹⁶ The lack of genotype–phenotype correlation remains unexplained, the causes of fulminant liver failure are not known, and the treatment of neurologic symptoms is only partially successful.¹¹⁷

TREATMENT

Wilson disease is universally fatal if untreated. D-Penicillamine, which mobilizes copper and forms copper–penicillamine complexes that are excreted in the urine, is very effective.¹¹⁸ Adults usually require 1 g/day divided in two doses, but it is best to start with small doses (125 mg/day) to avoid hypersensitivity reactions. Twenty-four-hour urinary excretion of copper should be monitored to achieve copper losses of 2 mg/day. Doses of D-penicillamine can be decreased after 1 or 2 years to achieve urinary losses of 1 mg/day. Alternative treatments include trientine, a copper chelator which can be used as a first-line treatment¹¹⁹ or in case of D-penicillamine intolerance. Prevention of reaccumulation of copper can be achieved by zinc salts, which block intestinal copper absorption by inducing metallothionein synthesis in the mucosal intestinal cells. Tetrathiomolybdate, a copper chelator, appears to be an excellent form of initial treatment in patients with neurologic symptoms and signs. In contrast to penicillamine therapy, initial treatment with tetrathiomolybdate can often be effective in preventing further neurologic deterioration.¹²⁰

Patients should be considered for liver transplantation when suitable medical therapy has failed or for the development of acute liver failure, when there is no time for other therapies to take effect. Patients with a combination of hepatic and neuropsychiatric conditions warrant careful neurologic assessment, but liver transplantation is contraindicated only in cases of severe neurologic impairment.¹⁰⁴

HEREDITARY FRUCTOSE INTOLERANCE

There are several disorders of fructose metabolism, secondary to deficiencies in aldolase B (OMIM #229600),

fructose-1-phosphate aldolase, and fructokinase, respectively.¹²¹ Aldolase B is found primarily in the liver and is involved in the metabolism of fructose so this sugar can be used as energy. Fructose-1-phosphate accumulation in liver cells results from enzyme dysfunction.

Hereditary fructose intolerance is an autosomal recessive disorder characterized by vomiting shortly after the intake of fructose. The disease can be associated with proximal tubule dysfunction and lactic acidosis. Kidney biopsies show discrete findings. Liver dysfunction, hepatomegaly, cirrhosis, and jaundice appear from prolonged exposure. Unfortunately, hypoglycemia is frequently absent. Continued ingestion of noxious sugars leads to hepatic and renal injury and growth retardation. The most common mutation has a prevalence of 1.3%, suggesting a frequency of 1 in 23,000 homozygotes.¹²² Because of the difficulty in making the diagnosis, genetic analysis is useful.¹²³

The pathophysiology of renal Fanconi syndrome is not clear, but could be related to vacuolar proton pump dysfunction in the proximal tubule, because a direct binding interaction between V-ATPase and aldolase was demonstrated for the regulation of the V-ATPase. This study showed that aldolase B was abundant in endocytosis zones of the proximal tubule, a subcellular domain also abundant in V-ATPase.¹²⁴ V-ATPases are essential for acidification of intracellular compartments and for proton secretion from the plasma membrane in kidney epithelial cells and osteoclasts. Perhaps the release of nonfunctional aldolase B in response to fructose ingestion impairs the coupling of the V-ATPase to glycolysis. For these reasons, mechanistic similarities in organelle dysfunction between Dent disease and hereditary fructose intolerance are apparent.

The management of hereditary fructose intolerance involves withdrawal of sucrose, fructose, and sorbitol from the diet.

INHERITED DISORDERS OF RENAL AMINO ACID TRANSPORT

Proteins ingested in the regular diet and degraded in the intestine are absorbed by the mucosa as amino acids and small oligopeptides. Intestinal apical and basolateral transporters carry the amino acids into the blood, where they are used for metabolic needs, but are also freely filtered by the kidneys as they are not significantly bound to proteins in the plasma (except for tryptophan, which is 60%–90% bound). The proximal tubule reabsorbs 95% to 99.9% of the filtered load (reviewed elsewhere¹²⁵). Thus the excretion of more than 5% of the filtered load of an amino acid is abnormal, except for histidine, which has a fractional excretion of 5%.¹²⁶ Renal amino acid reabsorption occurs in the proximal tubule through a variety of transporters. Most amino acids are reabsorbed by more than one transporter. Amino acids can share transporters with low affinity but high transport capacity and have a specific transporter for one amino acid that has a high affinity and low maximal transport capacity. Common carriers have been divided into five groups, which transport neutral and cyclic amino acids, glycine and imino acids, cystine and dibasic amino acids, dicarboxylic amino acids, and β -amino acids.¹²⁶ The transport of amino acids is coupled to the sodium gradient established by the basolateral Na^+/K^+ -ATPase.

Aminoaciduria occurs when a renal transport defect of the proximal tubule decreases the reabsorptive capacity for one or several amino acids or when the threshold for reabsorbing an amino acid is exceeded when its plasma concentration is elevated as a result of a metabolic defect (overflow aminoaciduria).¹²⁷ This latter category is not discussed in this section. Theoretically, renal aminoacidurias can be secondary to defects in brush border or basolateral transporters and intracellular trafficking of amino acids (see also Chapter 8). They are usually detected by newborn urine screening programs, and the most frequent abnormalities identified (apart from phenylketonuria, now normally detected by blood screening) are cystinuria, histidinemia, Hartnup disease, and iminoglycinuria (Fig. 44.5).¹²⁸ Clinically, the most significant renal aminoaciduria is cystinuria (Table 44.4). In the past 10 years, all major apical neutral amino acid transporters have been identified on a molecular level.¹²⁹ This has considerably improved our understanding of inherited aminoacidurias and helped explain some of the features of Fanconi syndrome.

CYSTINURIA

Cystinuria (OMIM #220200) is the most frequent and best known of the aminoacidurias.¹³⁰ Worldwide, the prevalence is approximately 1 in 7000 and varies according to geographic location, being 1 in 15,000 in the United States,¹³¹ 1 in 2000 in England,¹³² 1 in 4000 in Australia,¹³³ and 1 in 2500 in Jews of Libyan origin.¹³⁴ Newborn screening programs worldwide now help identify cases. The existence of cystinuria has been recognized since 1810,^{135,136} first suspected in two patients with bladder stones, hence the name—cystic oxide and cystine to characterize their chemical composition. It is an autosomal recessive disorder associated with defective apical membrane transport of cystine and of the dibasic amino acids, ornithine, lysine, and arginine. It involves the epithelial cells of the renal tubule and gastrointestinal tract (Fig. 44.5). The formation of cystine calculi in the urinary tract is the hallmark of the disorder. Cystine is the least soluble of the naturally occurring amino acids, particularly at low pH.

PATHOGENESIS

Cystinuria is caused by mutations in the genes that encode the two subunits that form a heterodimer—neutral and basic amino acid transport protein rBAT and b^(0,+)-type amino acid transporter 1—of the amino acid transport system b^(0,+).¹³⁰ By itself, the subunit b^(0,+)AT is sufficient to catalyze transmembrane amino acid exchange (i.e., the exchange of dibasic amino acids for neutral amino acids).¹³⁷ A model for the reabsorption of cystine and dibasic amino acids is shown in Fig. 44.6. Cystinuria also manifests as defective intestinal absorption of dibasic amino acids, implying that there is a transporter defect in the gut similar to that in the renal proximal tubule cells.¹³⁰ Why defective intestinal amino acid transport does not lead to more serious metabolic problems is not known, but the reason could be either the ability of the intestine to absorb small (di- and tri-) peptides or the presence of other transporters of dibasic amino acids in the gut.

CLINICAL PRESENTATION

Cystinuria is classified according to the gene responsible, but this classification currently has no clinical utility. Type

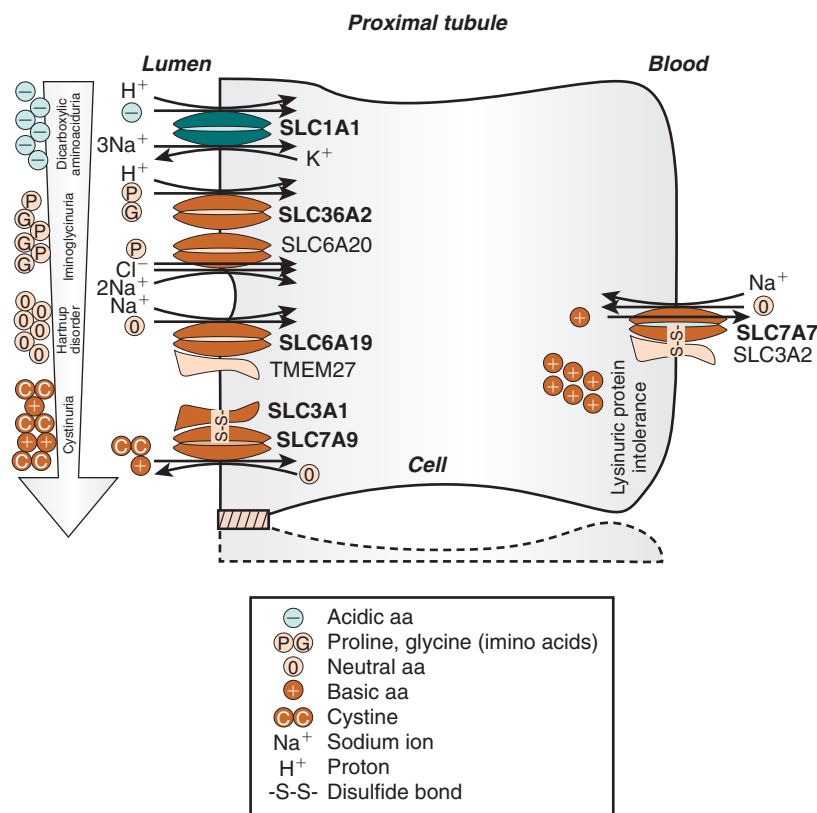


Fig. 44.5 A model of the renal proximal tubule, illustrating the principal epithelial transporters involved in amino acid reabsorption, which are mutated in human aminoacidurias. A cross section of the proximal convoluted tubule is represented. Four of the aminoacidurias—dicarboxylic aminoaciduria (DA), iminoglycinuria, Hartnup disorder, and cystinuria—manifest at the apical surface of the renal tubule, whereas lysinuric protein intolerance manifests at the basolateral surface. Mutations in the high-affinity glutamate and aspartate transporter SLC1A1 are responsible for DA.¹⁷⁶ Iminoglycinuria results from complete inactivation of SLC36A2, a proline and glycine transporter, or from additional modifying mutations in the high-affinity proline transporter SLC6A20 when SLC36A2 is not completely inactivated.¹⁷⁵ (From Broer S, Bailey CG, Kowalczyk S, et al. Iminoglycinuria and hyperglycinuria are discrete human phenotypes resulting from complex mutations in proline and glycine transporters. *J Clin Invest.* 2008;118:3881–3892.) Mutations in the neutral amino acid transporter SLC6A19 are responsible for Hartnup disorder.^{173,353} The neutral amino acid transport defect can also be exacerbated by a kidney-specific loss of heterodimerization of mutant SLC6A19 with TMEM27.³⁵⁴ Cystinuria has a heterogeneous phenotype and arises from mutations in individual or both subunits of the disulfide bridge-linked heterodimer comprising the type II membrane protein SLC3A1³⁵⁵ and the cystine and basic amino acid transporter SLC7A9.³⁵⁶ Lysinuric protein intolerance^{161,167} results from mutations in the basolaterally expressed basic amino acid transporter SLC7A7, which forms a disulfide bridge-linked heterodimer with type II membrane protein SLC3A2. aa, Amino acid. (Modified from Bailey CG, Ryan RM, Thoeng AD, et al. Loss-of-function mutations in the glutamate transporter SLC1A1 cause human dicarboxylic aminoaciduria. *J Clin Invest.* 2011;121:446–453.)

A cystinuria (also called type I; OMIM #220100) is caused by mutations in the *SLC3A1* gene (chromosome 2). It is a completely recessive disease because heterozygote parents excrete normal amounts of cystine (0–100 mmol/g creatinine). Jejunal uptake of cystine and dibasic amino acids is absent, and there is no plasma response to an oral cystine load. The risk of nephrolithiasis is very high. The *SLC3A1* gene encodes the renal proximal tubule S3 segment and intestinal dibasic rBAT amino acid transporter (Fig. 44.6).¹³⁸ More than 100 different mutations have been reported to date.¹³⁰ The most common point mutation, the M467T and its relative M467K, results in retention of the transporter in the ER with impaired maturation and export to the plasma membrane.¹³⁹

Type B cystinuria (also called non-type I cystinuria; OMIM #600918) is an incompletely recessive form caused by mutations in the *SLC7A9* gene encoding BAT1 and located on chromosome 19q13. Both parents excrete intermediate amounts of cystine (100–600 mmol/g creatinine) but may

also have a normal pattern. More than 90 mutations have been identified.¹³⁰ BAT1 is a subunit linked to the rBAT through a disulfide bond. It belongs to a family of light subunits of amino acid transporters, expressed in the kidney, liver, small intestine, and placenta. Cotransfection of b^{0,+}AT and rBAT brings the latter to the plasma membrane, and results in the uptake of L-arginine in vitro.

Type AB cystinuria is caused by one mutation in *SLC3A1* and one mutation in *SLC7A9*, but this digenic inheritance is exceptional. The observed prevalence of AB cystinuria is low because patients present with a mild phenotype and escape detection.

The only known manifestation of cystinuria is nephrolithiasis. Cystine stones account for 1% to 2% of all kidney stones (5% in children), and cystinuria should be suspected in all staghorn calculi.¹⁴⁰ Clinical expression of the disease frequently starts between the first and third decades of life but may occur any time from the first year up to the ninth decade. The disease occurs equally in both sexes, but male

Table 44.4 Classification of the Aminoacidurias

Aminoaciduria	Gene	Gene Name	Protein	Chromosome	Hallmark (Amino Acid[s] Elevated in Urine)
Cystinuria A	<i>SLC3A1</i>	Solute carrier family 3 (cystine, dibasic, and neutral amino acid transporters, activator of cystine, dibasic, and neutral amino acid transport), member 1	rBAT	2p21	Cystine, lysine, arginine, ornithine
Cystinuria B	<i>SLC7A9</i>	Solute carrier family 7 (cationic amino acid transporter, y^+ system), member 9	b^0+ AT	19q13.11	Cystine, lysine, arginine, ornithine
Cystinuria AB	<i>SLC3A1/SLC7A9</i>				Cystine, lysine, arginine, ornithine
Lysinuric protein intolerance	<i>SLC7A7</i>	Solute carrier family 7 (cationic amino acid transporter, y^+ system), member 7	y^+ LAT1	14q11.2	Lysine, arginine, ornithine
Hartnup disorder	<i>SLC6A19</i>	Solute carrier family 6 (neutral amino acid transporter), member 19	B^0 AT1	5p15.33	Neutral amino acids
Iminoglycinuria	<i>SLC36A2</i>	Solute carrier family 36 (imino and glycine transporter), member 2	PAT2	5q33.1	Proline, hydroxyproline, glycine
Dicarboxylic aminoaciduria	<i>SLC1A1</i>	Solute carrier family 1	EAAT ₃	9p24	Aspartate, glutamate

Modified from Tanzi RE, Petrukhin K, Chernov I, et al. The Wilson disease gene is a copper transporting ATPase with homology to the Menkes disease gene. *Nat Genet.* 1993;5:344–350.

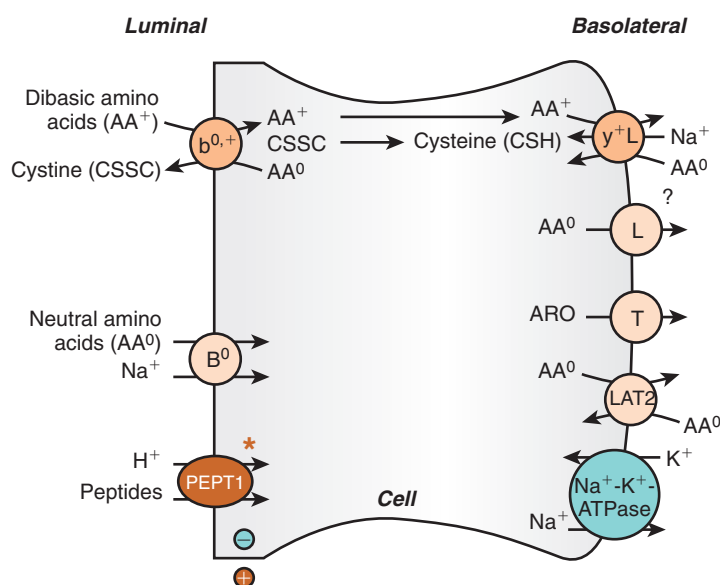


Fig. 44.6 Trafficking of cystine and dibasic amino acids in epithelial cells of the renal proximal tubule or small intestine. The apical transport system b^0+ mediates influx of AA^+ and CSSC in exchange for AA^0 . The Na^+ -dependent transporter B^0 is believed to be the major apical contributor to the high intracellular content of AA^0 exploited by b^0+ . In the small intestine (but not in kidney), uptake of dipeptides and tripeptides through apical H^+ -dependent solute carrier family 15 member 1 (PEPT1) compensates the defective absorption of AA^+ and CSSC. AA^+ exits the basolateral membrane through system y^+L , mutations in which cause lysinuric protein intolerance. The basolateral efflux of CSH and AA^0 are not well understood. The general AA^0 exchanger LAT2 and the aromatic transporter T have been suggested to participate in the excretion of CSH and aromatic amino acids (ARO). In addition, an unidentified transporter L for unidirectional efflux of AA^0 is believed to have a role in AA^0 reabsorption. *Only in intestinal cells. (Modified from Chillaron J, Font-Llitjos M, Fort J, et al. Pathophysiology and treatment of cystinuria. *Nat Rev Nephrol.* 2010;6:424–434.)

patients tend to be more severely affected. Cystine stones are made of a yellow-brown substance, are very hard, and appear radiopaque on radiographs because of their sulfur molecules. Stones are frequently multiple and staghorn, and they tend to be smoother than calcium stones. Magnesium ammonium phosphate and calcium stones can also form as a result of infection.

Diagnosis can be made through the analysis of a simple urine sample in which typical hexagonal crystals appear. Acidification of concentrated urine with acetic acid can also precipitate crystals not initially visible. Diagnosis is ultimately made from the measurement of cystine excretion in the urine usually performed in specialized centers. Molecular diagnosis is not necessary.

TREATMENT

Cystine stone formers have significantly higher procedure rates than other stone formers, but procedure rates decline with time. The lower creatinine clearance suggests that the need for procedures is high and that close follow-up and treatment to prevent recurrences are important.¹⁴¹

Unfortunately, there have been few advances in the treatment. A regularly followed medical program based on high diuresis and alkalization with a second-line addition of thiols slows down stone formation and precludes the need for urologic procedures in more than half of patients.¹⁴² Patients who are poorly compliant with hyperdiuresis remain at risk for recurrence.

Diet

Cystine production arises from the metabolism of methionine. Previous attempts at reducing methionine in the diet have been both uncomfortable and of limited usefulness.^{143,144} Reducing sodium in the diet results in lower urine cystine.^{145–147}

Decreasing Urine Cystine Saturation

A combination of increasing both fluid intake and urine pH usually decreases urine cystine saturation. Fluid intake should ideally reach 4 L/day, because many cystinuric patients excrete 1 g/day or more of cysteine.¹⁴² It is also important for patients to drink at bedtime and during sleep (using an alarm clock helps) to prevent supersaturation during periods of reduced urine output.¹⁴⁸ Cystine solubility can be increased by alkalization of the urine with potassium citrate or bicarbonate, but the solubility of cystine does not increase until the pH reaches 7.0 to 7.5. Use of citrate is the preferred method because alkalization lasts longer. The requirements for alkali often reach 3 to 4 mmol/kg/day.

Penicillamine

Patients who are unable to comply with a regimen of high fluid intake and urine alkalization or in whom adequate treatment fails may be given D-penicillamine in doses of 30 mg/kg per day up to a maximum of 2 g. Through a disulfide exchange reaction, D-penicillamine can form the disulfide cysteine–penicillamine, which is much more soluble than cystine. D-Penicillamine is associated with frequent side effects such as rash, fever, and, more rarely, arthralgias and medullary aplasia, leading to a significant proportion of patients discontinuing therapy.^{149–152} Other reactions include proteinuria and membranous nephropathy, epidermolysis, and loss of taste. Inhibition of pyridoxine by D-penicillamine is also a potential side effect.¹⁵³

Another drug that may be useful in cystinuria is mercaptopropionylglycine, with recommended dosing ranges of 400 to 1200 mg/day in three divided doses.^{154,155} Identical in mechanism of action to D-penicillamine, mercaptopropionylglycine is as effective in reducing urine cystine excretion. Side effects are similar to D-penicillamine. Finally, captopril has been advocated as a potential treatment for cystinuria, but its efficacy is controversial.

Surgical Management

Patients with cystinuria frequently require stone-removing procedures.¹⁵⁶ It is important to achieve a stone-free state because recurrent stone activity has been demonstrated to

be higher in those with residual calculi.¹⁵⁷ The type of procedure does not impact recurrence rates. The introduction of extracorporeal shockwave lithotripsy has not been of great benefit to cystinuric patients. Cystine stones are hard and have proved difficult to pulverize. Consequently, percutaneous lithotripsy is more effective and is the preferred approach. Progress in urologic treatment of kidney stones has decreased the need for open surgery.¹⁵⁸ Urinary alkalization, as well as direct irrigation of the urinary tract with D-penicillamine, N-acetylpenicillamine, or tromethamine to form disulfide compounds, has resulted in the dissolution of stones, but this approach requires irrigation for several weeks with a risk for potential complications of catheterization and has been largely abandoned.

Transplantation is sometimes necessary for patients with terminal renal failure from chronic obstruction or infection (or both). A kidney from an unaffected donor will not form cystine stones once transplanted.

LYSINURIC PROTEIN INTOLERANCE

PATHOGENESIS

Lysinuric protein intolerance (LPI; OMIM #222700) is a very rare recessively inherited dibasic amino acid transport disorder, mostly reported in Finland,^{159,160} but also present in Japan and Italy. It is caused by defective basolateral membrane efflux of the cationic amino acids lysine, arginine, and ornithine in intestinal, hepatic, and renal tubular epithelia from inactivating mutations in *SLC7A7*.^{161,162} *SLC7A7* encodes a 511-amino acid protein, y⁺LAT-1, predicted to harbor 12 membrane-spanning domains, with both amino and carboxy termini located intracellularly. This protein is thought to be part of the y⁺L multimeric unit. Cationic amino acid transport occurs through five different systems: y⁺, y⁺L, b⁺, b⁰⁺, and B⁰⁺. Defective system y⁺L transport explains the abnormality in cationic amino acid transport as it mediates sodium-independent high-affinity transport of cationic amino acids and the transport of zwitterionic amino acids with low affinity. It is responsible for renal reabsorption and intestinal absorption of dibasic amino acids at the basolateral membranes. y⁺L transport is induced by a cell surface glycoprotein heavy chain (4F2hc) that represents the heavy-chain subunit of a disulfide-linked heterodimer.¹⁵⁹ The amino acid deficiency in LPI leads to impaired urea cycle and postprandial hyperammonemia.

CLINICAL PRESENTATION

Infants with LPI are usually asymptomatic while breastfeeding. After weaning, they show protein aversion, a delay in bone growth, and prominent osteoporosis, hepatosplenomegaly, muscle hypotonia, and sparse hair. Jaundice, hyperammonemia, coma, and metabolic acidosis occur. Micronodular cirrhosis develops from protein malnutrition, and pulmonary alveolar proteinosis is an occasional finding.^{163,164} Several immunologic abnormalities have been described¹⁶⁵ and are possibly secondary to low levels of arginine, the substrate for nitric oxide synthase and nitric oxide production.¹⁶⁶

Various renal disorders, including immunoglobulin A (IgA) nephropathy, are associated with LPI.¹⁶⁷ The urinary excretion of lysine and all cationic amino acids is increased, and plasma levels are decreased. The biochemical diagnosis can be uncertain, requiring confirmation by DNA testing. So far,

approximately 50 different mutations have been identified in the *SLC7A7* gene.¹⁶⁸

TREATMENT

The current treatment of LPI involves moderate protein restriction as well as supplementation with 3 to 8 g of citrulline daily during meals and lysine.¹⁶⁹ Citrulline is transported by a different pathway from that of dibasic amino acids and can be converted to ornithine and arginine in the liver. Lysine cannot be made from citrulline.

HARTNUP DISORDER

PATHOGENESIS

Described initially in 1956 in the Hartnup family, Hartnup disorder (OMIM #234500) is an autosomal recessive and usually benign condition, consisting of excessive urinary excretion of the monoamino, monocarboxylic (neutral) amino acids alanine, asparagine, glutamine, histidine, isoleucine, leucine, methionine, phenylalanine, serine, threonine, tryptophan, tyrosine, and valine. Its incidence has been estimated at 1 in 26,000 in newborn screening programs.¹⁷⁰ Hartnup disorder is caused by mutations in the neutral amino acid transporter B⁰AT1 (*SLC6A19*).¹⁷¹ The transporter is found in the kidney and intestine, where it is involved in the resorption of all neutral amino acids.¹²⁹ Knockout *SLC6A19* mice have improved glycemic control and pharmacological inhibitors could be a suitable target to treat type 2 diabetes.¹⁷²

CLINICAL PRESENTATION

Most newborns identified prospectively by genetic screening programs as having Hartnup disorder have been completely asymptomatic. In affected individuals there is also a decreased intestinal absorption of neutral amino acids, particularly tryptophan. The clinical features of this disorder, if any, are caused by deficiency of nicotinamide, which is partly derived from tryptophan. They include a photosensitive erythematous rash (pellagra-like) clinically identical with that seen in niacin deficiency, intermittent cerebellar ataxia, and, rarely, mental retardation. Emotional instability, psychosis, and depression have been rarely noted, particularly during episodes of ataxia. Although in the Hartnup family there were several cases with mental retardation, most affected subjects described subsequently have not had mental retardation. Hartnup disorder should be suspected in all subjects with pellagra and unexplained intermittent ataxia. Siblings of affected patients should be screened as well. Clinical manifestations can be triggered by periods of inadequate dietary intake or increased metabolic needs. For example, a young woman presenting with pellagra precipitated by prolonged lactation and increased activity was diagnosed with Hartnup disorder.¹⁷³

The diagnosis is easily made with a urinary aminogram, which shows increased excretion of neutral amino acids but not of glycine, cystine, or dibasic, dicarboxylic, or imino amino acids. Thus any confusion with renal Fanconi syndrome is avoided through a complete evaluation of amino acids in the urine using one of several chromatographic methods. The pattern of amino acid excretion, rather than the total amount, is the determining factor. The reabsorption defect involves 12 amino acids, and most patients with Hartnup disorder have the same pattern of aminoaciduria. The

levels of amino acids in the monoamino and dicarboxylic group (such as glutamic acid and aspartic acid), and basic group (lysine, ornithine, arginine) are normal or slightly increased. The excretion of proline, hydroxyproline, and cystine is also normal. In spite of the defect, substantial reabsorption of the involved amino acids occurs through other transporters.

TREATMENT

Treatment of symptomatic cases involves the administration of nicotinamide in doses of 50 to 300 mg/day. The value of treating asymptomatic cases is not known, but given the harmlessness of the treatment, this might be a rational choice.

IMINOGLYCINURIA

Familial iminoglycinuria (OMIM #242600) is a benign autosomal recessive disorder with no clinical symptoms, in which the main interest is that it suggests the presence of a common carrier for the imino acids, proline and hydroxyproline, as well as glycine.¹⁷⁴ Iminoglycinuria was discovered after the application of chromatographic methods to the investigation of disorders of amino acid metabolism. *SLC36A2*, the gene encoding proton amino acid transporter 2 (PAT2), has been identified as the major gene responsible for iminoglycinuria.¹⁷⁵ Mutations in *SLC36A2* that retain residual transport activity result in the iminoglycinuria phenotype when combined with mutations in the gene encoding the imino acid transporter *SLC6A20*. Additional mutations have been identified in the genes encoding the putative glycine transporter SLC6A18 (XT2) and the neutral amino acid transporter SLC6A19 (B⁰AT1) in families with either iminoglycinuria or hyperglycinuria, suggesting that mutations in the genes encoding these transporters may also contribute to these phenotypes.

The diagnosis is usually suggested by increased urinary excretion of imino acids and glycine. Newborns and infants usually excrete detectable amounts of imino acids and glycine for up to 3 months. Thus the presence of increased urinary excretion of imino acids and glycine in infants older than 6 months can be considered abnormal.

DICARBOXYLIC AMINOACIDURIA

Dicarboxylic aminoaciduria (OMIM #222730) is an autosomal recessive disorder with, in a limited number of cases, an association with mental retardation. It is secondary to loss-of-function mutations in the glutamate transporter SLC1A1, a high-affinity anionic amino acid transporter expressed in the kidney and a wide variety of epithelial tissues, the brain, and the eye.¹⁷⁶

INHERITED DISORDERS OF RENAL PHOSPHATE TRANSPORT

Inherited disorders of renal phosphate transport are hypophosphatemic conditions caused by a reduction in renal tubule reabsorption of phosphate¹⁷⁷ with frequent metabolic bone disease manifesting as rickets in childhood and osteomalacia in adults.

RENAL PHOSPHATE EXCRETION

Determinants of inorganic phosphate (Pi) homeostasis are ingestion, intestinal absorption, and renal excretion. In a steady state, urine Pi excretion reflects dietary intake. Normal phosphate intake in adults varies from 800 to 1600 mg/day, and average serum phosphate levels remain normal over a wide range of intake. Unlike dietary calcium, ingested phosphate is generally efficiently absorbed (65%–90% in children) from the gastrointestinal tract, although complex plant phosphate (phytate) is almost totally excreted. Most dietary phosphate is absorbed by passive concentration-dependent processes, but the active metabolite of vitamin D increases intestinal Pi absorption marginally. Contrary to active calcium absorption, which is greatest in the duodenum with a lower rate in the jejunum, ileum, and colon, active Pi absorption is highest in the jejunum and ileum with a lower rate in the duodenum and colon.¹⁷⁸

Inorganic phosphate is filtered by the glomerulus and reabsorbed in the proximal tubule. The difference between the amounts of Pi filtered and reabsorbed determines the net appearance of Pi in the urine. Phosphate reabsorption by the proximal tubule occurs by a Tm-limited active process. The fractional reabsorption of filtered phosphate is usually estimated by the tubular reabsorption of Pi (TRP). There is

a simple equation to assess the renal tubular phosphate transport, as follows:

$$\text{TRP} = \frac{(1 - U_P \times P_{Cr})}{U_{Cr} \times P_P}$$

where P_{Cr} and P_P are plasma creatinine and phosphate concentrations, respectively; and U_{Cr} and U_P are urine creatinine and phosphate concentrations, respectively. Given normal renal function and normal diet, the TRP is usually above 85%. A more precise way of estimating the tubular reabsorption of Pi is to calculate the theoretical threshold, as follows:

$$\frac{T_{m_P}}{\text{GFR}} = P_P - \frac{(U_P \times P_{Cr})}{U_{Cr}}$$

RENAL PHOSPHATE TRANSPORTERS

Inorganic phosphate is reabsorbed almost exclusively in the renal proximal tubule through a transcellular pathway.¹⁷⁹ The limiting step of this transepithelial transport system is the entry of phosphate at the apical domain of proximal tubular cells. This process requires sodium–phosphate cotransporters that use the inward sodium gradient established and maintained by the activity of $\text{Na}^+\text{--K}^+\text{--ATPase}$ (Fig. 44.7). See Chapter 7 for more detailed discussions. There are three types of

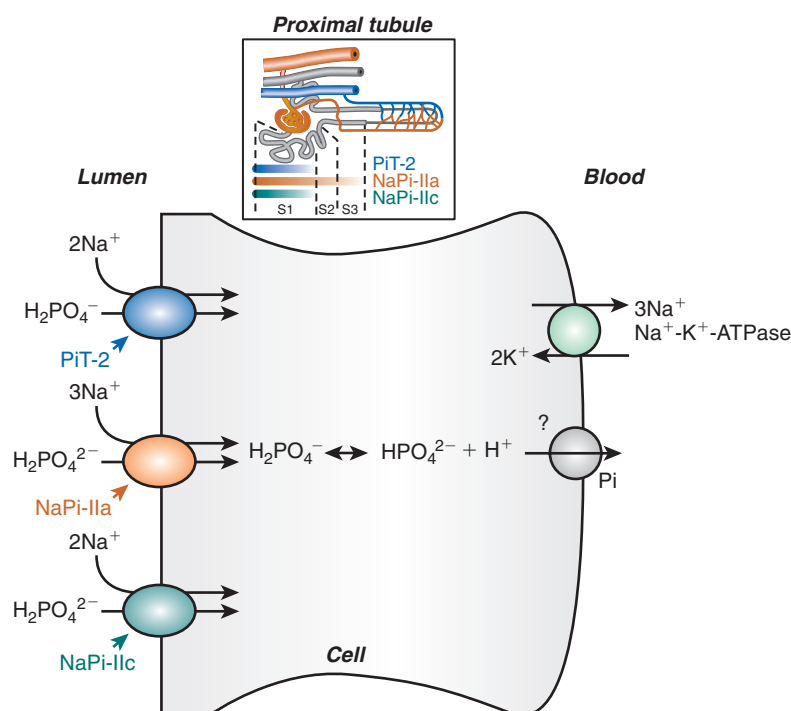


Fig. 44.7 Cross section of mammalian kidney and schematic representation of renal tubular phosphate reabsorption in segments of the proximal tubule (S1, S2, and S3) through sodium–phosphate cotransporters. This task is accomplished by two distinct families of sodium-dependent phosphate transporters, one being NaPi-IIa/NaPi-IIc and the other PiT-2, which are expressed in the luminal membrane of proximal tubular cells. The number of sodium–phosphate cotransporter units of NPT2a/c expressed at the membrane is regulated by parathyroid hormone (PTH) and fibroblast growth factor-23 (FGF-23). NaPi-IIa and NaPi-IIc prefer transport divalent P_i (HPO_4^{2-}), whereas PiT-2 prefers monovalent P_i (H_2PO_4^-). The basolateral exit pathway remains unknown. The basolaterally localized $\text{Na}^+\text{--K}^+\text{--ATPase}$ maintains an inwardly directed Na^+ gradient to drive cotransport. *ATPase*, Adenosine triphosphatase. (Modified from Biber J, Hernando N, Forster I. Phosphate transporters and their function. *Annu Rev Physiol*. 2013;75:535–550.)

sodium–phosphate cotransporters (members of the SLC34 and SLC20 families) at the apical (brush border) renal proximal tubular cells.¹⁷⁸ The putative proteins responsible for basolateral P_i flux have not been identified. The transport mechanism of the two kidney-specific SLC34 proteins (NaPi-IIa and NaPi-IIc) and of the ubiquitously expressed SLC20 protein (PiT-2) has been studied by heterologous expression to reveal important differences in kinetics, stoichiometry, and substrate specificity. NaPi-IIa is central to renal phosphate reabsorption and phosphate balance. It is found almost exclusively in the apical membrane of renal proximal tubular cells. The amount of NaPi-IIa protein at the brush border membrane determines the capacity of the proximal tubule to reabsorb phosphate; this finding explains why NaPi-IIa/SLC34A1-deficient mice have increased urinary phosphate excretion and marked hypophosphatemia.¹⁸⁰ NaPi-IIa is the target of the two main hormones that control renal phosphate reabsorption, PTH and fibroblast growth factor-23 (FGF-23), both of which decrease the amount of NaPi-IIa at the brush border membrane (reviewed elsewhere¹⁸¹).

The contribution of NaPi-IIa to renal phosphate balance was established by Magen and coworkers when they reported autosomal recessive hypophosphatemic rickets with renal Fanconi syndrome secondary to a loss-of-function mutation in NaPi-IIa.¹⁸²

Membrane sorting of NaPi-IIa requires sodium–hydrogen exchanger regulatory factor 1 (NHERF-1), a multifunctional intracellular protein with two structural domains, the PSD95, Discs-large, ZO-1 (PDZ1) domain and the PSD95, Discs-large, ZO-2 (PDZ2) domain. These domains can interact with specific sequences of the carboxy terminus of various membrane proteins, among them NaPi-IIa,^{182,183} and the PTH type 1 receptor (PTH1R).^{184,185} Disruption of *NHERF1* in mice results in a phenotype that is very similar to that of *NaPi-IIa* knockout mice, including increased urinary phosphate excretion and hypophosphatemia because of decreased NPT2a in brush border membranes.¹⁸⁶

Two other bone-specific proteins, PHEX [see discussion of **X-linked hypophosphatemic rickets (XLH)**] and dentin matrix protein 1 (DMP1), appear to be necessary for limiting the expression of FGF-23, thereby allowing sufficient renal conservation of phosphate.¹⁸⁷ A reduction in serum phosphate levels also leads to increased $1,25(\text{OH})_2\text{D}_3$ levels from greater

activity of the 1α -hydroxylase. The decrease in serum phosphate levels also inhibits bone deposition, and the rise in serum $1,25(\text{OH})_2\text{D}_3$ increases bone resorption, thus favoring a net shift of phosphate from bone. Higher serum levels of $1,25(\text{OH})_2\text{D}_3$ also increase intestinal phosphate and calcium absorption. As a consequence, serum calcium levels rise and inhibit PTH secretion. The reduction in PTH levels does not lead to a further increase in phosphate reabsorption by the kidney because the proximal tubule is insensitive to the action of PTH in states of phosphate deprivation. As a result, one should predict that a renal phosphate leak will lead to raised serum $1,25(\text{OH})_2\text{D}_3$ levels, decreased PTH, and induction of hypercalciuria.

X-LINKED HYPOPHOSPHATEMIC RICKETS

PATHOGENESIS

XLH (OMIM #307800) is the most common inherited hypophosphatemic disorder, accounting for more than 50% of cases of familial phosphate wasting (Table 44.5). XLH is caused by mutations in the *PHEX* gene (phosphate-regulating gene with homologies to endopeptidases on the X chromosome). *PHEX* encodes an M13 zinc metalloprotease expressed in osteoblasts and odontoblasts but not in the kidney. Although it is not immediately apparent how loss of PHEX function leads to a decrease in renal P_i reabsorption, mutations in *PHEX* (in bone tissue) indirectly alter the degradation and production of FGF-23, causing increased circulating levels of this phosphatonin.^{187,188} FGF-23 synthesis and secretion appear reduced through yet unknown mechanisms that indirectly involve PHEX.¹⁸⁹

CLINICAL PRESENTATION

XLH phenotype is characterized by hypophosphatemia secondary to renal phosphate wasting, inappropriately low concentrations of $1,25$ dihydroxyvitamin D, and high circulating levels of FGF-23. Early in life, patients with XLH demonstrate short stature (growth retardation), femoral and/or tibial bowing, and histomorphometric evidence of rickets and osteomalacia. Male patients are usually more severely affected than female patients, with variable penetrance. Serum phosphate levels are lower than 0.8 mmol/L (2.5 mg/dL) and the Tm_P/GFR is lower than 0.56 mmol/L (1.8 mg/dL).

Table 44.5 Inherited Hypophosphatemias – Mutated Proteins and Laboratory Findings

Disease	Protein	Laboratory Values			
		Serum Ca^{2+}	$1,25(\text{OH})_2\text{D}$	FGF-23	PTH
Hypophosphatemic rickets, X-linked dominant (XLH)	PHEX	Normal	Low/Normal	High/Normal	Normal
Hypophosphatemic rickets, autosomal recessive (ARHR)	DMP1, ENPP1	Normal	Normal	Normal	High/Normal
Hypophosphatemic rickets, autosomal dominant (ADHR)	FGF-23	Normal	Normal	High	Normal
Nephrolithiasis/osteoporosis, hypophosphatemic 2	NHERF-1	Normal	High	Normal	Normal
Hypophosphatemic rickets and hyperparathyroidism	KLOTHO	High	High	High	High
Nephrolithiasis/osteoporosis, hypophosphatemic 1	SLC34A1 (?)	High	n.d.	High	n.d.
Hypophosphatemic rickets with hypercalciuria (HHRH)	SLC34A3	High	Low	High	High

DMP1, Dentin matrix protein 1; FGF-23, fibroblast growth factor-23; n.d., not determined; NHERF-1, sodium-hydrogen exchanger regulatory factor 1; PHEX, M13 zinc metalloprotease encoded by PHEX; PTH, parathyroid hormone; SLC, solute carrier.

Modified from Amatschek S, Haller M, Oberbauer R. Renal phosphate handling in human—what can we learn from hereditary hypophosphatemia? *Eur J Clin Invest.* 2010;40:552–560.

The $1,25(\text{OH})_2\text{D}_3$ levels are normal or near normal. There is apparently no correlation between the serum levels of phosphate and the severity of the disease. Affected children tend to have higher serum phosphate and $\text{Tm}_\text{P}/\text{GFR}$ values than affected adults, as is the case with normal subjects. The earliest sign of the disease in children can be increased serum alkaline phosphatase levels.¹⁸⁹

TREATMENT

Early therapy with $1,25(\text{OH})_2\text{D}_3$ (1.0–3.0 mcg/day) and phosphate (1–2 g/day in divided doses) has a beneficial effect on growth, bone density, and deformations.¹⁹⁰ Nephrocalcinosis caused by vitamin D and phosphate therapy can lead to deterioration of renal function.

AUTOSOMAL DOMINANT HYPOPHOSPHATEMIC RICKETS

Autosomal dominant hypophosphatemic rickets (ADHRs; OMIM #193100) is a very rare cause of rickets (Table 44.5) characterized by hypophosphatemia, phosphaturia, inappropriately low or normal $1,25(\text{OH})_2\text{D}$ levels, and bone mineralization defects that can result in bone pain, fracture, rickets, osteomalacia, lower extremity deformities, and muscle weakness.¹⁹¹ These features are similar to those of XLH. However, ADHR is far less common than XLH and has incomplete penetrance with a variable age of onset.

The gene responsible for ADHR encodes FGF-23, a 251-amino acid peptide that is secreted and processed to amino- and carboxy-terminal peptides at a consensus proprotein convertase (furin) site, RHTR (ArgHisThrArg),¹⁹² by a subtilisin-like proprotein convertase. Missense mutations in FGF-23 are identified in all families with ADHR abrogate peptide processing.¹⁹³ The mutant FGF-23 is predicted to have a longer circulating half-life than the wild type and to be associated with higher serum concentrations. Iron deficiency may trigger late-onset ADHR (in pregnancy and adolescence, for example) by increasing FGF-23 production.¹⁹⁴ Treatment of ADHR involves phosphate supplementation and calcitriol.

AUTOSOMAL RECESSIVE HYPOPHOSPHATEMIC RICKETS

Autosomal recessive hypophosphatemic rickets (ARHR1; OMIM #241520) is a very rare form of hypophosphatemic rickets caused by inactivating mutations of the *DMP1* gene (*DMP1*) that result in secondary elevation of FGF-23 concentrations.¹⁹⁵ The *DMP1* gene is located on chromosome 4q21 and is one of a cluster of genes encoding a class of tooth and bone noncollagenous matrix proteins known as SIBLINGs (small integrin-binding ligand, N-linked glycoproteins). ARHR2 (OMIM #613312) is caused by mutations in *ENPP1* (ectonucleotide pyrophosphatase/phosphodiesterase family member 1), a regulator of extracellular pyrophosphate, and has previously been linked to the development of generalized arterial calcification of infancy (*GACI*; OMIM #208000).¹⁹⁶ The biochemical features are similar to those of XLH – hypophosphatemia, phosphaturia, and inappropriately low $1,25(\text{OH})_2\text{D}_3$. The clinical presentation is usually found not at birth but later, during childhood and even in adulthood.

HEREDITARY HYPOPHOSPHATEMIC RICKETS WITH HYPERCALCIURIA

Hereditary hypophosphatemic rickets (OMIM #241530) associated with hypercalciuria is a rare autosomal recessive disease caused by mutations in *SLC34A3*, the gene encoding the renal sodium–phosphate cotransporter NaPi-IIc.^{196,197} Affected subjects appear to have a phenotype restricted to the renal phosphate leak with an appropriate response to hypophosphatemia, because serum levels of calcitriol are increased. Consequently, intestinal calcium absorption is enhanced, resulting in hypercalciuria. Serum calcium levels are normal, and other features include variable degrees of rickets and short stature, and suppressed PTH. The condition responds to daily oral phosphate (1–2.5 g/day), which leads to an increase in serum phosphate and decreases in serum $1,25(\text{OH})_2\text{D}_3$, calcium, and alkaline phosphatase. Growth rate can be restored, and the clinical manifestations of rickets and osteomalacia disappear.

FAMILIAL TUMORAL CALCINOSIS

Familial tumoral calcinosis (OMIM #211900) is a severe autosomal recessive metabolic disorder that manifests as hyperphosphatemia and massive calcium deposits in the skin and subcutaneous tissues.¹⁹⁸ It is caused by loss-of-function mutations in *FGF23*¹⁹⁹ as well as in *GALNT3*²⁰⁰ and *KL*,²⁰¹ leading to inadequate concentrations or action of intact FGF-23. Affected individuals report recurrent painful, calcified subcutaneous masses of up to 1 kg, often resulting in secondary infection and incapacitating mutilation. The *GALNT3* gene encodes a glycosyltransferase responsible for initiating mucin-type O-glycosylation, whereas Klotho (encoded by *KL*) is a coreceptor for FGF-23.²⁰² Treatment involves phosphate chelators and acetazolamide.

HEREDITARY SELECTIVE DEFICIENCY OF $1\alpha,25(\text{OH})_2\text{D}_3$

A rare form of autosomal recessive vitamin D responsive rickets, hereditary selective deficiency of $1\alpha,25(\text{OH})_2\text{D}_3$ (OMIM #264700) is not a disease of tubule transport per se, but a 1α -hydroxylation deficiency. It results from inactivating mutations in the *CYP27B1* gene.^{202,203} Vitamin D is metabolized by sequential hydroxylations in the liver (25-hydroxylation) and the kidney (1α -hydroxylation). Hydroxylation of 25-hydroxyvitamin D_3 is mediated by 25(OH) D_3 1α -hydroxylase in the kidney. Patients usually appear normal at birth, but muscle weakness, tetany, convulsions, and rickets start to develop at 2 months of age. Serum calcium levels are low; PTH levels are high with low to undetectable $1,25(\text{OH})_2\text{D}_3$. Serum levels of 25(OH) D_3 are normal or slightly increased. Once recognized, this rare disorder is easily treated with physiologic doses of calcitriol, which leads to healing of rickets and restoration of the plasma calcium, phosphate, and PTH levels.

HEREDITARY GENERALIZED RESISTANCE TO $1\alpha,25(\text{OH})_2\text{D}_3$

Hereditary vitamin D-resistant rickets is a rare monogenic autosomal recessive disorder caused by mutations in the

vitamin D receptor and is similar to selective deficiency of $1\alpha,25(\text{OH})_2\text{D}_3$. Its salient features are increased serum levels of $25(\text{OH})\text{D}_3$ and $1\alpha,25(\text{OH})_2\text{D}_3$, and the disease does not respond to doses of $1\alpha,25(\text{OH})_2\text{D}_3$ and $1\alpha,(\text{OH})\text{D}_3$. In addition, approximately half of the patients described with the disease have alopecia. In a subset of affected kindreds, premature stop codons in the vitamin D receptor gene lead to the absence of the ligand-binding domain.²⁰⁴ Administration of very high doses of calcium (≤ 3 g/day) and $1,25(\text{OH})_2\text{D}$ (≤ 30 – 60 mcg/day) may be necessary to normalize calcemia and mineralize depleted bones.

RESISTANCE TO PARATHORMONE ACTION

Pseudohypoparathyroidism (PHP) is associated with biochemical hypoparathyroidism (i.e., hypocalcemia, hyperphosphatemia) caused by resistance to, rather than deficiency of PTH (reviewed elsewhere²⁰⁴). Patients with PHP type 1a have a generalized form of hormone resistance and a constellation of developmental defects termed *Albright hereditary osteodystrophy*. Within PHP type 1a families, some individuals show osteodystrophy but have normal hormone responsiveness, a variant phenotype termed *pseudo-PHP*. By contrast, patients with PHP type 1b manifest only PTH resistance and lack features of osteodystrophy. These various forms of PHP are caused by defects in the imprinted *GNAS1* gene that lead to decreased expression or activity of the α subunit of the stimulatory G protein ($G_{\alpha s}$). Tissue-specific genomic imprinting of *GNAS1* accounts for the variable physical and endocrine phenotypes (parathyroid, thyroid, gonads, and pituitary gland) of patients with *GNAS1* defects.²⁰⁵

INHERITED DISORDER OF URATE TRANSPORT

FAMILIAL RENAL HYPOURICEMIA

Renal hypouricemia is an autosomal recessive disorder characterized by impaired urate handling in the renal tubules.²⁰⁶ Type 1 (>90% of cases) is caused by a loss-of-function mutation in the *SLC22A12* gene²⁰⁷ (OMIM #220150) encoding the apical urate/anion exchanger URAT1 particularly prevalent in Japan and in Ashkenazi Jews, whereas type 2 is caused by defects in the *SLC2A9* gene (OMIM #612076) encoding glucose transporter 9 (GLUT9), a high-capacity urate transporter belonging to the facilitated GLUT family that is responsible for most of the basolateral urate transport.^{207,208} The disorder is associated with exercise-induced acute renal failure and nephrolithiasis, particularly for carriers of *SLC2A9* mutations who show very high fractional excretion of urate (100%–150%), reflecting absent reabsorption but retained urate secretion. Carriers of *SLC22A12* mutations have a milder phenotype because of the presence of other urate anion exchangers OAT4 (SLC22A11) and OAT10 (SLC22A13).²⁰⁹ Hyperuricosuria, combined with dehydration or exercise, results in acute uric acid nephropathy and causes an obstructive acute renal failure. This can be prevented by forced hydration with bicarbonate or saline solutions. Affected subjects typically have very high urate fractional excretion ($\geq 50\%$) and their parents have intermediate levels.²¹⁰ Interestingly, polymorphisms in *SLC2A9*

explain 1.7% to 5.3% of the variance in serum uric acid concentrations.²¹¹

Hereditary xanthinuria, a recessively inherited disorder, is also characterized by very low serum urate levels resulting from a deficiency of xanthine oxidase.²¹² However, urine urate excretion is very low, and xanthine excretion is elevated, contrary to renal hypouricemia. Type I disease results from mutations in xanthine dehydrogenase (XDH), and type II results from a dual deficiency of XDH and aldehyde oxidase.

INHERITED DISORDERS OF RENAL GLUCOSE TRANSPORT

Under normal conditions, glucose is almost completely reabsorbed by the proximal tubule. Thus very small amounts of glucose are present in the urine. Glucose urinary excretion (500 mg or 2.75 mmol/day in adults) is caused most often by hyperglycemia (overload glucosuria) and rarely by abnormal handling of glucose by the kidney (Box 44.1). Renal glucosuria may be part of a generalized dysfunction of the proximal tubule (Fanconi syndrome) or may manifest as an isolated defect.

RENAL GLUCOSURIA

Familial renal glucosuria (FRG; OMIM #233100) is an inherited renal tubular disorder characterized by persistent isolated glucosuria in the absence of hyperglycemia. It is usually a benign clinical condition. Mutations in *SLC5A2* encoding the sodium–glucose cotransporter SGLT2 are responsible for the disorder.²¹³ Some Japanese patients have mutations in *GLUT2*.²¹⁴

SGLT2 and SGLT1 mediate apical glucose uptake in the S1 and S3 segments of the proximal tubule, respectively. SGLT2 inhibitors are now used in diabetic patients to control hyperglycemia by increasing urinary glucose excretion.²¹⁵ Of interest, this class of drugs also seems to show benefit in slowing the progression of diabetic nephropathy and cardiovascular complications of diabetes, with the former attributed to enhancement of tubuloglomerular feedback–mediated reduction of diabetic hyperfiltration injury^{216,217} (see also Chapter 39).

Box 44.1 Causes of Glucosuria

Hyperglycemia

Diabetes mellitus
iatrogenic:
 Glucocorticoids
 Catecholamines
 Angiotensin I–converting enzyme inhibitors
 Dextrose intravenous solutions
 Total parenteral nutrition

Renal Glucosuria

Idiopathic
Glucose–galactose malabsorption
Fanconi syndrome
Pregnancy

FRG is transmitted as a codominant trait with incomplete penetrance. Homozygotes can show glucosuria of more than 60 g/day, evidence of renal sodium wasting, mild volume depletion, and raised basal plasma renin and serum aldosterone levels.²¹⁸

The definition of “glucosuria” is arbitrary, and different investigators have proposed different guidelines to distinguish abnormal from normal glucosuria. A currently accepted stringent definition of glucosuria proposes the following criteria:

- The oral glucose tolerance test result and the levels of plasma insulin and free fatty acids and of glycosylated hemoglobin should all be normal.
- The amount of glucose in the urine (10–100 g/day) should be relatively stable except during pregnancy, when it may increase.
- The degree of glucosuria should be largely independent of diet but may fluctuate according to the amount of carbohydrates ingested. All specimens of urine should contain glucose.
- The carbohydrate excreted should be glucose. Other sugars are not found (fructose, pentoses, galactose, lactose, sucrose, maltose, and heptulose).
- Subjects with renal glucosuria should be able to store and use carbohydrates normally.

GLUCOSE–GALACTOSE MALABSORPTION

Glucose–galactose malabsorption (OMIM #606824) is a rare autosomal recessive disease resulting from a selective defect in the intestinal transport of glucose and galactose. Mutations in *SLC5A1*—which couples transport of sugar to sodium gradients across the intestinal brush border²¹⁸—cause absence of the transporter in the intestine and the kidney plasma membranes, leading to glucose–galactose malabsorption. Neonatal onset of severe watery and acidic diarrhea results in death unless these sugars are removed from the patient’s diet.²¹⁹ Significant weight loss from hyperosmolar dehydration and metabolic acidosis is frequent. The disease occurs occasionally in adults.²²⁰ The acidic diarrhea results from bacterial metabolism of sugar in the stools. Normally, lactose in milk is broken down into glucose and galactose by lactase, an ectoenzyme on the brush border, and the hexoses are transported into the cell by SGLT1.

The disease is usually suspected from the clinical history and the presence of glucosuria despite normal serum glucose levels. Dramatic improvement occurs after withdrawal of glucose and galactose from the patient’s diet. The acidic diarrhea can be improved with antibacterial treatment.

INHERITED DISORDERS OF ACID–BASE TRANSPORTERS

A typical Western diet generates an acid load of approximately 1 mmol of mineral acid per kilogram of body weight, which must be excreted by the kidney. In addition, the kidney filters approximately 4000 mmol of bicarbonate daily and must reclaim most of the filtered load to maintain acid–base balance. Excretion of the ingested acid load and reabsorption

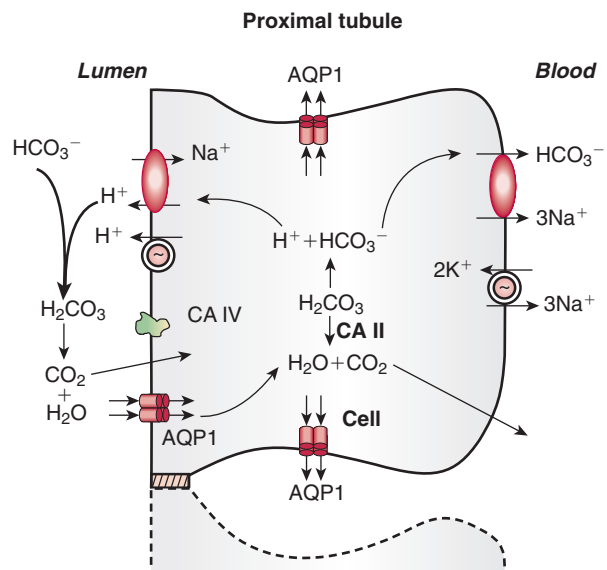


Fig. 44.8 Mechanisms of renal acidification in the proximal tubule. AQP1, Aquaporin-1; CA II/IV, carbonic anhydrase II/IV.

of filtered bicarbonate are accomplished by complex processes requiring coordinated actions of transport and enzymatic activities in the apical and basolateral membranes (Fig. 44.8).

In the proximal tubule, filtered bicarbonate (HCO_3^-) is almost completely reabsorbed by an indirect mechanism. H^+ and HCO_3^- are generated by intracellular hydration of CO_2 by carbonic anhydrase II (CA II). H^+ secretion occurs across the apical membrane via NHE3, the Na^+/H^+ exchanger, and an H^+ -ATPase, and HCO_3^- is transferred via a basolateral $\text{Na}^+/\text{HCO}_3^-$ cotransporter. The secreted hydrogen ions react with filtered HCO_3^- to form H_2CO_3 , which is rapidly converted to CO_2 and H_2O by CA IV present in the apical membrane. The CO_2 and H_2O then diffuse into the cell. The result is the removal of a filtered HCO_3^- and its replacement by another in the plasma, but the process is neutral for net urinary H^+ excretion because the secreted hydrogen ions are used to reabsorb filtered HCO_3^- . CAs are zinc metallo-enzymes that catalyze the reversible hydration of CO_2 to form HCO_3^- and protons, according to the following reaction:



The first reaction is catalyzed by CA, and the second reaction occurs instantaneously. Net urinary elimination of H^+ depends on its buffering and excretion as titratable acid (mainly phosphate: $\text{HPO}_4^{2-} + \text{H}^+ \leftrightarrow \text{H}_2\text{PO}_4^-$), and excretion as NH_4^+ . The production of NH_4^+ from glutamine by the proximal tubule and its secretion generate new plasma HCO_3^- . This process is stimulated in metabolic acidosis.

In the distal and collecting tubule, the connecting segment and type A intercalated cells secrete H^+ into the lumen via a vacuolar Mg^{2+} -dependent H^+ -ATPase and an exchanger, H^+/K^+ -ATPase (Fig. 44.9). The generation of H^+ is catalyzed by CA II and HCO_3^- is transported across the basolateral membrane through the anion exchanger 1 (AE1), or $\text{Cl}^-/\text{HCO}_3^-$ exchanger. Luminal H^+ is trapped by urinary buffers, including ammonium secreted by the proximal tubule, and phosphate.

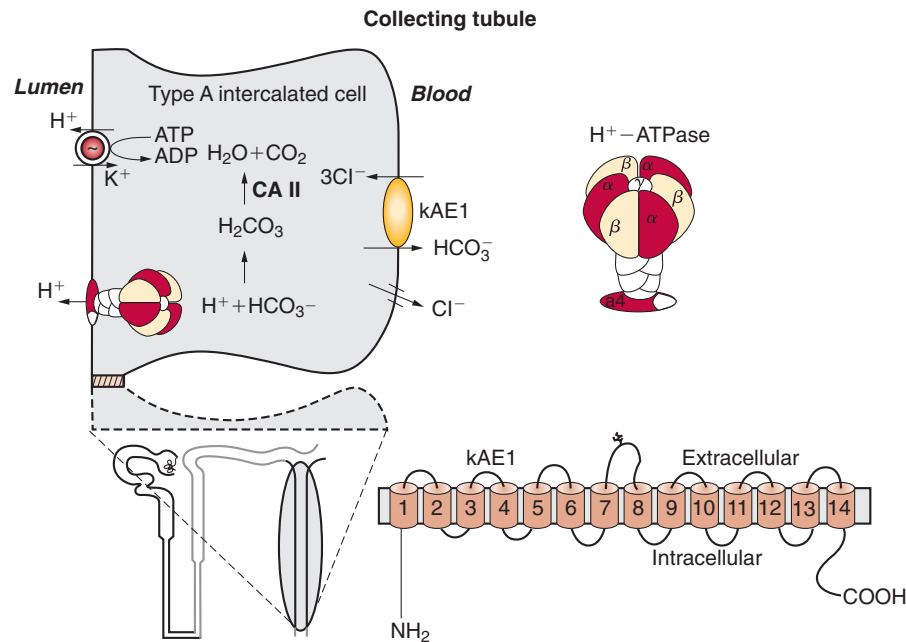


Fig. 44.9 Mechanisms of renal acidification in the collecting tubule. Dominant distal renal tubular acidosis (RTA) is caused by mutations in the gene *SLC4A1* encoding the chloride–bicarbonate exchanger AE1.^{232,357} The *AE1* gene (chromosome 17) encodes both the erythroid (eAE1) and the kidney (kAE1) isoforms of the band 3 protein.³⁵⁸ Mutations in the gene encoding the β_1 -subunit of H^+ -ATPase (*ATP6V1B1*; chromosome 2p13) cause recessive distal renal tubular acidosis with sensorineural deafness.²³⁵ Distal RTA with preserved hearing is secondary to mutations in *ATP6V0A4*, which encodes the α_4 subunit of the proton pump.³⁵⁹ Both H^+ and H^+ - K^+ -ATPases are represented. The H^+ -ATPase is schematically represented according to the proposed structure of the F-type F_1 -ATPase of the inner mitochondrial membrane.³⁶⁰ F_1 is represented as a flattened sphere 80 Å high and 100 Å across. The three α - and three β -subunits are arranged alternately like the segments of an orange around a central α -helix 90 Å long. Mutations in the β -subunit cause autosomal recessive distal RTA. Autosomal recessive distal RTA has also been found, in small kindred, for the *SLC4A1* mutation G701D.²³⁴ ADP, Adenosine diphosphate; ATP, adenosine triphosphate; ATPase, Adenosine triphosphatase; RTA, renal tubular acidosis.

RTA is a clinical syndrome characterized by hyperchloremic (normal anion gap) metabolic acidosis secondary to abnormal urine acidification.²²¹ It can be identified from inappropriately high urine pH, bicarbonaturia, and reduced net acid excretion. Clinical and functional studies allow classification into four types, historically numbered in the order of discovery: proximal (type 2), classical distal (type 1), hyperkalemic distal (type 4), and combined proximal and distal (type 3). Rare forms of hereditary proximal and dRTA have been identified^{221–223} and are discussed here (Table 44.6).

PROXIMAL RENAL TUBULAR ACIDOSIS

pRTA usually occurs as part of the spectrum of Fanconi syndrome, in which the excretion of glucose, amino acids, urate, and phosphate is also increased. Primary, isolated hereditary pRTA is an extremely rare disorder that may be inherited as an autosomal recessive or dominant trait.²²⁴ The diagnosis of pRTA rests on an appropriately acid urine pH (pH <5.5) in patients with acidosis and a high fractional excretion of bicarbonate (>10%–15%) during intravenous loading with sodium bicarbonate ($NaHCO_3$). The underlying defect is a failure of proximal bicarbonate reabsorption. It results in an abnormally low threshold for renal bicarbonate reabsorption because the distal nephron is unable to compensate and reabsorb the large bicarbonate load presented to it. However, distal acidification mechanisms are intact, and acid urine can be produced. The metabolic acidosis is

generally mild and associated with hypokalemia, and metabolic bone disease is common. Because of a lowering of the tubular threshold for bicarbonate reabsorption, once the plasma bicarbonate is reduced, the threshold can be reached and a steady state maintained at a serum concentration of approximately 15 mM. By contrast, levels can fall to less than 10 mM in dRTA. Administration of large amounts of alkali (10–20 mmol/kg/day) may be required to normalize serum bicarbonate in the patient with pRTA.

SODIUM-BICARBONATE SYMPORTER MUTATIONS

Inactivating mutations in *SLC4A4*, the gene coding for the Na^+ - HCO_3^- (NBC1) symporter, cause autosomal recessive pRTA with various ocular abnormalities such as band keratopathy, glaucoma, and cataracts (OMIM #604278).²²⁵ The Na^+ - HCO_3^- symporter is expressed in multiple ocular tissues,²²⁶ thus explaining the abnormalities. Pancreatitis can be associated with mutations in NBC1, which is expressed in the pancreas.²²⁷

CARBONIC ANHYDRASE II DEFICIENCY

Recessive mixed proximal–distal (type 3) RTA accompanied by osteopetrosis and mental retardation (OMIM #259730) is caused by inactivating mutations in the cytoplasmic CA II gene.²²⁸ The pathogenesis of the mental subnormality and cerebral calcification is poorly understood. More than 50 cases have been described, predominantly from the Middle East and Mediterranean regions. The disorder is discovered

Table 44.6 Classifications, Features, and Underlying Molecular Transport Defect(s) in Inherited Renal Tubular Acidoses

	Clinical Features	Protein	Gene
Proximal RTA			
Autosomal recessive PRTA with ocular abnormalities	Band keratopathy, glaucoma, cataracts, short stature, mental retardation, dental enamel defects, pancreatitis, basal ganglia calcification	NBC1	SLC4A4
Autosomal recessive PRTA with osteopetrosis and cerebral calcification (inherited carbonic anhydrase II deficiency)	Mental retardation, osteopetrosis, cerebral calcification	CA II	CA2
Autosomal dominant PRTA	Short stature, osteomalacia	Unknown	Unknown
Distal RTA			
Autosomal dominant dRTA	Complete or incomplete dRTA, hypercalciuria, nephrocalcinosis, nephrolithiasis, hypokalemia, short stature, osteomalacia, rickets	AE1	SLC4A1
Autosomal recessive dRTA	Complete or incomplete dRTA, other features as above	H ⁺ -ATPase (A4 subunit)	ATP6V0A4
	Other features as above Reported in Asian populations in association with ovalocytosis	AE1	SLC4A1
Autosomal recessive dRTA with progressive sensorineural deafness	Complete or incomplete dRTA	H ⁺ -ATPase (B1 subunit)	ATP6V1B1
	As above, but with late-onset sensorineural deafness	H ⁺ -ATPase (A4 subunit)	ATP6V0A4

AE1, Ion exchanger 1; CA II, cytosolic carbonic anhydrase; d, distal; NBC1, Na⁺-HCO₃⁻ cotransporter; P, proximal; RTA, renal tubular acidosis.

Modified from Laing CM, Tøye AM, Capasso G, et al. Renal tubular acidosis: developments in our understanding of the molecular basis. *Int J Biochem Cell Biol.* 2005;37:1151–1161.

late in infancy or early in childhood because of developmental delay, short stature, fracture, weakness, cranial nerve compression, dental malocclusion, and/or mental subnormality. Typical radiographic features of osteopetrosis are present, and histopathologic study of the iliac crest reveals unresorbed calcified primary spongiosa. The radiographic findings are unusual, however, in that cerebral calcification appears by early childhood and the osteosclerosis and skeletal modeling defects may gradually resolve by adulthood. Patients are usually not anemic. A hyperchloremic metabolic acidosis, sometimes with hypokalemia, is caused by RTA, which may be a proximal, distal, or combined type.²²⁹ Bilateral recurrent renal stones, hypercalciuria, and medullary nephrocalcinosis have been described.²³⁰ There is no established medical therapy, and the long-term outcome remains to be characterized. Treatment involves alkali supplementation for the acidosis and, potentially, bone marrow transplantation for osteopetrosis.²³¹

DISTAL RENAL TUBULAR ACIDOSIS

Hereditary dRTA is a genetically heterogeneous disorder with dominant and recessive forms caused by dysfunction of type A intercalated cells^{221,222} (Fig. 44.9). Implicated transporters include the AE1 (Cl⁻/HCO₃⁻) exchanger of the basolateral membrane and at least two subunits of the apical membrane V-ATPase, the V₁ (head) subunit B1 (associated with deafness) and the V₀ (stalk) subunit A4. Clinical features include inability

to acidify urine, variable hyperchloremic hypokalemic metabolic acidosis, hypercalciuria, nephrocalcinosis, and nephrolithiasis. Patients with recessive dRTA present with either acute illness or growth failure at a young age, sometimes accompanied by deafness. Dominant dRTA is usually a milder disease and involves no hearing loss.

CHLORIDE-BICARBONATE EXCHANGER MUTATIONS

Mutations in the *SLC4A1* gene encoding AE1 can lead to dominant (OMIM #179800) or recessive dRTA.²³² AE1 is the basolateral Cl⁻/HCO₃⁻ exchanger located in alpha intercalated cells of the collecting duct.²³³ The renal AE1 contributes to urinary acidification by providing the major exit route for HCO₃⁻ across the basolateral membrane. dRTA results from aberrant targeting of AE1.

The dominant form is usually a mild disorder that can be discovered incidentally after a kidney stone episode. Serum bicarbonate concentrations are usually between 14 and 25 mmol/L, and serum potassium levels between 2.1 and 4.2 mmol/L. Minimum urine pH following an acid load varies from 5.95 to 6.8 (normal <5.30). Nephrocalcinosis and kidney stones are present in approximately 50% of subjects. Deafness is usually absent.

The recessive form of dRTA (OMIM #109270) is diagnosed at a younger age, often before 1 year. It is found in Southeast Asia (Thailand, Papua–New Guinea, and Malaysia), where it is associated with ovalocytosis.²³⁴ Affected subjects present with vomiting, dehydration, failure to thrive, or delayed

growth. Nephrocalcinosis, kidney stones, or both are frequent, and rickets can be present. Severe metabolic acidosis with serum pH less than 7.30 and serum bicarbonate less than 15 mmol/L is common. Serum potassium levels are also lower than in autosomal dominant dRTA.

PROTON ATPASE SUBUNIT MUTATIONS

Mutations in ATP6V1B1, the B₁-subunit of the apical proton pump ATP6B1 that mediates distal nephron acid secretion (OMIM #267300), cause dRTA with sensorineural deafness in a significant proportion of families.^{234,235} In type A intercalated cells, the H⁺-ATPase pumps protons against an electrochemical gradient. Active proton secretion is also necessary to maintain proper endolymph pH. These findings implicate ATP6B1 in endolymph pH homeostasis and in normal auditory function, because nearly all patients with ATP6V1B1 mutations also have sensorineural hearing loss.

Mutations in the ATP6V0A4 gene on chromosome 7 (OMIM #602722) also give rise to recessive dRTA,²³⁶ but hearing is preserved. ATP6V0A4 encodes a kidney-specific A4 isoform of the proton pump's 116-kDs accessory a subunit. The treatment of dRTA involves the correction of dehydration, electrolyte, and bicarbonate anomalies, which will improve symptoms. In adults, administration of alkali 1 to 3 mmol/kg/day usually corrects the metabolic abnormality. In children, up to 5 mmol/kg per day may be required. Potassium supplementation may be needed even after correction of the acidosis.

BARTTER AND GITELMAN SYNDROMES

In 1962, Bartter and coworkers described two patients with hypokalemic metabolic alkalosis, hyperreninemic hyperaldosteronism, normal blood pressure, as well as hyperplasia and hypertrophy of the juxtaglomerular apparatus.²³⁷ Since then, familial hypokalemic, hypochloremic metabolic alkalosis has been recognized as not a single entity but, rather, a set of closely related disorders.²³⁸ Although Bartter syndrome and Bartter mutations are used commonly as diagnoses, it is likely, as explained by Jeck and colleagues, that the two patients with a mild phenotype originally described by Bartter had Gitelman syndrome, a thiazide-like salt-losing tubulopathy with a defect in the distal convoluted tubule.²³⁷ As a consequence, salt-losing tubulopathy of the furosemide type is a more physiologically appropriate definition for Bartter syndrome. Bartter syndrome is a genetically heterogeneous disorder affecting the loop of Henle, where 30% of the filtered sodium chloride is reabsorbed, that typically manifests during the neonatal period and is associated with hypercalciuria and nephrocalcinosis (Fig. 44.10). By contrast, Gitelman syndrome is a disorder affecting the distal tubule²³⁹ that is usually diagnosed at a later stage and is associated with hypocalciuria and hypomagnesemia, with predominant muscular signs and symptoms (Fig. 44.11).²⁴⁰

BARTTER SYNDROME

PATHOGENESIS

Bartter syndrome (OMIM #601678, #241200, #607364, and #602522) is an autosomal recessive disorder affecting the function of the thick ascending limb (TAL) of the loop of

Henle, giving a clinical picture of salt wasting and hypokalemic metabolic alkalosis. It is caused by inactivating mutations in one of at least four genes encoding membrane proteins (Bartter syndrome types 1 through 4)^{241–245}: type 1, the Na⁺-K-2Cl[−] cotransporter (*SLC12A1* encoding NKCC2); type 2, the apical inward-rectifying potassium channel (*KCNJ1* encoding ROMK); type 3, a basolateral chloride channel (*CLCNK* encoding ClC-Kb); and type 4, *BSND*, encoding Barttin, a protein that acts as an essential activator β-subunit for ClC-Ka and ClC-Kb chloride channels (Fig. 44.10). Gain-of-function mutations in the extracellular calcium-sensing receptor (CaSR) cause a variant of Bartter syndrome^{241,246} with hypocalcemia. In this regard, it is of interest that a Bartter-like syndrome has been described in patients treated with aminoglycosides such as gentamicin and amikacin, characterized by transient hypokalemia, metabolic alkalosis, hypomagnesemia with urinary magnesium wasting, and hypercalciuria, which resolve weeks after drug termination. Drugs in this class are polycations acting as calcimimetics and stimulating the CaSR. This disorder can be thought of as an acquired form of type 5 Bartter syndrome. Alternatively, direct drug-induced tubular damage may be the etiologic mechanism.²⁴⁷

An additional type of Bartter syndrome has been shown to be a digenic disorder that is attributable to loss-of-function mutations in the genes that encode the chloride channels ClC-Ka and ClC-Kb.^{242,245}

CLINICAL PRESENTATION

Most cases of Bartter syndrome present antenatally or in neonates. Polyhydramnios and premature labor are common findings. Postnatal findings include polyuria, polydipsia, failure to thrive, growth retardation, dehydration, low blood pressure, muscle weakness, seizures, tetany, paresthesias, and joint pain from chondrocalcinosis.²⁴⁸ In contrast to patients with Gitelman syndrome, those with Bartter syndrome are virtually always hypercalciuric and normomagnesemic.

Nephrocalcinosis occurs in almost all patients with Bartter syndrome with NKCC2 (type 1) and ROMK (type 2) mutations but in only 20% of those with ClC-Kb mutations. This finding could be attributable to lower urine calcium excretion. Patients with ROMK mutations may show hyperkalemia at birth, which converts to hypokalemia within the first weeks of life.²⁴⁹ Thus they can be misdiagnosed with pseudohypoaldosteronism type I (PHA I), vide infra. This pattern could be explained by the fact that ROMK, in addition to being required for sodium reabsorption in the TAL, is also expressed in the collecting duct. Patients do not need important K⁺ supplementation, contrary to other patients with Bartter syndrome, but still demonstrate hypokalemia because of reduced reabsorption in the TAL and possibly to K secretion by maxi-K channels in the late distal tubule.²⁵⁰ The type 3 Bartter syndrome (ClC-Kb) phenotype is highly variable and may manifest as either a typical antenatal variant or a “classic” Bartter variant characterized by an onset in early childhood and lower severity or absence of hypercalciuria and nephrocalcinosis. *BSND* mutations (type 4 Bartter syndrome) are usually associated with an extremely severe phenotype with intrauterine onset, profound renal salt and water wasting, renal failure, sensorineural deafness, and motor retardation.²⁵¹ Sensorineural deafness is specific for Barttin (type 4) because it is an essential subunit of chloride channels in the inner

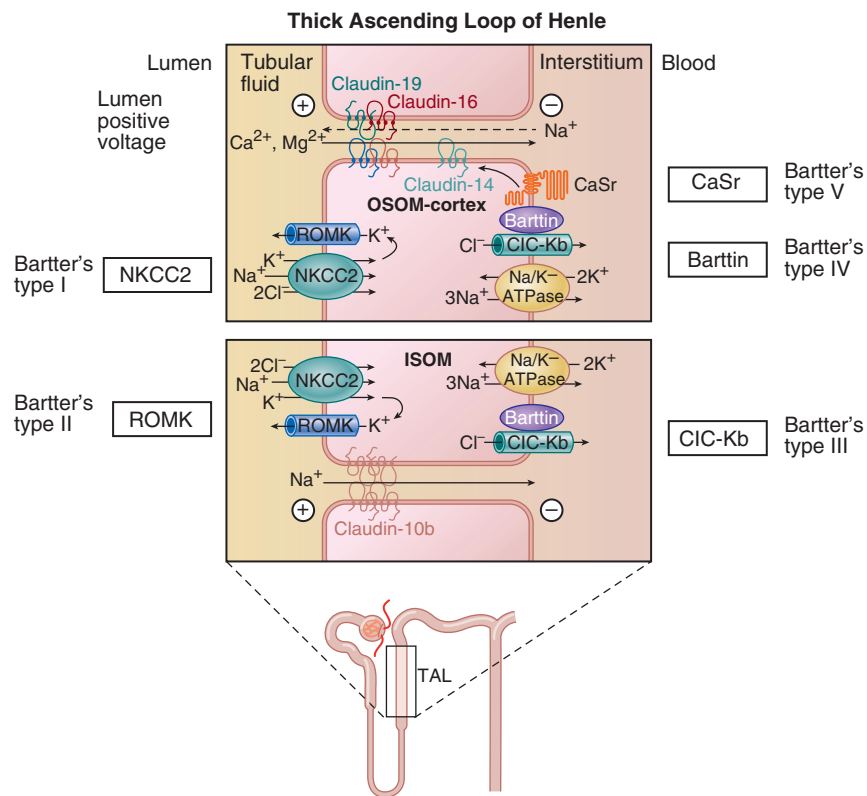


Fig. 44.10 Schematic representation of transepithelial salt reabsorption in a cell of the thick ascending limb (TAL) of the loop of Henle. Filtered NaCl is reabsorbed through Na⁺-K⁺-2Cl⁻ cotransporter type 2 (NKCC2), which uses the sodium gradient across the membrane to transport chloride and potassium into the cell. The potassium ions are recycled through the apical membrane by the potassium channel ROMK. Sodium leaves the cell actively through the basolateral Na⁺-K⁺-ATPase. Chloride diffuses passively through two basolateral channels, CIC-Ka and CIC-Kb. Both of these chloride channels must bind to the β-subunit of Barttin to be transported to the cell surface. The recycling of potassium maintains a lumen-positive voltage that drives calcium and magnesium reabsorption paracellularly. Claudin-16 and claudin-19 are necessary for the paracellular transport of calcium and magnesium. Four types of Bartter syndrome (types I, II, III, and IV) are attributable to recessive mutations in the genes that encode NKCC2, ROMK, CIC-Kb, and Barttin, respectively. *CIC-Kb*, Basolateral chloride channel b; *NKCC2*, Na⁺-K⁺-2Cl⁻ cotransporter 2; *ROMK*, potassium channel; (Modified from Bichet DG, Fujiwara TM. Reabsorption of sodium chloride—lessons from the chloride channels. *N Engl J Med*. 2004;350:1281–1283. With permission; Olinger E, Houillier P, Devuyst O. Claudins: a tale of interactions in the thick ascending limb. *Kidney Int*. 2018;93:535–537.)

ear, necessary for generating the endocochlear potential.²⁵² The severity of type 4 Bartter syndrome would be consistent with contributions of both CIC-Ka and CIC-Kb to basolateral chloride exit in the TAL.

TREATMENT

Treatment of Bartter syndrome usually involves potassium and magnesium supplements, spironolactone, and nonsteroidal antiinflammatory drugs. Indomethacin has been widely used, for which elevations of urinary prostaglandin E₂ have provided a rationale.²³⁸ Angiotensin I-converting enzyme inhibitors have been used successfully in conjunction with potassium supplements.^{252,253} Therapy should lead to catch-up growth in infants.²⁵⁴

GITELMAN SYNDROME

PATHOGENESIS

Gitelman syndrome (OMIM #263800), a milder disorder than Bartter syndrome,²⁵⁵ is usually diagnosed in adolescents and adults.²⁵⁵ It is an autosomal recessive trait caused by inactivating mutations in the *SLC12A3* gene encoding the thiazide-sensitive Na⁺-Cl⁻ cotransporter (NCC).²³⁹ Rare cases

are caused by mutations in the *CLCNKB* gene, which encodes the renal chloride channel CLC-Kb, located in basolateral membrane of cells of the TAL and the distal tubules. These patients appear to have a Gitelman and not the expected Bartter phenotype.²⁴³ Gitelman syndrome results in a thiazide-like effect, consisting of sodium and chloride wasting with secondary hypovolemia and metabolic alkalosis. Activation of the renin-angiotensin-aldosterone system from volume depletion, plus increased sodium load to the cortical collecting duct, leads to increased sodium reabsorption by the epithelial sodium channel (ENaC), which is counterbalanced by potassium and hydrogen excretion, resulting in hypokalemia and metabolic alkalosis. Enhanced passive Ca²⁺ transport in the proximal tubule rather than active Ca²⁺ transport in the distal convoluted tubule explains the hypocalciuria. Downregulation of TRPM6 (epithelial Mg²⁺ channel transient receptor potential channel subfamily M, member 6) explains the hypomagnesemia.²⁵⁶

CLINICAL PRESENTATION

The carrier state of *SLC12A3* mutations occurs in 1% of the general population, suggesting a prevalence of Gitelman syndrome of 25 per million population and making it the

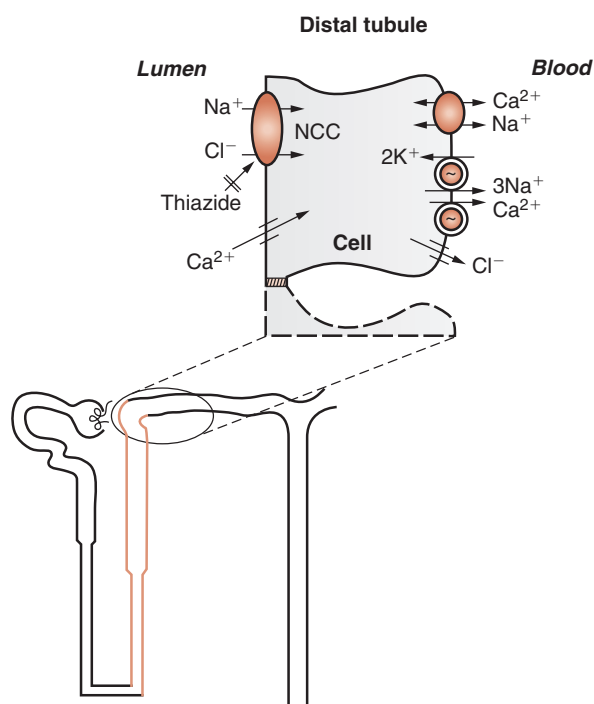


Fig. 44.11 Gitelman syndrome: loss-of-function mutations of the thiazide-sensitive Na–Cl cotransporter (NCC).

most common inherited renal tubule disorder.^{256,257} Contrary to patients with Bartter syndrome, those with Gitelman syndrome are usually asymptomatic in the neonatal period and often discovered incidentally (Fig. 44.12).²³⁸ Subjects have hypokalemic metabolic alkalosis, but unlike those with Bartter syndrome, they are hypocalciuric and hypomagnesemic and do not have signs of overt volume depletion.²⁵⁸ Polyuria and polydipsia are not features of Gitelman syndrome either. Patients suffer from arthritis because of chondrocalcinosis in several joints,²⁵⁸ possibly secondary to hypomagnesemia. Urinary prostaglandin E_2 levels are normal,²⁵⁹ a finding compatible with the poor response observed to prostanoid synthetase inhibition. The major conditions in the differential diagnosis of Gitelman syndrome are diuretic abuse, laxative abuse, and chronic vomiting. A careful history, as well as measurement of urinary chloride and detection of diuretics, should help differentiate among these conditions.

TREATMENT

The treatment of Gitelman syndrome includes potassium supplementation and spironolactone.²⁶⁰ Nonsteroidal anti-inflammatory drugs are usually not helpful because prostaglandin levels are normal.

INHERITED DISORDERS WITH HYPERTENSION AND HYPOKALEMIA

Most patients with hypokalemia and hypertension have essential hypertension associated with the use of diuretics, secondary aldosteronism from renal artery stenosis, or primary hyperaldosteronism.²⁶¹ Hereditary causes of hypertension and hypokalemia include excess secretion of aldosterone or other mineralocorticoids and abnormal sensitivity to miner-

alocorticoids. They are primarily characterized by low or low-normal plasma renin concentration and salt-sensitive hypokalemic hypertension, suggesting enhanced mineralocorticoid activity.²⁶² The molecular basis for several of these traits has been elucidated.

CONGENITAL ADRENAL HYPERPLASIA

Inherited abnormalities in steroid biosynthesis cause hypertension in some cases of congenital adrenal hyperplasia. These autosomal recessive disorders arise from deficiencies of key enzymes of the steroid biosynthesis pathway (Fig. 44.13).^{263,264} The decrease in cortisol production causes an increase in adrenocorticotrophic hormone (ACTH) secretion and subsequent hyperplasia of the adrenal glands. The phenotypes are determined by deficiencies as well as by overproduction of steroids unaffected by the enzymatic defect. Hypertension is observed in only two of the three major subtypes of congenital adrenal hyperplasia (11 β -hydroxylase and 17 α -hydroxylase deficiencies), because metabolic blockade distal to 21 α -hydroxylase allows the formation of 21-hydroxyl groups necessary for mineralocorticoid precursor biosynthesis. Other clinical manifestations depend on the consequences of the enzymatic defect on androgen biosynthesis with either an increase (11 β -hydroxylase) or a decrease (17 α -hydroxylase) in production. In both deficiencies, overproduction of cortisol precursors that are metabolized to mineralocorticoid agonists or that have intrinsic mineralocorticoid activity induce volume and salt-dependent forms of hypertension. The elevated zona fasciculata deoxycorticosterone (DOC) produces mineralocorticoid hypertension with suppressed renin and reduced potassium concentrations. Aldosterone, the most important mineralocorticoid, regulates electrolyte excretion and intravascular volume mainly through its effects on renal distal convoluted tubules and cortical collecting ducts.

11 β -HYDROXYLASE DEFICIENCY

Inactivating mutations in the gene encoding 11 β -hydroxylase^{263,265} cause the second most common form of congenital adrenal hyperplasia (OMIM #20210), representing 5% of cases (90% are caused by 21-hydroxylase deficiency). This disorder is associated with excess production of DOC, 18-deoxycortisol, and androgens. By virtue of the significant intrinsic mineralocorticoid activity of DOC, subjects with mutations in both alleles of the gene exhibit hypokalemic hypertension. Because the androgen pathway is unaffected, prenatal masculinization occurs in female patients and postnatal virilization in both sexes. The diagnosis of 11 β -hydroxylase deficiency is established by means of an ACTH test showing elevation of DOC and 11-deoxycortisol and marked suppression of plasma renin activity.²⁶³ The treatment consists of exogenous corticoids that inhibit ACTH secretion.

17 α -HYDROXYLASE DEFICIENCY

17 α -Hydroxylase deficiency (OMIM #202110) results in reduced conversion of pregnenolone to progesterone and androgens and absence of sex hormone production.^{262,266} The absence of sex hormone formation in both the adrenal glands and the gonads, which causes hypogonadism and male pseudohermaphroditism, is usually detected at adolescence because of failure to undergo puberty. Patients have

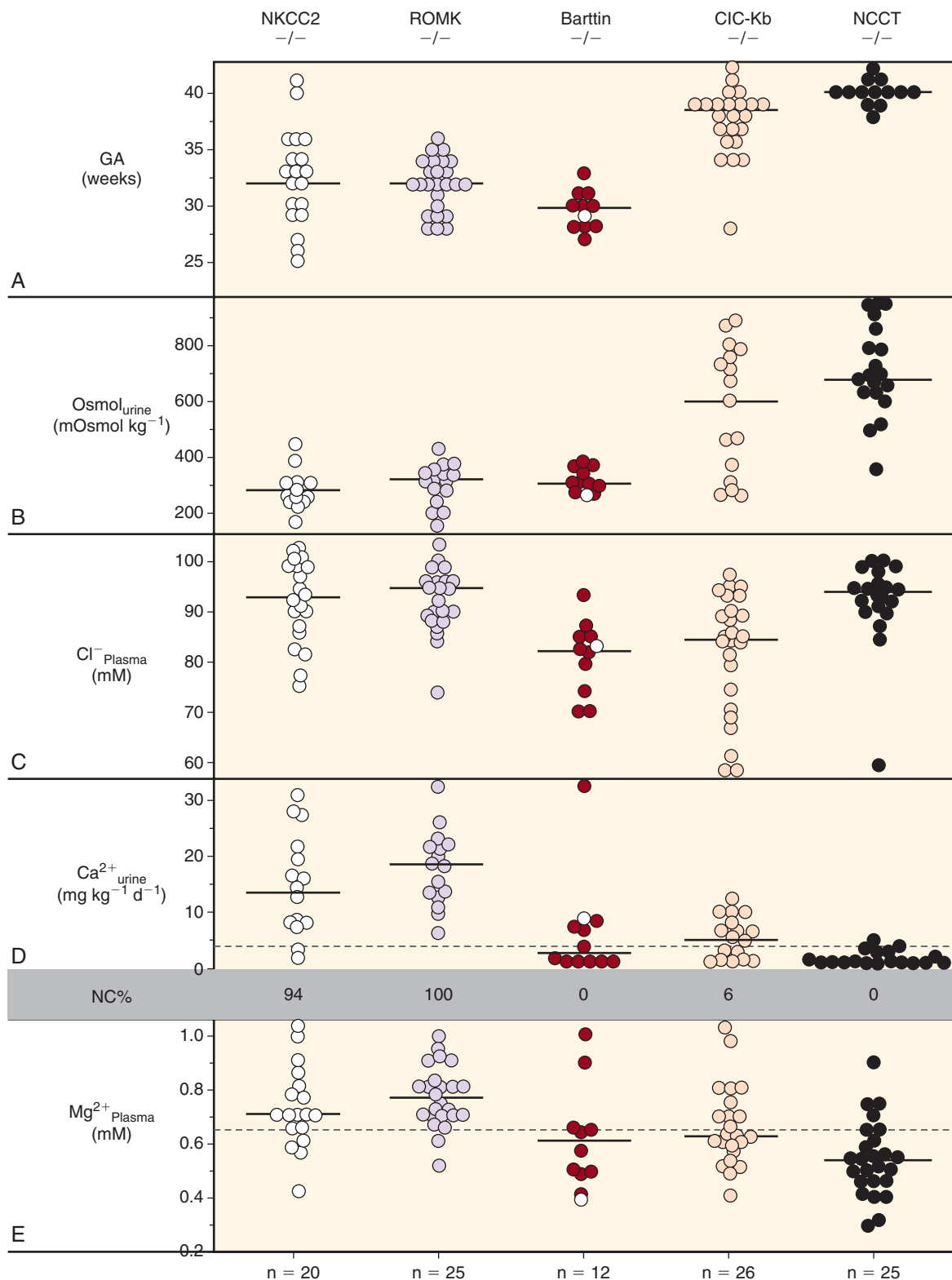


Fig. 44.12 Genotype-phenotype correlation in untreated salt-losing tubulopathies. (A) Age of gestation at birth (GA). (B) Maximal urine osmolality in random morning urine samples. (C) Minimal plasma Cl^- concentration. (D) Maximal urinary Ca^{2+} excretion (dashed line indicates the upper normal limit, about $4 \text{ mg} \cdot \text{kg}^{-1} \times \text{day}^{-1}$) and percentage of medullary nephrocalcinosis (NC%). (E) Minimal plasma Mg^{2+} concentration (dashed line indicates the lower normal limit, 0.65 mM). Horizontal lines indicate the median; the open symbol in the Barttin group indicates the digenic CIC-Ka/CIC-Kb disorder. CIC-Kb, Basolateral chloride channel b; NCCT, Na-Cl transporter; NKCC2, $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransporter 2; ROMK, potassium channel. (From Jeck N, Schlingmann KP, Reinalter SC, et al. Salt handling in the distal nephron: lessons learned from inherited human disorders. *Am J Physiol Regul Integr Comp Physiol*. 2005;288:R782-R795. With permission.)

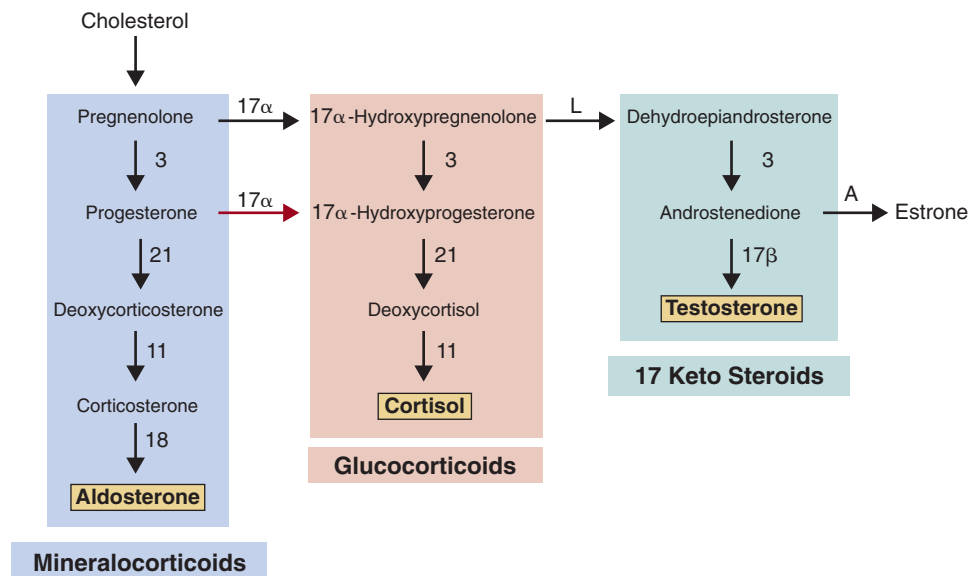


Fig. 44.13 Key enzymes of the steroid biosynthesis pathway.

a decreased ability to synthesize cortisol, leading to elevated ACTH, which increases serum levels of DOC and, especially, corticosterone, resulting in low-renin hypertension, hypokalemia, and metabolic alkalosis. The clinical features vary depending on the enzymatic activity affected. In severe 17 α -hydroxylase deficiency, both the 17 α -hydroxylase and 17,20-lyase activities are reduced or absent. This deficiency results in high mineralocorticoid activity and hypertension and produces a female phenotype in all subjects because of the absence of sex steroid production in both the adrenal and gonads. Partial 17 α -hydroxylase deficiency leads to sexual ambiguity in male patients without hypertension. Corticosteroid replacement corrects ACTH levels and hypertension. Women usually require hormonal therapy. Genetic male patients reared as female patients also require estrogen replacement. Genetic male patients reared as male patients require surgical correction of their external genitalia and androgen replacement therapy.

LIDDLE SYNDROME

PATHOGENESIS

Liddle syndrome (OMIM #177200) is an autosomal dominant form of hypertension characterized by hypokalemia and low levels of plasma renin and aldosterone, resulting from either premature termination or frameshift mutations in the carboxy-terminal tail of the ENaC β - or γ -subunits.²⁶⁷ The amiloride-sensitive epithelial Na⁺ channel is a tetramer formed by the assembly of three homologous subunits, α , β , and γ , with the α subunit being present in two copies (Fig. 44.14).²⁶⁸ The NH₂ and carboxy-terminal terminal segments are cytoplasmic and contain potential regulatory segments that are able to modulate the activity of the channel. Mutations in the β - and γ -subunits of ENaC lead to channel hyperactivity by deleting or altering a conserved proline-rich amino acid sequence referred to as the PY motif. *SCNN1B* β -subunit or *SCNN1G* γ -subunit mutations could lead to an increase in the number of channels in the membrane or

in their “openness.” The identification of specific binding domains for Nedd4 (for all subunits) and α -spectrin (for the α -subunit only) within the cytosolic carboxy-terminal region of the ENaC subunits suggests that interactions with cytoskeletal elements control the expression of ENaC at the apical membrane.²⁶⁹ Therefore Nedd4 (neural precursor cell expressed developmentally downregulated protein 4) and α -spectrin appear to play a role in the assembly, insertion, and/or retrieval of the ENaC subunits in the plasma membrane.²⁷⁰

CLINICAL PRESENTATION

Liddle syndrome is characterized by inappropriate renal sodium reabsorption, blunted sodium excretion, and low-renin hypertension.²⁶⁷ The features of this syndrome were described by Liddle and colleagues in 1963 in a large pedigree.²⁷¹ Affected subjects are at increased risk of cerebrovascular and cardiovascular events. Liddle syndrome can be differentiated from other rare mendelian forms of low-renin hypertension with the use of urinary/plasma hormonal profiles. Glucocorticoid-remediable aldosteronism is associated with increased production of 18-hydroxycortisol and aldosterone metabolites. Apparent mineralocorticoid excess (AME) is associated with an elevated ratio of urinary cortisol (tetrahydrocortisol) to cortisone (tetrahydrocortisone) metabolites (Table 44.7).

TREATMENT

Hypertension is not improved by spironolactone but can be corrected by a low-salt diet and an ENaC antagonist (amiloride or triamterene).

APPARENT MINERALOCORTICOID EXCESS

PATHOGENESIS

The syndrome of AME (OMIM #207765) is a rare autosomal recessive disorder that results in hypokalemic hypertension with low serum levels of renin and aldosterone.^{271,272} AME is

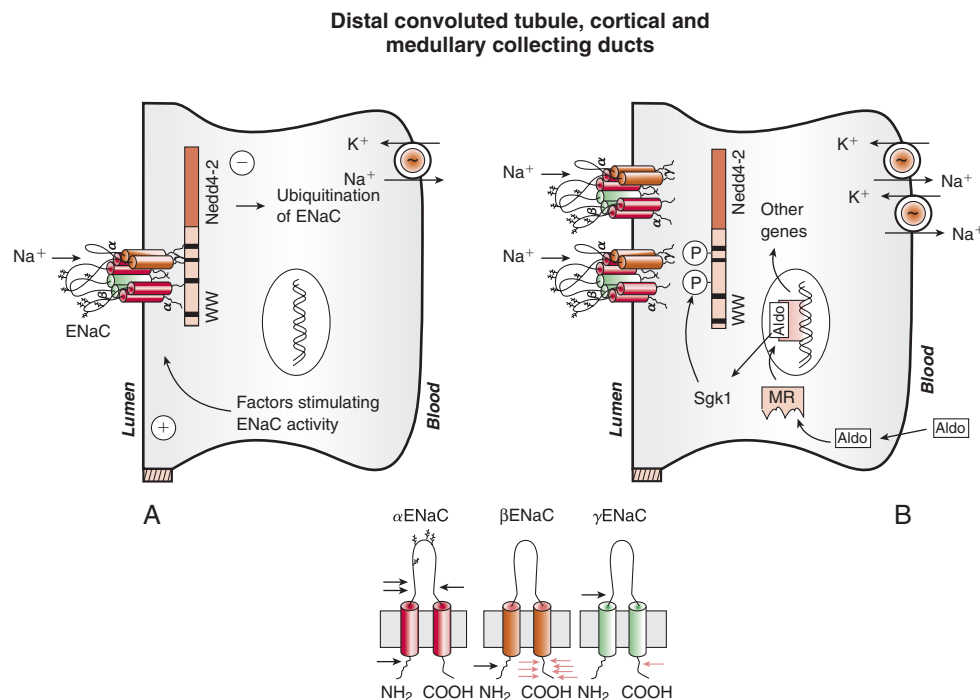


Fig. 44.14 Disorders of the epithelial Na channel (ENaC). (Top panel) The mechanism of ENaC expression in an aldosterone-sensitive epithelial cell is represented. (A) In a resting state, few ENaCs, which facilitate sodium reabsorption in a rate-limiting fashion, are resident in the apical membrane. Factors known to enhance ENaC surface expression and activity are counterbalanced by retrieval of these channels from the membrane through the ubiquitin pathway mediated by ubiquitin–protein ligase Nedd4-2. (B) Shortly after aldosterone exposure and binding to the mineralocorticoid receptor (MR), transcriptional stimulation of serum and glucocorticoid-regulated kinase 1 (Sgk1) leads to phosphorylation of Nedd4-2, which subsequently disrupts ENaC/Nedd4-2 interactions. In this situation, ubiquitination of ENaCs is reduced, thus favoring both their residence in the apical membrane and enhanced sodium reabsorption. (Bottom panel) The ENaC is composed of two α -, one β -, and one γ -subunits surrounding the channel pore²⁶⁷. Each subunit has two transmembrane domains with short cytoplasmic amino and carboxy termini and a large ectocytosolic loop. Mutations in subunits of ENaC cause either Liddle syndrome (pink arrows; β - or γ -subunits) or the autosomal recessive form of pseudohypoaldosteronism type I [PHA I (black arrows)] (α -, β -, or γ -subunits).^{268,284} (From Rossier BC. The epithelial sodium channel (ENaC): new insights into ENaC gating. *Pflugers Arch* 2003;446:314–316.)

Table 44.7 Urinary Steroid Profiles in Mendelian Forms of Low-Renin Hypertension

	Liddle Syndrome	GRA	AME
Aldosterone	↓↓↓	↑↑	↓
TH-Aldo	↓↓↓	↑↑	↓
18-OH-TH-Aldo	↓↓↓	↑↑	↓
18-OH F	–	↑↑↑	–
TH-F	nl	nl	↑↑
TH-E	nl	nl	↓
TH-F/TH-E ratio	nl	nl	↑↑↑

Aldo, Aldosterone; AME, apparent mineralocorticoid excess; E, cortisone; F, cortisol; GRA, glucocorticoid-remediable aldosteronism; nl, normal; TH-, tetrahydro-; –, not usually detected.

Modified from Warnock DG: Liddle syndrome: an autosomal dominant form of human hypertension. *Kidney Int.* 1998;53:18–24.

caused by a deficiency in 11 β -hydroxysteroid dehydrogenase type 2 enzymatic activity (11 β -HSD2), which is responsible for the conversion of cortisol to the inactive metabolite cortisone, therefore protecting the mineralocorticoid receptors from cortisol intoxication. In AME, cortisol acts as a

potent mineralocorticoid and causes salt retention, hypertension, and hypokalemia with a suppression of the renin–angiotensin–aldosterone system. A milder phenotype, or type 2 variant, also results from abnormal activity of the enzyme.²⁷³ Cushing syndrome and extremely high cortisol levels can overcome the ability of 11 β -HSD2 to convert cortisol to cortisone.

CLINICAL PRESENTATION

AME is associated with severe juvenile low-renin hypertension, hypokalemic alkalosis, low birth weight, failure to thrive, poor growth, nephrocalcinosis, and variable degrees of polyuria. The urinary metabolites of cortisol demonstrate an abnormal ratio, with predominance of cortisol metabolites (i.e., tetrahydrocortisol plus 5 α -tetrahydrocortisol/tetrahydrocortisone in the range 6.7–33, the normal ratio being 1.0). The milder form of AME (type 2) lacks the typical urinary steroid profile (i.e., biochemical analysis reveals a moderately elevated ratio of cortisol to cortisone metabolites).²⁷³ The heterozygote state is phenotypically normal but is associated with subtle defects in cortisol metabolism.

TREATMENT

The treatment of AME is sodium restriction and triamterene, amiloride, or spironolactone. Additional antihypertensive agents may be used as needed.

AUTOSOMAL DOMINANT EARLY ONSET HYPERTENSION WITH SEVERE EXACERBATION DURING PREGNANCY

Autosomal dominant early onset hypertension with severe exacerbation during pregnancy is a rare autosomal dominant disorder (OMIM #605115) described in a family that is associated with activating mutations in the mineralocorticoid receptor. By screening the mineralocorticoid receptor in 75 patients with early onset of severe hypertension, Geller and colleagues identified a 15-year-old boy with severe hypertension, suppressed plasma renin activity, low aldosterone, and no other underlying cause of hypertension, who had a heterozygous missense mutation (S810L) in the mineralocorticoid receptor gene.²⁷⁴ Of 23 relatives evaluated, 11 had been diagnosed with severe hypertension before age 20, whereas the remaining 12 had unremarkable blood pressures. Two L810 carriers had undergone five pregnancies, all of which had been complicated by marked exacerbation of hypertension with suppressed aldosterone levels. The S810L mutation alters a conserved amino acid, resulting in a constitutively active and altered mineralocorticoid receptor. In addition, progesterone and other steroids lacking 21-hydroxyl groups, normally mineralocorticoid receptor antagonists, become potent agonists. Spironolactone is also a potent agonist of L810, so its use is contraindicated in L810 carriers.

GLUCOCORTICOID-REMEDIALABLE HYPERALDOSTERONISM

PATHOGENESIS

Glucocorticoid-remediable hyperaldosteronism (GRA) is also known as familial hyperaldosteronism type I (OMIM #103900), aldosteronism sensitive to dexamethasone, glucocorticoid-suppressible hyperaldosteronism, and syndrome of ACTH-dependent hyperaldosteronism. It is an autosomal dominant hypertensive disorder caused by a chimeric gene duplication arising from unequal crossover between genes encoding aldosterone synthase and 11 β -hydroxylase,²⁷⁵ two highly similar genes with the same transcriptional orientation lying 45,000 base pairs apart on chromosome 8. Humans have two isozymes with 11 β -hydroxylase activity that are required for synthesis of cortisol and aldosterone, respectively. CYP11B1 (11 β -hydroxylase) is expressed at high levels and is regulated by ACTH, whereas CYP11B2 (aldosterone synthase) is normally expressed at low levels and is regulated by angiotensin II. In addition to 11 β -hydroxylase activity, the latter enzyme has 18-hydroxylase and 18-oxidase activities and can synthesize aldosterone from DOC.²⁷⁶ Thus with the unequal crossover between the two genes, the aldosterone synthase gene is under the control of regulatory promoter sequences of the 11 β -hydroxylase. The chimeric gene product is expressed at high levels in both the zona glomerulosa and zona fasciculata and is controlled by ACTH, leading to increased production of 18-hydroxycortisol and aldosterone metabolites.

CLINICAL PRESENTATION

The phenotype of GRA is highly variable.²⁷⁷ Affected individuals may have mild hypertension and normal biochemistry and may be clinically indistinguishable from patients with essential hypertension. However, some subjects have early onset severe hypertension, hypokalemia, and metabolic alkalosis. In a study of 376 patients from 27 genetically proven

GRA pedigrees, 48% of all GRA families and 18% of all patients with GRA had cerebrovascular complications, findings similar to the frequency of aneurysm in adult polycystic kidney disease.²⁷⁸

The diagnosis is usually established by measuring 18-hydroxycortisol or 18-oxocortisol metabolites in the urine or with the dexamethasone suppression test.²⁷⁸ In addition, because patients with GRA secrete aldosterone in response to ACTH, glucocorticoid administration can suppress excessive aldosterone secretion.²⁷⁹ The dexamethasone suppression test is a variably reliable method for establishing the diagnosis. Patients without the disease (i.e., subjects with an aldosterone-producing adenoma or with idiopathic hyperaldosteronism) can suppress aldosterone secretion.²⁸⁰ The diagnosis of GRA can be definitively established by demonstration of the chimeric gene by molecular techniques.

TREATMENT

Simple glucocorticoid replacement is the treatment for GRA. Salt restriction combined with either spironolactone or ENaC inhibition is also effective.

FAMILIAL HYPERALDOSTERONISM TYPE II

Familial hyperaldosteronism type II (FH-II; OMIM #605635) is characterized by hypersecretion of aldosterone because of adrenocortical hyperplasia, an aldosterone-producing adenoma, or both. In contrast to familial hyperaldosteronism type I, FH-II is not suppressible by dexamethasone. Stowasser and colleagues reported five families with this phenotype with a segregation pattern supporting dominant inheritance.²⁸¹ Analysis of an extended kindred has revealed linkage between FH-II and markers on chromosome 7p22, but the gene implicated has not been clearly identified.²⁸²

PSEUDOHYPOALDOSTERONISM

PSEUDOHYPOALDOSTERONISM TYPE I

PATHOGENESIS

PHA I is a rare disorder characterized by salt wasting, hypotension, hyperkalemia, metabolic acidosis, and failure to thrive in infants. There are two subtypes of PHA I.²⁸³ The autosomal recessive form (OMIM #264350 and #177735) leads to severe manifestations that persist in adulthood and is caused by inactivating mutations in any of the three subunits (α , β , or γ) of ENaC. The autosomal dominant form (OMIM #177735) is associated with milder manifestations that remit with age and is caused by mutations in the mineralocorticoid receptor gene²⁸⁴ that result in haploinsufficiency or in dominant negative actions. Homozygous mineralocorticoid receptor mutations are probably lethal in humans, because knockout mice show a severe salt-wasting syndrome and die a few days after birth.²⁸⁵

CLINICAL PRESENTATION

The clinical contrast between PHA I because of ENaC and that as a result of mineralocorticoid receptor mutations is striking.²⁸³ Autosomal recessive PHA I manifests neonatally or in childhood as renal salt wasting, hypotension, hyperkalemia, metabolic acidosis, and, on occasion, failure to thrive. Other biologic features include hyponatremia, high plasma

and urinary aldosterone levels despite hyperkalemia, and elevated plasma renin activity. Autosomal dominant PHA I is associated with milder manifestations and remits with age. PHA I must be differentiated from aldosterone synthase deficiency, salt-wasting forms of congenital adrenal hyperplasia, and adrenal hypoplasia congenita, all of which cause aldosterone deficiency and are associated with hyponatremia, hyperkalemia, hypovolemia, elevated plasma renin activity, and occasionally shock and death.²⁸⁶ Bartter syndrome type 2 (ROMK gene mutations) can also manifest in the neonatal period with a similar (transient) clinical picture.

TREATMENT

Treatment of PHA I consists of salt supplementation, which can greatly improve hyponatremia, hyperkalemia, and growth. Administration of aldosterone, fludrocortisone, or DOC is not helpful. Patients with the recessive form usually need lifelong treatment for salt wasting and hyperkalemia, whereas for those with the dominant form, treatment can usually be withdrawn in adulthood.

PSEUDOHYPOALDOSTERONISM TYPE II

PATHOGENESIS

Pseudohypoaldosteronism type II (PHA II; OMIM #145260), also known as familial hyperkalemia and hypertension or

Gordon syndrome, is a volume-dependent, low-renin hypertension characterized by persistent hyperkalemia despite a normal renal GFR. Variants within at least four genes—the with-no-lysine-(K) kinases, *WNK1* and *WNK4*, *Cullin3*, and *KLHL3* (encoding the Kelch-like 3 protein)—can cause the phenotype.^{286,287} Details are still emerging for some of these genes, but it is likely that they all cause a gain of function in thiazide-sensitive NCC and, hence, salt retention. Hypertension is attributable to increased renal salt reabsorption, and hyperkalemia to reduced renal K^+ excretion. Reduced renal H^+ secretion is also commonly seen, resulting in metabolic acidosis. The features of PHA II are chloride dependent, because they are corrected when infusion of sodium sulfate or sodium bicarbonate is substituted for sodium chloride.²⁸⁸

WNK1 and *WNK4* function as molecular switches, eliciting coordinated effects on diverse ion transport pathways to maintain homeostasis during physiologic perturbation. In PHA II, mutations that appear to activate *WNK1* and inactivate *WNK4* have been proposed to result in increased thiazide-sensitive cotransporter (NCC) activity and decreased ROMK activity (Fig. 44.15).²⁸⁹ Kelch-like 3 and Cullin3 regulate electrolyte homeostasis via ubiquitination and degradation of *WNK1* and *WNK4*.^{289,290} On one hand is the overactivity of NCC and on the other is the increased paracellular reabsorption of Cl^- , also known as the *chloride shunt* hypothesis.

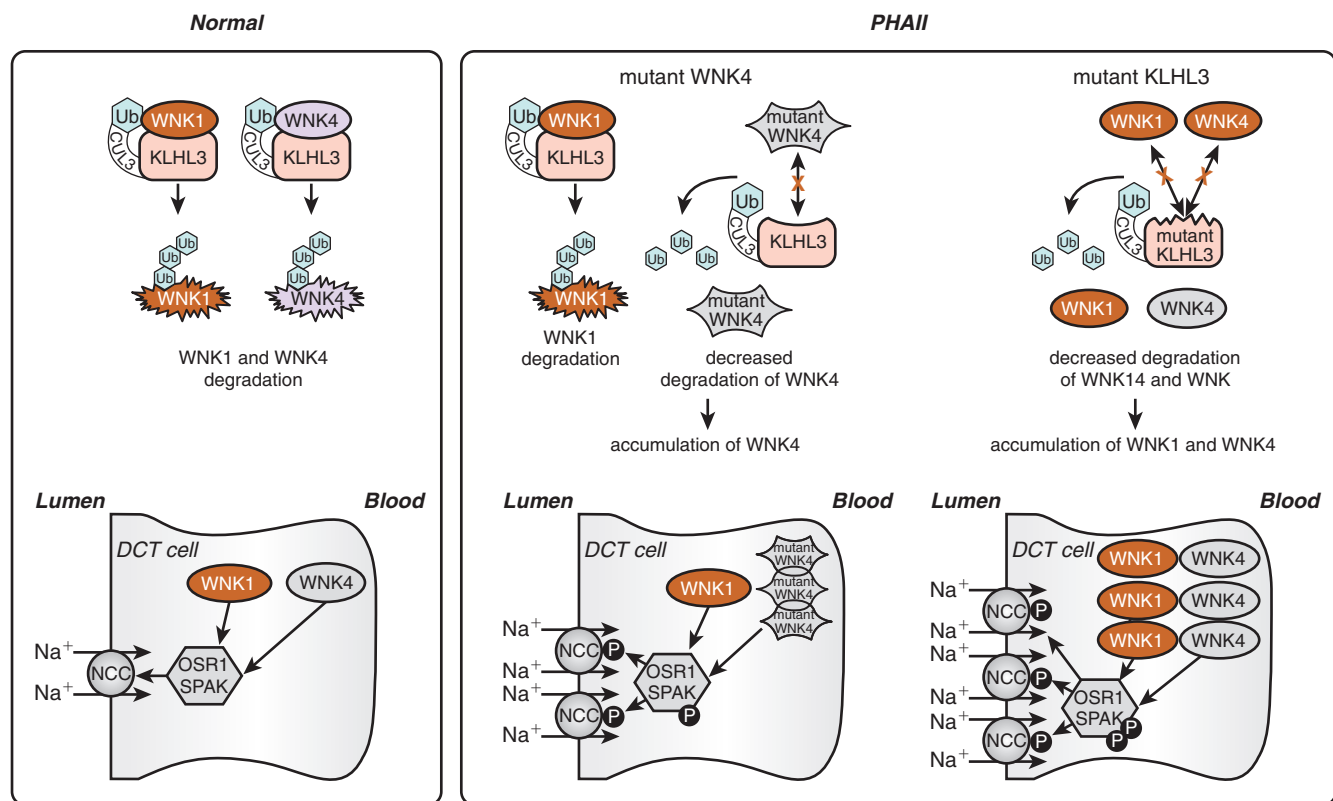


Fig. 44.15 Molecular mechanism of pseudohypoaldosteronism type II in the distal convoluted tubule. (Left) Normally kinases WNK1 and WNK4 can stimulate Na–Cl cotransporter (NCC) trafficking to the plasma membrane and phosphorylation of NCC by Ste20-related proline alanine-rich kinase (SPAK) and oxidative stress–responsive kinase (OSR1). This is the WNK–OSR1/SPAK–NCC phosphorylation signaling cascade, recognized now as the major pathogenic mechanism of pseudohypoaldosteronism type II (PHA II; shown at right). WNK1 and WNK4 are substrates of KLHL3 (Kelch-like 3)–Cullin3 E3 ligase–mediated ubiquitination. DCT, Distal convoluted tubule; Ub, ubiquitin. (From Susa K, Sohara E, Rai T, et al. Impaired degradation of WNK1 and WNK4 kinases causes PHAII in mutant KLHL3 knock-in mice. *Hum Mol Genet.* 2014;23:5052–5060; Pathare G, Hoenderop JG, Bindels RJ, et al. A molecular update on pseudohypoaldosteronism type II. *Am J Physiol Renal Physiol.* 2013;305:F1513–F1520.)

The action of these kinases may serve to increase salt reabsorption and intravascular volume in volume-depleted states and decrease potassium secretion in K depletion, by enabling the distal nephron to allow either maximal NaCl reabsorption or maximal K⁺ secretion in response to hypovolemia or hyperkalemia, respectively.

CLINICAL PRESENTATION

PHA II is usually diagnosed in adults but can also be seen neonatally.²⁹¹ Unexplained hyperkalemia is the usual presenting feature and occurs before the onset of hypertension. The severity of hyperkalemia varies greatly and is influenced by prior intake of diuretics and salt.²⁹² Causes of spurious elevation of potassium should be ruled out before this diagnosis is made. In its most severe form, PHA II is associated with muscle weakness (from hyperkalemia), short stature, and intellectual impairment. Mild hyperchloremia, metabolic acidosis, and suppressed plasma renin activity are findings variably associated with the trait. Aldosterone levels vary from low to high depending on the level of hyperkalemia. Urinary concentrating ability, acid excretion, and proximal tubular function are all normal.

TREATMENT

Thiazides reverse all biochemical abnormalities. Lower than average doses can be given if overcorrection occurs. Loop diuretics may also be used.

INHERITED DISORDERS OF RENAL MAGNESIUM PROCESSING

Magnesium, the second most abundant intracellular cation, plays an important role as a cofactor in energy metabolism, nucleotide and protein synthesis, neuromuscular excitability, and oxidative phosphorylation and as a regulator of sodium, potassium, and calcium channels. Under normal conditions, extracellular magnesium concentration is maintained at nearly constant values (0.70–1.1 mmol/L). Hypomagnesemia results from decreased dietary intake but more commonly is caused by intestinal malabsorption, renal losses, or use of drugs including cyclosporine, omeprazole, cetuximab, and cisplatin.²⁹³

Primary hypomagnesemia is composed of a heterogeneous group of disorders characterized by renal and intestinal Mg wasting often associated with hypercalciuria (Table 44.8).^{293–298} The genetic basis and cellular defects of a number of primary hypomagnesemias have been elucidated. These inherited conditions affect different nephron segments and different cell types, leading to variable but increasingly distinguishable phenotypic presentations.

FAMILIAL HYPOMAGNESEMIA WITH HYPERCALCIURIA AND NEPHROCALCINOSIS

The syndrome of renal hypomagnesemia with hypercalciuria and nephrocalcinosis (OMIM #248250), or familial hypomagnesemia with hypercalciuria and nephrocalcinosis (FHHNC), is a rare autosomal recessive trait characterized by profound Mg wasting that results in severe hypomagnesemia not correctable by oral or intravenous magnesium supplementation. The disorder is caused by mutations in claudin-16

(*CLDN16*), previously known as paracellin-1,²⁹⁴ a protein located in tight junctions of the TAL and related to the claudin family of tight junction proteins (see Fig. 44.10). A study of nine additional families with severe hypomagnesemia identified mutations in *CLDN19* (OMIM #248190) that share the same renal phenotype as FHHNC. *CLDN19* encodes claudin-19, a tight junction protein expressed in renal tubules and the eye.²⁹⁹

Every patient with FHHNC has hypomagnesemia with inappropriately high urinary Mg excretion (Mg²⁺ fractional excretions >10%). Renal calcium wasting is present in every case initially and leads to parenchymal calcification (nephrocalcinosis) and renal failure, often requiring dialysis. The progression rate of renal insufficiency correlates with the severity of nephrocalcinosis. Other clinical findings are polyuria, polydipsia, ocular abnormalities, recurrent urinary tract infections, and renal colic with stone passage. Serum PTH levels are abnormally high. Serum levels of calcium, phosphorus, and potassium and urinary excretion of uric acid and oxalate are normal. In contrast to patients with a *CLDN16* defect, affected individuals with a *CLDN19* mutation have ocular symptoms that include severe visual impairment, macular colobomata, horizontal nystagmus, and marked myopia.²⁹⁹ Long-term oral Mg administration does not normalize serum Mg²⁺ levels. Thiazides are effective to reduce urinary Ca excretion.³⁰⁰ After kidney transplantation, tubular handling of Mg²⁺ and Ca is normalized.

FAMILIAL HYPOMAGNESEMIA WITH SECONDARY HYPOCALCEMIA

Familial hypomagnesemia with secondary hypocalcemia (OMIM #602014) is an autosomal recessive disease that results in electrolyte abnormalities shortly after birth and is caused by mutations in *TRPM6* (Fig. 44.16). *TRPM6* holds 39 exons that code for a protein of 2022 amino acids. Affected individuals show extremely low serum Mg²⁺ levels (0.1–0.3 mmol/L) and hypocalcemia, causing muscular and neurologic complications including seizures in early infancy that can lead to neurologic damage or cardiac arrest if left untreated. The disorder is caused by impaired intestinal uptake and renal Mg²⁺ wasting.³⁰¹ Restoring the concentrations of serum magnesium to normal values with high-dose magnesium supplementation can overcome the apparent defect in magnesium absorption and in serum concentrations of calcium. Lifelong magnesium supplementation is required to overcome the defect in the seizures and magnesium handling in these individuals.

ISOLATED DOMINANT HYPOMAGNESEMIA WITH HYPOCALCIURIA

A rare autosomal dominant disorder, isolated dominant hypomagnesemia with hypocalciuria (OMIM #154020) is caused by a dominant negative mutation of the *FXD2* gene resulting in a trafficking defect of the γ -subunit of the Na⁺–K⁺–ATPase at the basolateral membrane of the distal convoluted tubule,³⁰² the main site of active renal Mg²⁺ reabsorption. The hypomagnesemia in patients with the disorder can be as low as 0.40 mmol/L, resulting in convulsions. Mutation of *FXD2* leads to misrouting of the Na⁺/K⁺–ATPase γ -subunit, so Mg²⁺ reabsorption is abnormal.

Table 44.8 Inherited Disorders of Magnesium Transport

Disease/OMIM ^a Entry	Gene/Inheritance	Protein	Key Clinical/Biochemical Symptoms
Gitelman syndrome/#263800	<i>SLC12A3</i> /AR	NCC	Muscle weakness/tetany Fatigue Chondrocalcinosis Hypomagnesemia Hypocalciuria
Familial hypomagnesemia with hypercalciuria and nephrocalcinosis/#248250/#248190	<i>CLDN16</i> /AR <i>CLDN19</i> /AR	Claudin-16 Claudin-19	Polyuria Renal stones/nephrocalcinosis Ocular abnormalities Severe hypomagnesemia Hypercalciuria
Autosomal dominant isolated renal Mg loss/#154020	<i>FXYD2</i> /AD	γ -Subunit sodium–potassium ATPase	Seizures Chondrocalcinosis Hypomagnesemia Hypocalciuria
Autosomal dominant hypomagnesemia, hypermagnesuria, and hypocalciuria/#137920	<i>HNF1B</i> /AD	Transcription factor hepatocyte nuclear factor 1 homeobox B (<i>HNF1B</i>) is linked to the regulation of <i>FXYD2</i>	Renal malformations Maturity-onset diabetes of the young (MODY)
Maturity-onset diabetes of the young (MODY) with hypomagnesemia and renal Mg ²⁺ loss/#126090	<i>PCBD1</i> /AR	Hepatocyte nuclear factor 1 homeobox A (<i>PCBD1</i>)	MODY
Familial hypomagnesemia with secondary hypocalcemia/#602014	<i>TRPM6</i> /AR	Epithelial magnesium channel TRPM6	Tetany/seizures Hypomagnesemia Hypocalcemia
Autosomal recessive isolated Mg loss/#611718	<i>EGF</i> /AR	Epidermal growth factor	Tetany/seizures Hypomagnesemia Normocalciuria
Autosomal dominant hypomagnesemia/#176260	<i>KCNA1</i> /AD	Voltage-gated potassium channel Kv1.1	Muscle cramps Tetany Tremor Muscle weakness Cerebral atrophy Myokymia
Epilepsy, ataxia, sensorineural deafness and tubulopathy (EAST)/#612780	<i>KCNJ10</i> /AR	K ⁺ channel Kir4.1	Polyuria Hypokalemic metabolic alkalosis Hypomagnesemia Hypocalciuria
Dominant and recessive hypomagnesemia with impaired brain development and seizures/#607803	<i>CNNM2</i> /AD/AR	CNNM2 Mg-ATP	Mental retardation Seizures

^aOMIM numbers. Available at <http://www.ncbi.nlm.gov/omim/OMIM>.

AD, Autosomal dominant; AR, autosomal recessive; ATP, adenosine triphosphate; *CLDN*, claudin; *CNNM2*, cyclin M2; *EGF*, epidermal growth factor; *FXYD2*, FXYP domain-containing ion transport regulator 2; *NCC*, Na–Cl cotransporter; *OMIM*, Online Mendelian Inheritance in Man; *SLC*, solute carrier; *TRPM6*, transient receptor potential channel 6.

Hypomagnesemic patients have lower urinary excretion of calcium, presumably as a consequence of increased reabsorption in the loop of Henle.³⁰³

The transcription factor hepatocyte nuclear factor 1 homeobox B (*HNF1B*) has been linked to the regulation of the *FXYD2* gene, and hypomagnesemia, hypermagnesuria, and hypocalciuria have been observed in 44% of *HNF1B* mutation carriers (OMIM #137920). Mutations have also been described in *PCBD1*, which encodes hepatocyte nuclear factor 1 homeobox A (*PCBD1*; see Fig. 44.16).³⁰⁴

CA²⁺/MG²⁺-SENSING RECEPTOR-ASSOCIATED DISORDERS

An important regulator of magnesium homeostasis is the Ca²⁺/Mg²⁺ (calcium)–sensing receptor (CaSR). CaSR senses ionized serum calcium and magnesium concentrations and is involved in renal calcium and magnesium reabsorption as well as in PTH secretion.³⁰⁵ Activating mutations of the *CaSR* gene were first described in families affected with autosomal dominant hypocalcemia.^{241,306} Affected individuals present

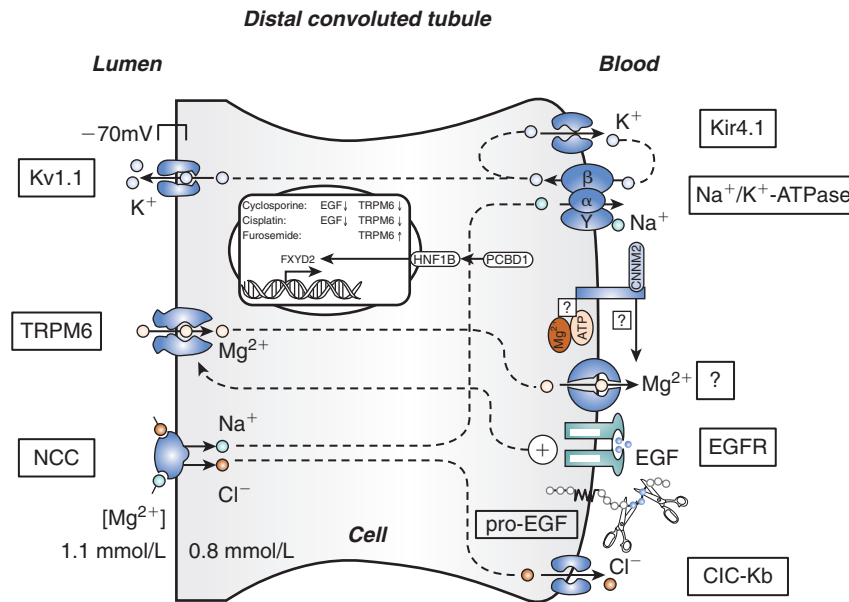


Fig. 44.16 Magnesium reabsorption in the distal convoluted tubule. Transient receptor potential 6 (TRPM6) channels, located in the luminal membrane, facilitate transport of Mg^{2+} from the proline into the cell, which is driven primarily by the luminal membrane potential established by the voltage-gated K^+ channel Kv1.1. Epidermal growth factor (EGF) and insulin function as magnesiotropic hormones, stimulating TRPM6 activity through activation of the PI3K–Akt pathway. Insulin can also act on TRPM6 via phosphorylation of cyclin-dependent kinase 5 (CDK5). The expression of TRPM6 in the distal convoluted tubule is affected by treatment with furosemide, cyclosporine A, and cisplatin; the last two have been shown to also downregulate EGF levels. The Mg^{2+} buffering and extrusion systems are not yet known. CNNM2, the gene encoding cyclin M2, is suggested as playing a role in the Mg^{2+} extrusion and can bind Mg–adenosine triphosphate (ATP), which might play a role in this process. Transcription factor HNF1B, together with its regulator PCBD1, is proposed to regulate the expression of FXRD2, which encodes the γ -subunit of the Na^+/K^+ -ATPase. The basolateral potassium channel, Kir4.1, recycles potassium to facilitate Na^+/K^+ -ATPase activity. ClC-Kb, Basolateral chloride channel b; EGFR, EGF receptor; NCC, Na–Cl cotransporter; PI3K, phosphoinositide 3-kinase. (Modified from van der Wijst J, Bindels RJ, Hoenderop JG. Mg^{2+} homeostasis: the balancing act of TRPM6. *Curr Opin Nephrol Hypertens*. 2014;23:361–369; Glaudemans B, Knoers NV, Hoenderop JG, et al. New molecular players facilitating Mg^{2+} reabsorption in the distal convoluted tubule. *Kidney Int*. 2010;77:17–22.)

with hypocalcemia, hypercalciuria, and polyuria, and approximately 50% of them have hypomagnesemia.²⁹⁶ Clinically, autosomal dominant hypocalcemia may be mistaken for primary hypoparathyroidism, because of the decreased PTH secretion in the setting of mild to moderate hypocalcemia. Most affected individuals have hypomagnesemia and renal magnesium wasting.

ISOLATED RECESSIVE HYPOMAGNESEμία WITH NORMOCALCIURIA

Isolated recessive hypomagnesemia (OMIM #131530) is a rare hereditary disease that was originally described in a consanguineous family.³⁰⁷ The affected individuals presented with symptoms of hypomagnesemia during early infancy. Isolated recessive hypomagnesemia is caused by a mutation of the epidermal growth factor (EGF) precursor protein pro-EGF, which is expressed in the gastrointestinal tract, the respiratory tract, and the basolateral membrane of the distal convoluted tubule. On membrane insertion, pro-EGF is processed by unknown proteases into a functional EGF peptide hormone, which activates EGF receptors on the basolateral membrane (Fig. 44.16). As a result, EGF stimulates the trafficking of TRPM6 channels to the luminal membrane, increasing the reabsorption of Mg^{2+} through TRPM6.²⁹³

DOMINANT AND RECESSIVE HYPOMAGNESEμία

Dominant and recessive hypomagnesemia with impaired brain development (recessive) and seizures secondary to *CNNM2* mutations (OMIM #607803). Mutations in the gene encoding cyclin M2 (*CNNM2*) have been found in two unrelated families with dominant isolated hypomagnesemia, mental retardation, and seizures.³⁰⁸ Recessive forms have also been identified.³⁰⁹

DIABETES INSIPIDUS

PATHOGENESIS

The conservation of water by the human kidney is regulated by the action of the neurohypophyseal antidiuretic hormone arginine vasopressin (AVP) on cells of the collecting tubules,³¹⁰ as reviewed in detail in Chapter 10.

AVP promotes urinary concentration by allowing water to be transported passively down an osmotic gradient between the tubular fluid and the surrounding interstitium.³¹²

THE AVP–AVPR2–AQP SHUTTLE PATHWAY

Water homeostasis in the kidney is regulated by three key proteins. AVP, secreted from the posterior pituitary, activates

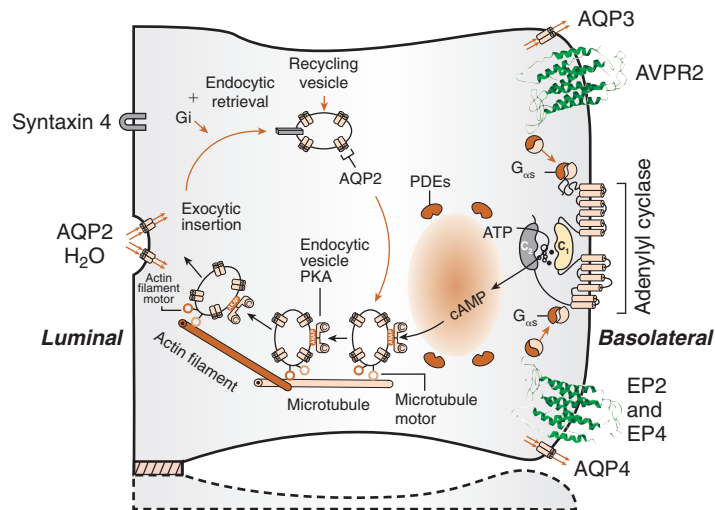


Fig. 44.17 Effect of vasopressin to increase water permeability in the principal cells of the collecting duct. Vasopressin (AVP) binds to the V_2 receptor (AVPR2, a G protein-coupled receptor) on the basolateral membrane. This activates adenylyl cyclase, increasing the intracellular concentration of cyclic AMP (cAMP). Protein kinase A (PKA) is the target of the generated cAMP. The binding of cAMP to the regulatory subunits of PKA induces a conformational change, causing these subunits to dissociate from the catalytic subunits. These activated subunits as shown here are anchored to an aquaporin-2 (AQP2)-containing endocytic vesicle via an A-kinase anchoring protein. The local concentration and distribution of the cAMP gradient is limited by phosphodiesterases (PDEs). Cytoplasmic vesicles carrying the water channels (represented as homotetrameric complexes) are fused to the luminal membrane in response to AVP, thereby increasing the water permeability of this membrane. The dissociation of the A-kinase anchoring protein from the endocytic vesicle is not represented. Microtubules and actin filaments are necessary for vesicle movement toward the membrane. When AVP is not available, AQP2 water channels are retrieved by an endocytic process, and water permeability returns to its original low rate. Aquaporin-3 (AQP3) and aquaporin-4 (AQP4) water channels are expressed constitutively at the basolateral membrane. Other G protein-coupled receptors, such as prostaglandin receptors EP2 and EP4, and the secretin receptor, may also contribute to an increase in intracellular cAMP. ATP, Adenosine triphosphate; Gi, inhibitory guanine nucleotide-binding protein.

the process of water excretion by binding to the vasopressin V_2 receptor (AVPR2; see Fig. 44.17) located on the basolateral membrane of collecting duct cells. This step activates the stimulatory G protein (G_s) and adenylyl cyclase, resulting in the production of cyclic AMP (cAMP) and stimulation of protein kinase A. The final step in the antidiuretic action of AVP is the exocytic insertion of a specific water channel, AQP2, into the luminal membrane, thereby increasing the water permeability of that membrane. These water channels are members of a superfamily of integral membrane proteins that facilitate water transport.^{313,314}

The short-term regulation of AQP2 by AVP involves the movement of AQP2 from the intracellular vesicles to the luminal membrane; in the long-term regulation, which requires a sustained elevation of circulating AVP for 24 hours or more, AVP increases the abundance of water channels. This increase is thought to be a consequence of increased transcription of the *AQP2* gene.³¹⁷ AQP3 and AQP4 are the water channels in basolateral membranes of renal medullary collecting ducts. In addition, vasopressin increases the water reabsorptive capacity of the kidney by regulating the urea transporter UT-A1, which is expressed in the inner medullary collecting duct, predominantly in its terminal part.³¹⁸ AVP also increases the permeability of principal collecting duct cells to sodium.³¹⁹

In summary, in the absence of AVP stimulation, collecting duct epithelia exhibit very low permeabilities to sodium, urea, and water. These specialized permeability properties permit the excretion of large volumes of hypotonic urine formed during intervals of water diuresis. By contrast, AVP stimulation

of the principal cells of the collecting ducts leads to selective increases in the permeabilities of the apical membrane to water (P_f), urea (P_{urea}), and sodium (P_{Na}).

In neurohypophyseal diabetes insipidus, termed “familial neurohypophyseal diabetes insipidus” (FNDI), levels of AVP are insufficient, and patients show a positive response to treatment with desmopressin (DDAVP). Growth retardation might be observed in untreated children with autosomal dominant FNDI (adFNDI).³²⁰

More than 85 variants in the prepro-arginine-vasopressin-neurophysin II *AVP* gene located on chromosome 20p13 have been reported in adFNDI. Autosomal dominant central diabetes insipidus is secondary to *AVP* mutations that result in misfolding and aggregation of the protein in the ER.³²¹ ER-associated degradation (ERAD) is a principal quality control mechanism in cells responsible for targeting misfolded ER proteins for cytosolic degradation. In mammals, the most well-characterized ERAD machinery is the highly conserved complex of suppressor-enhancer of lin-12-like and hydroxymethylglutaryl-CoA reductase degradation protein 1 (SEL1L-HRD1), which consists of the E3 ubiquitin ligase HRD1 and its adaptor protein SEL1L. WT proAVP aggregates with mutant proAVP to form complexes that are retrotranslocated and destroyed. Moreover, WT proAVP and mutant proAVP were recently shown to be substrates for Sel1L, which links these proAVPs to Hrd1 and ERAD-mediated destruction and eliminates expression of the normal allele, explaining the vasopressin deficiency before magnocellular cell death.^{322,323} Recessive FNDI has also been described.^{324,325}

In nephrogenic diabetes insipidus (NDI), AVP values are normal or elevated, but the kidney is unable to concentrate urine. The clinical manifestations of polyuria and polydipsia can be present at birth and must be immediately recognized to avoid severe episodes of dehydration. Most (>90%) of patients with congenital NDI have X-linked mutations in the *AVPR2* gene, the Xq28 gene coding for the vasopressin V₂ (antidiuretic) receptor. In less than 10% of the families studied, congenital NDI has an autosomal recessive inheritance, and more than 45 mutations have been identified in the *AQP2* gene (*AQP2*), located in chromosome region 12q13. For the *AVPR2* gene, more than 250 putative disease-causing mutations in 326 unrelated families with X-linked NDI have now been published.³²⁶ When studied in vitro, most *AVPR2* mutations lead to receptors that are trapped intracellularly and are unable to reach the plasma membrane.³²⁷

A minority of the mutant receptors reach the cell surface but are unable to bind AVP or to trigger an intracellular cAMP signal. Similarly, *AQP2* mutant proteins are trapped intracellularly and cannot be expressed at the luminal membrane. This *AQP2* trafficking defect is correctable, at least in vitro, by chemical chaperones. Other inherited disorders with mild, moderate, or severe inability to concentrate urine include Bartter syndrome (MIM 601678),^{248,328} *MAGED2* mutations with transient Bartter and severe polyhydramnios,³²⁹ cystinosis, autosomal dominant hypocalcemia, nephronophthisis, and AME.³³⁰

CLINICAL PRESENTATION AND HISTORY OF X-LINKED NEPHROGENIC DIABETES INSIPIDUS

X-linked NDI (OMIM #304800) is secondary to *AVPR2* mutations, which result in a loss of function or dysregulation of the V₂ receptor.³²⁶ Male patients who have an *AVPR2* mutation have a phenotype characterized by early dehydration episodes, hypernatremia, and hyperthermia as early as the first week of life. Dehydration episodes can be so severe that they lower arterial blood pressure to a degree that is not sufficient to sustain adequate oxygenation to the brain, kidneys, and other organs. Mental and physical retardation and renal failure are the classic “historical” consequences of a late diagnosis and lack of treatment. Heterozygous female patients may exhibit variable degrees of polyuria and polydipsia because of skewed X chromosome inactivation. The “historical” clinical characteristics include hypernatremia, hyperthermia, mental retardation, and repeated episodes of dehydration in early infancy.³³¹ Mental retardation, a consequence of repeated episodes of dehydration, was prevalent in the Crawford and Bode study, which found that only nine of 82 patients (11%) had normal intelligence.³³¹

Early recognition and treatment of X-linked NDI with an abundant intake of water allow a normal life span with normal physical and mental development.³³² Two characteristics suggestive of X-linked NDI are the familial occurrence and the confinement of mental retardation to male patients. It is then tempting to assume that the family described in 1892 by McIlraith and discussed by Reeves and Andreoli was a family with X-linked NDI.^{333,334} Lacombe and Weil described a familial form of diabetes insipidus with autosomal transmission and without any associated mental retardation.^{335,336} The descendants of the family originally described by Weil were later found to have neurohypophyseal adfNDI (OMIM

#192340).³³⁷ Patients with adfNDI retain some limited capacity to secrete AVP during severe dehydration, and the polyuropolydipsic symptoms usually appear after the first year of life, when the infant’s demand for water is more likely to be understood by adults.

The severity in infancy of NDI was clearly described by Crawford and Bode.³³¹ The first manifestations of the disease can be recognized during the first week of life. The infants are irritable, cry almost constantly, and although eager to suck, will vomit milk soon after ingestion unless prefed with water. The history given by the mothers often includes persistent constipation, erratic unexplained fever, and failure to gain weight. Even though the patients characteristically show no visible evidence of perspiration, increased water loss during fever or in warm weather exaggerates the symptoms. Unless the condition is recognized early, children experience frequent bouts of hypertonic dehydration, sometimes complicated by convulsions or death. Mental retardation is a common consequence of these episodes. The intake of large quantities of water, combined with the patient’s voluntary restriction of dietary salt and protein intake, lead to hypocaloric dwarfism beginning in infancy. Affected children frequently have lower urinary tract dilation and obstruction, probably secondary to the large volume of urine produced. Dilation of the lower urinary tract is also seen in patients with primary polydipsia and in patients with neurogenic diabetes insipidus.^{338,339}

Chronic renal insufficiency may occur by the end of the first decade of life and could be the result of episodes of dehydration with thrombosis of the glomerular tufts. Almost 30 years ago the authors’ group observed that the administration of desmopressin, a V₂ receptor agonist, caused an increase in plasma cAMP concentrations in normal subjects but had no effect in 14 male patients with X-linked NDI.³⁴⁰

Intermediate responses were observed in obligate carriers of the disease, possibly corresponding to half of the normal receptor response. On the basis of these results, our group predicted that the defective gene in these patients with X-linked NDI was likely to code for a defective V₂ receptor (Fig. 44.18).³⁴⁰

X-linked NDI is a rare disease, with an estimated prevalence of approximately 8.8 per million male live births in the province of Quebec (Canada).³⁴¹ In defined regions of North America, the prevalence is much higher. Our group estimated the incidence in Nova Scotia and New Brunswick (Canada) to be 58 per million male live births because of shared ancestry.³⁴²

An additional example has been identified in a Mormon pedigree whose members reside in Utah (Utah families). This pedigree was originally described by Cannon.³⁴² The “Utah mutation” is a nonsense mutation (L312X) predictive of a receptor that lacks transmembrane domain 7 and the intracellular COOH terminus.³⁴²

The largest known kindred with X-linked NDI is the Hopewell family, named after the Irish ship “Hopewell,” which arrived in Halifax, Nova Scotia, in 1761. Aboard the ship were members of the Ulster Scot clan, descendants of Scottish Presbyterians who migrated to Ulster province in Ireland in the 17th century and left Ireland for the New World in the 18th century. Although families arriving with the first emigration wave settled in northern Massachusetts in 1718, the members of a second emigration wave, passengers

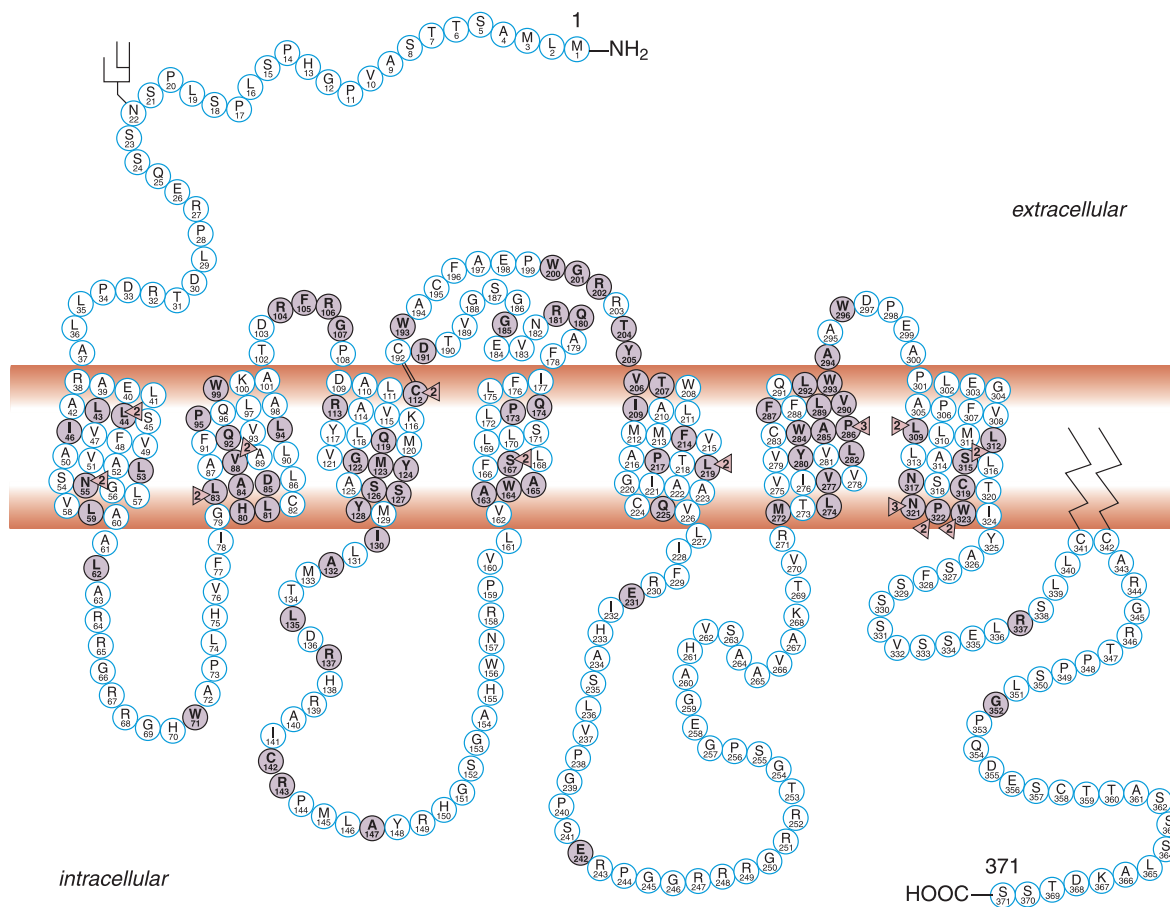


Fig. 44.18 Schematic representation of the V_2 receptor (AVPR2) and identification of 193 putative disease-causing AVPR2 mutations. Predicted amino acids are shown as their one-letter amino acid codes. A solid symbol indicates a codon with a missense or nonsense mutation; a number (within a triangle) indicates more than one mutation in the same codon; other types of mutations are not indicated on the figure. There are 95 missense, 18 nonsense, 46 frameshift deletion or insertion, seven in-frame deletion or insertion, four splice-site, and 22 large deletion mutations, and one complex mutation.

of the “Hopewell,” settled in Colchester County, Nova Scotia. According to the “Hopewell hypothesis,”³⁴³ most patients with NDI in North America are progeny of female carriers of the second emigration wave. This assumption is based mainly on the high prevalence of NDI among descendants of the Ulster Scots residing in Nova Scotia. In two villages with a total of 2500 inhabitants, 30 patients have been diagnosed, and the carrier frequency has been estimated at 6%.

Given the numerous mutations found in North American X-linked NDI families, the Hopewell hypothesis cannot be upheld in its originally proposed form. However, among X-linked NDI patients in North America, the W71X mutation (the Hopewell mutation) is more common than the other AVPR2 mutation. It is a null mutation (W71X), predictive of an extremely truncated receptor consisting of the extracellular NH_2 terminus, the first transmembrane domain, and the NH_2 -terminal half of the first intracellular loop. Because the original carrier cannot be identified, it is not clear whether the Hopewell mutation was brought to North America by “Hopewell” passengers or by other Ulster Scot immigrants. The diversity of AVPR2 mutations found in many ethnic groups (Europeans, Japanese, African Americans, and Africans) and the low frequency of the disease are consistent with an X-linked recessive disease that in the past was lethal for male

patients and was balanced by recurrent mutations. In X-linked NDI, loss of mutant alleles from the population occurs because of the higher mortality of affected male patients compared with healthy male patients, whereas gain of mutant alleles occurs by mutation. If affected male patients with a rare X-linked recessive disease do not reproduce and if mutation rates are equal in mothers and fathers, then, at genetic equilibrium, one-third of new cases in affected male patients will be caused by new mutations. Our group has described ancestral mutations, de novo mutations, and potential mechanisms of mutagenesis. These data are reminiscent of those obtained from patients with late-onset autosomal dominant retinitis pigmentosa. In one-fourth of patients, the disease is caused by mutations in the light receptor rhodopsin. Here, too, many different mutations (approximately 100) spread throughout the coding region of the rhodopsin gene have been found.³⁴⁴

The basis of loss of function or dysregulation of 28 different mutant V_2 receptors (including nonsense, frameshift, deletion, and missense mutations) has been studied with the use of in vitro expression systems. Most of the mutant V_2 receptors tested were not transported to the cell membrane and were retained within the intracellular compartment. Our group also demonstrated that misfolded AVPR2 mutants could be

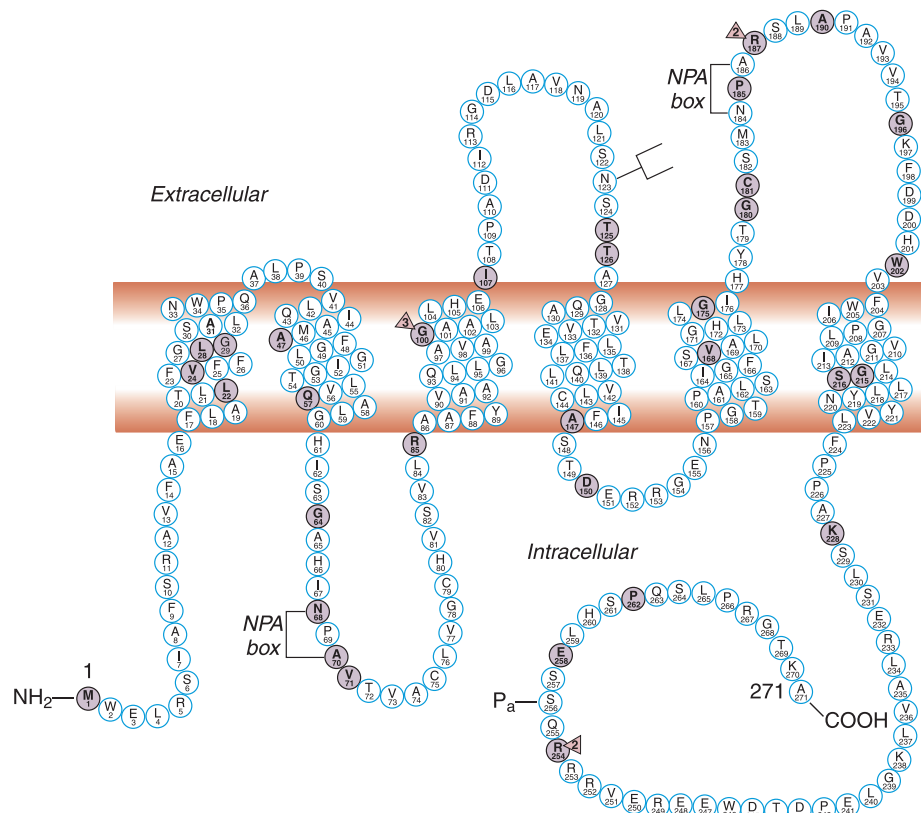


Fig. 44.19 A representation of the aquaporin-2 (AQP2) protein and identification of 46 putative disease-causing AQP2 mutations. Predicted amino acids are shown as their one-letter amino acid codes. A monomer is represented with six transmembrane helices. The location of the protein kinase A (PKA) phosphorylation site (Pa) is indicated. The extracellular, transmembrane, and cytoplasmic domains are defined according to Deen et al. 1994.³⁶¹ Triangles are indicating amino acids with more than one mutation in the same codon. Solid symbols indicate locations of the mutations: M1I; L22V; V24A; L28P; G29S; A47V; Q57P; G64R; N68S; A70D; V71M; R85X; G100X; G100V; G100R; I107D; 369delC; T125M; T126M; A147T; D150E; V168M; G175R; G180S; C181W; P185A; R187C; R187H; A190T; G196D; W202C; G215C; S216P; S216F; K228E; R254Q; R254L; E258K; and P262L. GenBank accession numbers – AQP2: AF147092, exon 1; AF147093, exons 2 through 4. NPA (asparagine–proline–alanine) motifs and the N-glycosylation site are also indicated by the fork-like symbol on amino acid N123.

rescued in vitro but also in vivo by nonpeptide vasopressin antagonists acting as pharmacologic chaperones.^{345,346} This new therapeutic approach could be applied to the treatment of several hereditary diseases resulting from errors in protein folding and kinesis.³⁴⁷

Only four *AVPR2* mutations (D85N, V88M, G201D, and P322S) have been associated with a mild phenotype.³⁴⁸ In general, the male infants bearing these mutations are identified later in life and the “classic” episodes of dehydration are less severe. This mild phenotype is also found in expression studies. The mutant proteins are expressed on the plasma membranes of cells transfected with these mutants and demonstrate a stimulation of cAMP for higher concentrations of agonists.

In contrast with NDI because of inactivating mutations in *AVPR2*, mutations which result in constitutive hormone-independent activity cause nephrogenic syndrome of inappropriate antidiuresis, often manifesting as hyponatremia,³⁴⁹ which is considered in greater depth in Chapter 15.

LOSS-OF-FUNCTION MUTATIONS OF AQP2 (OMIM #107777)

The *AQP2* gene is located on chromosome region 12q12-q13. Approximately 10% of NDI cases are caused by autosomal

mutations in *AQP2*. Forty-six mutations have been reported, which are either autosomal recessive (32 mutations reported) or autosomal dominant (8 mutations reported).³⁵⁰ Both male and female patients who are affected with congenital NDI have been described as homozygous for a mutation in the *AQP2* gene or carry two different mutations (Fig. 44.19). Autosomal recessive mutations give rise to misfolded proteins that are retained in the ER and are eventually degraded. Autosomal dominant mutations are believed to be restricted to the carboxy-terminal end of the AQP2 protein and to operate through a dominant negative effect whereby the mutant protein associates with functional AQP2 proteins within intracellular stores, thus preventing normal targeting and function.³⁵¹

Oocytes of the African clawed frog (*Xenopus laevis*) have provided a useful system for studying the behavior of various AQP2 mutants on membrane water permeability. Functional expression studies showed that *Xenopus* oocytes injected with mutant complementary RNA (cRNA) had abnormal coefficient of water permeability, whereas *Xenopus* oocytes injected with both normal cRNA and mutant cRNA had coefficient of water permeability similar to that of normal constructs alone. These findings provide conclusive evidence that NDI can be caused by homozygosity for mutations in the *AQP2* gene.³⁵⁰

POLYURIA, POLYDIPSIA, ELECTROLYTE IMBALANCE, AND DEHYDRATION IN CYSTINOSIS

Polyuria may be as mild as persistent enuresis or so severe as to contribute to death from dehydration and electrolyte abnormalities in infants with cystinosis who have acute gastroenteritis.⁴⁴

POLYURIA IN HEREDITARY HYPOKALEMIC SALT-LOSING TUBULOPATHIES

Patients with polyhydramnios, hypercalciuria, and hyposthenuria or isosthenuria have been found to bear *KCNJ1* (ROMK) and *SLC12A1* (NKCC2) mutations.^{238,248} In a recent analysis of 115 patients bearing *CLCNKB* mutations, 29.5% had early manifestations with polyhydramnios or early manifestations in the first month of life.³²⁸ Patients with polyhydramnios, profound polyuria, hyponatremia, hypochloremia, metabolic alkalosis, and sensorineural deafness have been found to bear *BSND* mutations.²⁴² The *MAGED2* protein is increasing the expression of NKCC2 and NCC of the thick ascending loop of Henle and the distal convoluted tubule.³²⁹ These studies demonstrate the critical importance of the proteins ROMK, NKCC2, and Barttin in transferring NaCl to the medullary interstitium and thereby generating, together with urea, a hypertonic milieu (see Fig. 44.10). Of clinical and diagnostic importance, polyhydramnios is never observed during the pregnancy of infants later found to have *AVPR2* or *AQP2* mutations.³²⁶

CARRIER DETECTION, PERINATAL TESTING, AND TREATMENT

We encourage physicians who observe families with X-linked and autosomal recessive diabetes insipidus to recommend mutation analysis before the birth of an infant because early diagnosis and treatment can avert the physical and mental retardation associated with episodes of dehydration. Diagnosis of X-linked NDI was accomplished by mutation testing of cultured amniotic cells or chorionic villus samples ($n = 17$), or of cord blood obtained at birth ($n = 65$) in 82 of our patients from 69 families. Some 36 male patients were found to bear mutant sequences, and 26 had normal sequences. Diagnosis of *AQP2* autosomal recessive mutants was made in four families for a total of six subjects, of whom three were found to be homozygous for the previously identified mutation, two were heterozygous, and one had a normal sequence on both alleles. The affected patients were immediately treated with abundant water intake, a low-sodium diet, and hydrochlorothiazide. They never experienced episodes of dehydration, and their physical and mental development was normal. Gene analysis is also important for the identification of nonobligatory female carriers in families with X-linked NDI. Most female patients heterozygous for a mutation in the V_2 receptor do not present with clinical symptoms, and a few are severely affected³⁴¹ (Bichet, unpublished observations). Mutational analysis of polyuric patients with cystinosis, hypokalemic salt-losing tubulopathy, nephronophthisis, and AME is also of importance for definitive molecular diagnosis.

All complications of congenital NDI can be prevented by an adequate water intake. Thus patients should be

provided with unrestricted amounts of water from birth to ensure normal development. In addition to a low-sodium diet, the use of diuretics (thiazides) or indomethacin may reduce urinary output. This advantageous effect must be weighed against the side effects of these drugs (thiazides: electrolyte disturbances; indomethacin: reduction of the GFR and gastrointestinal symptoms). Many affected infants frequently vomit because of an exacerbation of physiologic gastroesophageal reflux. These young patients often improve with the absorption of a histamine H_2 blocker and with metoclopramide, which could induce extrapyramidal symptoms, or domperidone, which seems to be efficacious and better tolerated.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Indicate the true propositions below:
- Renal Fanconi syndrome is a generalized dysfunction of both proximal and distal tubules.
 - The genetic forms of Fanconi syndrome are associated with multisystemic disorders including a) accumulation of toxic metabolites (e.g. Cystinosis), b) disruption of energy provision (e.g. mitochondrial cytopathies), c) disruption of endocytosis and intracellular transport (e.g. Dent disease).
 - Diseases associated with the accumulation of toxic metabolites are potentially reversible.
 - In Fanconi syndrome, all amino acids are excreted in excess but these losses do not lead to specific deficiencies.

Answer: b, c, d

2. Which of the following statements are useful for the clinical and molecular characterization of Dent disease?
- Dent disease, X-linked hypophosphatemic rickets and X-linked recessive nephrolithiasis are clinical manifestations of the same disease.
 - Clinical features include primary Fanconi syndrome, low molecular weight proteinuria, hypercalciuria with calcium nephrolithiasis, nephrocalcinosis, rickets and progressive renal failure.
 - Dent disease 1 is caused by mutations in the *CLCN5* gene.
 - Mutations in *OCRL1* gene are found in patients with Lowe syndrome also inducing a Dent-like phenotype.
 - CLCN5* mutations include defects in receptor-mediated endocytosis and endosome acidification.

Answer: all of the above

3. Which of the following statements are true?
- Cystinuria is an autosomal recessive disease of the lysosomal transport of the disulfide amino acid cystine.
 - Cystinuria is now treated with the cysteine-depleting drug cysteamine.
 - Cystinosis is caused by mutations of the genes encoding neutral and basic transport proteins rBAT and b^(O,+)

Answer: none of the above

4. Four of the five following statements are true
- Gitelman syndrome is a thiazide-like salt losing tubulopathy with a defect in the distal convoluted tubule.
 - Bartter syndrome is a genetically heterogeneous disorder affecting the loop of Henle where 30% of the filtered sodium chloride is reabsorbed, with manifestations during the neonatal period.
 - Most cases of Bartter syndrome present late in life.
 - Most cases of Bartter syndrome present antenatally or in neonates. Polyhydramnios and premature labor are common manifestations.

Answer: c

5. All the following statements except one apply to hereditary nephrogenic diabetes insipidus:
- X-linked nephrogenic diabetes insipidus is secondary to mutations in the *AVPR2* gene coding for the vasopressin V2 receptor.
 - Autosomal dominant or recessive nephrogenic diabetes insipidus is secondary to mutations in the *AQP2* gene coding for aquaporin2.
 - The first manifestations, polyuria and polydipsia are present during the first days of life and affected infants are at risk of severe dehydration with hypernatremia.
 - dDAVP administration will rapidly augment urine osmolality.

Answer: d